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## V18G019 - Bill of Materials



| Section       | Bubble Number  | Mech / Elec | Part Number | Parameter | Part Description                                       | Assembly Drawing | Qty      |
|---------------|----------------|-------------|-------------|-----------|--------------------------------------------------------|------------------|----------|
| <b>00</b>     |                |             |             |           | <b>Layouts &amp; Schemes</b>                           |                  |          |
| <b>00-010</b> | <b>910G482</b> |             |             |           | <b>Layouts &amp; Schemes K5-Expert 2900HD (Dahlia)</b> | <b>429G980</b>   | <b>1</b> |
|               | 1              | M           | 329G378     |           | General Arrangement K5 Expert-2900HD (Dahlia)          |                  | 1        |
|               | 2              | M           | 430G096     |           | Foundation Assembly K5-Expert 2900 HD (Dahlia)         |                  | 1        |
|               | 3              | M           | 329G379     |           | Mech Layout K5 Expert-2900HD (Dahlia)                  |                  | 1        |
|               | 4              | M           | 329G380     |           | Vacuum Schematic K5 Expert-2900HD (Dahlia)             |                  | 1        |
|               | 5              | M           | 329G381     |           | Drive Schematic K5 Expert-2900HD (Dahlia)              |                  | 1        |
|               | 6              | M           | 329G382     |           | Water Schematic K5 Expert-2900HD (Dahlia)              |                  | 1        |
|               | 7              | M           | 329G383     |           | Pneumatic Schematic K5 Expert-2900HD (Dahlia)          |                  | 1        |
|               | 8              | M           | 329G384     |           | Installation Drawing K5 Expert-2900HD (Dahlia)         |                  | 1        |
|               | 9              | M           | 329G385     |           | Gas Schematic O2 System K5 Expert-2900HD (Dahlia)      |                  | 1        |
|               | 10             | M           | 329G386     |           | Machine Lubrication Drawing K5 Exp-2900HD(Dahlia)      |                  | 1        |
|               | 11             | M           | 421G192     |           | O2 Exhaust Duct Extraction System User Guide           |                  | 1        |
|               | 12             | M           | 421G193     |           | O2 Exhaust Duct Instructions (non nitrogen dilute)     |                  | 1        |
|               | 13             | M           | 421G194     |           | O2 Exhaust Duct Instructions (Vent-tech)               |                  | 1        |
|               | 14             | M           | 328G366     |           | MACHINE COOLING SYSTEM WATER QUALITY SPEC.             |                  | 1        |
|               | 15             | M           | 328G587     |           | Cable Route                                            |                  | 1        |
|               | 16             | M           | 245G390     |           | K5 Expert - Power Schematic - Dahlia 2900              |                  | 1        |
|               | 17             | M           | 329G517     |           | Foundation Cable Tray Requirement                      |                  | 1        |
| <b>01</b>     |                |             |             |           | <b>Vessel External</b>                                 |                  |          |
| <b>01-010</b> | <b>910G170</b> |             |             |           | <b>Vessel Assembly with Roof Tracks K5-Expert</b>      | <b>428G780</b>   | <b>1</b> |
|               | B001           | M           | 239250      |           | Seal O Ring RM0793-57 Nitrile                          |                  | 2        |
|               | B002           | M           | 239204      |           | Seal O Ring RM3393-57 Nitrile                          |                  | 1        |
|               | B003           | E           | 6131478G    |           | Swifts Cable ladder hold down bracket                  |                  | 6        |
|               | M001           | M           | 428G781     |           | Vacuum Vessel Assy K5000-2900                          |                  | 1        |
|               | M002           | M           | 419G478     |           | 2900 Roof Track Beam                                   |                  | 1        |
|               | M003           | M           | 419G479     |           | 2900 Roof Track Beam                                   |                  | 1        |
|               | M004           | M           | 402G660     |           | Roof Track End Plate                                   |                  | 1        |
|               | M005           | M           | 403G770     |           | Runway Beam Packer                                     |                  | 16       |
|               | M006           | M           | 432G568     |           | 4" Vent Diffuser                                       |                  | 1        |
|               | M007           | M           | 303G349     | IT.C      | Electrical Leadthrough Flange                          |                  | 2        |
|               | M009           | M           | 411G663     |           | Roof Track End Plate                                   |                  | 1        |
|               | M010           | M           | 412G277     |           | COVER PLATE                                            |                  | 2        |
|               | M011           | M           | 419G997     |           | TRANSPORT BRACKET                                      |                  | 2        |
|               | M012           | M           | 420G364     | IT1       | TRANSPORT BRACKET                                      |                  | 1        |
|               | M013           | M           | 420G364     | IT2       | TRANSPORT BRACKET                                      |                  | 1        |
|               | M014           | M           | 422G946     |           | Blanking Plate                                         |                  | 1        |
|               | M015           | M           | 423G930     |           | Cable Ladder Bracket                                   |                  | 3        |
|               | M016           | M           | 427G928     | IT1       | Cable Tray Bracket                                     |                  | 1        |
|               | M017           | M           | 427G928     | IT2       | Cable Tray Bracket                                     |                  | 2        |
|               | S001           | M           | 421575      |           | Bolt HEXHD M12x35 GRADE 8.8                            |                  | 8        |
|               | S002           | M           | 421574      |           | Bolt HEXHD M20x50                                      |                  | 48       |
|               | S003           | M           | 421030      |           | PLAIN WASHER M12                                       |                  | 8        |
|               | S004           | M           | 421070      |           | PLAIN WASHER M20                                       |                  | 80       |
|               | S006           | M           | 421072      |           | NUT FULL HEXAGON M20                                   |                  | 16       |
|               | S007           | M           | 4211299     |           | SOCKET CAP SCREW M20 x 90 GRADE 12.9                   |                  | 8        |
|               | S008           | M           | 4211428     |           | SOCKET CAP SCREW M20 x 120 GRADE 12.9                  |                  | 8        |
|               | S009           | M           | 421019      |           | Socket Head Capscrew M6 x 20 GRADE 12.9                |                  | 16       |
|               | S010           | M           | 421034      |           | WASHER PLAIN M6                                        |                  | 20       |
|               | S011           | M           | 421022      |           | Washer Spring M6                                       |                  | 8        |
|               | S013           | M           | 421033      |           | SOCKET CAP SCREW M6 x 12 GRADE 12.9                    |                  | 4        |
|               | S014           | M           | 421122      |           | SOCKET CAP SCREW M10 x 30 GRADE 12.9                   |                  | 12       |
|               | S015           | M           | 421029      |           | PLAIN WASHER M10                                       |                  | 30       |
|               | S016           | M           | 421035      |           | SOCKET CAP SCREW M10 x 25 GRADE 12.9                   |                  | 12       |
|               | S017           | M           | 421027      |           | NUT FULL HEX M10                                       |                  | 6        |
| <b>01-020</b> | <b>908G633</b> |             |             |           | <b>Source Sight Glass Assembly</b>                     | <b>421G665</b>   | <b>2</b> |
|               | B001           | M           | 422003      |           | Hinge Chrome Dirak 217-9110                            |                  | 2        |
|               | B002           | M           | 235182      |           | Spring Compression RS751-607                           |                  | 4        |
|               | B003           | M           | 422080G     |           | Toggle Clamp WDS 4070-201                              |                  | 2        |
|               | B004           | M           | 239862G     |           | Seal O Ring 8.4c/sx536mm mean dia Nitrile              |                  | 1        |
|               | B005           | M           | 421039      |           | Screw Shoulder Screw 8mm (M6) x 25lg                   |                  | 4        |
|               | B006           | M           | 2441916     |           | Potted Magnet                                          |                  | 2        |
|               | B007           | M           | 268224G     |           | Loctite GlassBond 3ml Tube                             |                  | 1        |
|               | B008           | M           | 240027G     |           | ADJUSTABLE BOLT COVER                                  |                  | 2        |
|               | M001           | M           | 421G666     | IT1       | Polycarbonate Window                                   |                  | 1        |
|               | M002           | M           | 421G667     |           | Window Clamp Block                                     |                  | 2        |
|               | M003           | M           | 319G635     |           | WINDOW HINGE CAR BLK                                   |                  | 2        |
|               | M004           | M           | 421G670     | IT660     | Glass Panel                                            |                  | 1        |

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|---------------|----------------|-------------|-------------|-----------|----------------------------------------------------|------------------|----------|
|               | M005           | M           | 421G672     |           | Bracket                                            |                  | 2        |
|               | S001           | M           | 4211900G    |           | Screw HEXHD Setscrew M5x30 GRADE 8.8               |                  | 2        |
|               | S002           | M           | 421239      |           | Screw Socket M5x12 CSKHD                           |                  | 4        |
|               | S003           | M           | 421023      |           | Screw Countersunk Head Socket M5x16                |                  | 4        |
|               | S004           | M           | 421147      |           | SOCKET CAP SCREW M4 x 12 GRADE 12.9                |                  | 8        |
|               | S005           | M           | 421054      |           | PLAIN WASHER M8                                    |                  | 4        |
|               | S006           | M           | 421164      |           | WASHER PLAIN M5                                    |                  | 2        |
|               | S007           | M           | 421194      |           | Washer Plain M4                                    |                  | 8        |
|               | S008           | M           | 421138      |           | SOCKET CAP SCREW M8 x 25 GRADE 12.9                |                  | 4        |
| <b>01-021</b> | <b>908G633</b> |             |             |           | <b>Source Sight Glass Assembly</b>                 | <b>421G665</b>   | <b>2</b> |
|               | B001           | M           | 422003      |           | Hinge Chrome Dirak 217-9110                        |                  | 2        |
|               | B002           | M           | 235182      |           | Spring Compression RS751-607                       |                  | 4        |
|               | B003           | M           | 422080G     |           | Toggle Clamp WDS 4070-201                          |                  | 2        |
|               | B004           | M           | 239863G     |           | Seal O Ring 8.4c/sx663mm mean dia Nitrile          |                  | 1        |
|               | B005           | M           | 421039      |           | Screw Shoulder Screw 8mm (M6) x 25lg               |                  | 4        |
|               | B006           | M           | 2441916     |           | Potted Magnet                                      |                  | 2        |
|               | B007           | M           | 268224G     |           | Loctite GlassBond 3ml Tube                         |                  | 1        |
|               | B008           | M           | 240027G     |           | ADJUSTABLE BOLT COVER                              |                  | 2        |
|               | M001           | M           | 421G666     | IT2       | Polycarbonate Window                               |                  | 1        |
|               | M002           | M           | 421G667     |           | Window Clamp Block                                 |                  | 2        |
|               | M003           | M           | 319G635     |           | WINDOW HINGE CAR BLK                               |                  | 2        |
|               | M004           | M           | 421G670     | IT860     | Glass Panel                                        |                  | 1        |
|               | M005           | M           | 421G672     |           | Bracket                                            |                  | 2        |
|               | S001           | M           | 4211900G    |           | Screw HEXHD Setscrew M5x30 GRADE 8.8               |                  | 2        |
|               | S002           | M           | 421239      |           | Screw Socket M5x12 CSKHD                           |                  | 4        |
|               | S003           | M           | 421023      |           | Screw Countersunk Head Socket M5x16                |                  | 4        |
|               | S004           | M           | 421147      |           | SOCKET CAP SCREW M4 x 12 GRADE 12.9                |                  | 8        |
|               | S005           | M           | 421054      |           | PLAIN WASHER M8                                    |                  | 4        |
|               | S006           | M           | 421164      |           | WASHER PLAIN M5                                    |                  | 2        |
|               | S007           | M           | 421194      |           | Washer Plain M4                                    |                  | 8        |
|               | S008           | M           | 421138      |           | SOCKET CAP SCREW M8 x 25 GRADE 12.9                |                  | 4        |
| <b>01-025</b> | <b>903G484</b> |             |             |           | <b>Rectangular Sightglass Assy (Mech Viewing)</b>  | <b>403G961</b>   | <b>2</b> |
|               | B001           | M           | 204117      |           | 32774 Nylon Flanged Bush                           |                  | 16       |
|               | B002           | M           | 239624G     |           | O RINGS 8.4MM CROSS SECTION 560MM MEAN DIAMETER    |                  | 1        |
|               | M001           | M           | 403G963     | IT.1      | Polycarbonate Window                               |                  | 1        |
|               | S001           | M           | 421018      |           | SOCKET CAP SCREW M10 x 35 GRADE 12.9               |                  | 16       |
|               | S002           | M           | 421029      |           | PLAIN WASHER M10                                   |                  | 16       |
| <b>01-026</b> | <b>903G484</b> |             |             |           | <b>Rectangular Sightglass Assy (Mech Viewing)</b>  | <b>403G961</b>   | <b>2</b> |
|               | B001           | M           | 204117      |           | 32774 Nylon Flanged Bush                           |                  | 20       |
|               | B002           | M           | 239623G     |           | O RINGS 8.4MM CROSS SECTION 687MM MEAN DIAMETER    |                  | 1        |
|               | M001           | M           | 403G963     | IT.2      | Polycarbonate Window                               |                  | 1        |
|               | S001           | M           | 421018      |           | SOCKET CAP SCREW M10 x 35 GRADE 12.9               |                  | 20       |
|               | S002           | M           | 421029      |           | PLAIN WASHER M10                                   |                  | 20       |
| <b>01-030</b> | <b>909G430</b> |             |             |           | <b>Arrangement of 400 Square Sightglass</b>        | <b>425G384</b>   | <b>2</b> |
|               | B001           | M           | 204117      |           | 32774 Nylon Flanged Bush                           |                  | 8        |
|               | B002           | M           | 239204      |           | Seal O Ring RM3393-57 Nitrile                      |                  | 1        |
|               | M001           | M           | 425G378     |           | Square Viewing Window                              |                  | 1        |
|               | REF1           | M           | 425G433     |           | Square Viewing Port Pad (REF ONLY)                 |                  | 1        |
|               | S001           | M           | 421114      |           | SOCKET CAP SCREW M10 x 40 GRADE 12.9               |                  | 8        |
|               | S002           | M           | 421029      |           | PLAIN WASHER M10                                   |                  | 8        |
| <b>01-040</b> | <b>908G442</b> |             |             |           | <b>DIFFUSION PUMP LID ASSEMBLY</b>                 | <b>424G123</b>   | <b>2</b> |
|               | B001           | M           | 210910      |           | Cylinder DNC-125-160-PPV-A                         |                  | 1        |
|               | B002           | M           | 211373      |           | REED SWITCH & CONNECTOR                            |                  | 2        |
|               | B003           | E           | 6131457G    |           | M8 Male - M8 Female 3 Pin Conn. Cable PUR/PVC 0.5M |                  | 2        |
|               | B004           | M           | 211358      |           | Swivel Flange                                      |                  | 1        |
|               | B005           | M           | 427142      |           | Rod Clevis with Pin & Locknut 14987                |                  | 1        |
|               | B006           | M           | 2061164G    |           | Linear Set LSG-M-20-DD R1065 220 00                |                  | 4        |
|               | B007           | M           | 2061163G    |           | Bearing Shaft 20 dia x 432 lg                      |                  | 2        |
|               | B008           | M           | 205233G     |           | SPHERICAL BEARING - SKF GE 25C                     |                  | 1        |
|               | B009           | M           | 200M569     |           | DEEP GROOVE BEARING 4208A                          |                  | 1        |
|               | B010           | M           | 239852G     |           | SEAL SIMMERRING BABSL45-62-7FKM                    |                  | 2        |
|               | B011           | M           | 233136      |           | CIRCLIP INTERNAL D1300-0620                        |                  | 1        |
|               | B012           | M           | 234005      |           | CIRCLIP EXTERNAL 1400-0400                         |                  | 2        |
|               | B013           | M           | 200762      |           | SPERICAL PLAIN BEARING GE 20 TXE-2RS               |                  | 1        |
|               | B014           | M           | 200763      |           | SPERICAL PLAIN BEARING GE 40 TXE-2LS               |                  | 1        |
|               | B015           | M           | 214025      |           | Nipple Grease M6                                   |                  | 1        |
|               | B016           | M           | 203269      |           | BUSH GLACIER MB 2530DU                             |                  | 2        |
|               | B017           | M           | 203089      |           | BUSH DRY MB3030DU                                  |                  | 1        |

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|               | B018           | M           | 239198      |           | Seal O Ring RM0745-30 Nitrile                      |                  | 1        |
|               | B019           | M           | 239255      |           | Seal O Ring RM0795-30 Nitrile                      |                  | 1        |
|               | B020           | M           | 239356      |           | Seal O Ring RM0895-30Nitrile                       |                  | 1        |
|               | B021           | M           | 239282      |           | Seal O Ring RM1295-30 Nitrile                      |                  | 1        |
|               | B022           | M           | 239786G     |           | Seal O Ring 8.4C/S x 813 mean dia Nitrile          |                  | 1        |
|               | B023           | M           | 427440G     |           | Reducing Bush 3/8"BSP to 1/2"BSP SS                |                  | 2        |
|               | B024           | M           | 225354      |           | ZINC PLATED CHAIN                                  |                  | 1        |
|               | B025           | M           | 239671G     |           | Seal O Ring RM0355-30                              |                  | 1        |
|               | B027           | M           | 200859G     |           | Bearing Inner Race 40-45-20.5                      |                  | 2        |
|               | B028           | M           | 4211989G    |           | CAPTIVE PANEL SCREW - M6x1                         |                  | 2        |
|               | M001           | M           | 424G125     |           | Diff Pump Lid Shaft                                |                  | 1        |
|               | M002           | M           | 420G937     |           | Diff Pump Lid Outer Seal Cap                       |                  | 1        |
|               | M003           | M           | 424G124     |           | Diff Pump Lid Bearing Seal Housing                 |                  | 1        |
|               | M004           | M           | 420G939     |           | Diff Pump Lid Retaining Plate                      |                  | 1        |
|               | M005           | M           | 420G940     |           | Diff Pump Lid Drive Lever                          |                  | 1        |
|               | M006           | M           | 408G795     |           | BEARING HOUSING                                    |                  | 1        |
|               | M007           | M           | 408G796     |           | CLAMP PLATE                                        |                  | 1        |
|               | M008           | M           | 420G941     |           | Diff Pump Lid Lever Arm                            |                  | 1        |
|               | M009           | M           | 410G315     |           | STUD CAM FOLLOWER                                  |                  | 1        |
|               | M010           | M           | 410G316     |           | Spacer                                             |                  | 1        |
|               | M011           | M           | 421G082     |           | Diff Pump Lid Mounting Plate                       |                  | 1        |
|               | M012           | M           | 421G089     |           | Diff Pump Lid Mounting Bracket                     |                  | 1        |
|               | M013           | M           | 421G075     |           | Diff Pump Lid Carriage                             |                  | 1        |
|               | M014           | M           | 410G317     |           | WEAR STRIP                                         |                  | 2        |
|               | M015           | M           | 421G085     |           | Diff Pump Lid Drive Pin                            |                  | 1        |
|               | M016           | M           | 421G076     |           | Diff Pump Lid Trunion                              |                  | 1        |
|               | M017           | M           | 421G053     |           | LID PLATE DIFF PUMP                                |                  | 1        |
|               | M018           | M           | 421G077     |           | Diff Pump Lid Hanger Bracket                       |                  | 1        |
|               | M019           | M           | 407G225     |           | Pivot Bracket                                      |                  | 1        |
|               | M020           | M           | 410G888     |           | Safety Pin                                         |                  | 1        |
|               | M021           | M           | 421G078     |           | Diff Pump Lid Tailshaft Bracket                    |                  | 4        |
|               | M022           | M           | 418G248     | IT15      | Key                                                |                  | 2        |
|               | M023           | M           | 426G774     |           | Diff Lid Cylinder Guard                            |                  | 1        |
|               | REF1           | M           | 421G055     |           | Diff pump lid handle                               |                  | 2        |
|               | S001           | M           | 421703      |           | SOCKET CAP SCREW M4 x 16 GRADE 12.9                |                  | 16       |
|               | S002           | M           | 421171      |           | SOCKET CAP SCREW M5 x 16 GRADE 12.9                |                  | 14       |
|               | S003           | M           | 421368      |           | SOCKET CAP SCREW M5 x 25 GRADE 12.9                |                  | 10       |
|               | S004           | M           | 421033      |           | SOCKET CAP SCREW M6 x 12 GRADE 12.9                |                  | 2        |
|               | S005           | M           | 421049      |           | SOCKET CAP SCREW M8 x 20 GRADE 12.9                |                  | 10       |
|               | S006           | M           | 421237      |           | SOCKET CAP SCREW M8 x 35 GRADE 12.9                |                  | 3        |
|               | S007           | M           | 421278      |           | SOCKET CAP SCREW M8 x 45 GRADE 12.9                |                  | 4        |
|               | S008           | M           | 421122      |           | SOCKET CAP SCREW M10 x 30 GRADE 12.9               |                  | 4        |
|               | S009           | M           | 421114      |           | SOCKET CAP SCREW M10 x 40 GRADE 12.9               |                  | 3        |
|               | S010           | M           | 421210      |           | SOCKET CAP SCREW M12 x 35 GRADE 12.9               |                  | 4        |
|               | S011           | M           | 421526      |           | SOCKET CAP SCREW M12 x 45 GRADE 12.9               |                  | 4        |
|               | S012           | M           | 421164      |           | WASHER PLAIN M5                                    |                  | 8        |
|               | S013           | M           | 421034      |           | WASHER PLAIN M6                                    |                  | 2        |
|               | S014           | M           | 421054      |           | PLAIN WASHER M8                                    |                  | 17       |
|               | S015           | M           | 421986      |           | Washer Large Heavy M10 Form C (Mild Steel)         |                  | 6        |
|               | S016           | M           | 421030      |           | PLAIN WASHER M12                                   |                  | 9        |
|               | S017           | M           | 421027      |           | NUT FULL HEX M10                                   |                  | 3        |
|               | S018           | M           | 421042      |           | NUT FULL HEX M12                                   |                  | 1        |
|               | S019           | M           | 245011G     |           | M12 x 30 GRUB SCREW                                |                  | 4        |
|               | S020           | M           | 421172      |           | SOCKET CAP SCREW M5 x 12 GRADE 12.9                |                  | 2        |
| <b>01-060</b> | <b>910G591</b> |             |             |           | <b>OPERATOR PANEL</b>                              | <b>430G738</b>   | <b>1</b> |
|               | B001           | E           | 605051G     |           | Hoffman Straight Coupling 26901134                 |                  | 1        |
|               | B002           | E           | 605045G     |           | Syspend 281-Max Tubes Length 1000mm                |                  | 1        |
|               | B003           | E           | 605046G     |           | Hoffman Reduction Elbow 26901133                   |                  | 1        |
|               | B004           | E           | 605040G     |           | Hoffman Syspend 281-Max Set Up Joint               |                  | 1        |
|               | B005           | M           | 239848G     |           | Neoprene rubber seal                               |                  | 1        |
|               | B008           | E           | 605047G     |           | Hoffman Steel Suspension tube Dia 48mm x 1000mm lg |                  | 1        |
|               | M001           | M           | 430G744     |           | OPERATOR ARM STUD                                  |                  | 4        |
|               | M002           | M           | 430G753     | ITEM.2    | SUSPENSION TUBE                                    |                  | 1        |
|               | M003           | M           | 423G981     |           | Operator Arm Stops                                 |                  | 2        |
|               | S001           | M           | 421121      |           | SOCKET CAP SCREW M6 x 16 GRADE 12.9                |                  | 8        |
|               | S002           | M           | 421034      |           | WASHER PLAIN M6                                    |                  | 8        |
|               | S003           | M           | 421049      |           | SOCKET CAP SCREW M8 x 20 GRADE 12.9                |                  | 6        |
|               | S004           | M           | 421054      |           | PLAIN WASHER M8                                    |                  | 2        |

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|---------|---------------|-------------|-------------|-----------|----------------------------------------------------|------------------|-----|
| 01-070  | 909G461       |             |             |           | <b>Diffusion Pump Access Panel</b>                 | 424G743          | 2   |
|         | B001          | M           | 239857G     |           | Seal O Ring 8.4c/sx880mm mean dia Nitrile          |                  | 1   |
|         | M001          | M           | 424G744     |           | Aluminium Access Panel                             |                  | 1   |
|         | S001          | M           | 421638      |           | Screw Countersunk Head Socket M10x30               |                  | 14  |
| 01-080  | 903G581       |             |             |           | <b>Chuck port blanking flanges</b>                 | 404G285          | 2   |
|         | B001          | M           | 239406      |           | Seal O Ring 5.7c/sx606mm mean dia Nitrile          |                  | 1   |
|         | M001          | M           | 411G156     | IT1       | Laser Cut Blind Flange                             |                  | 1   |
|         | S001          | M           | 421059      |           | Washer Plain M16                                   |                  | 6   |
|         | S002          | M           | 4211433     |           | SOCKET CAP SCREW M16 x 30 GRADE 12.9               |                  | 6   |
| 01-090  | 908G661       |             |             |           | <b>WATER PIPING ASSY (VESSEL END)</b>              | 421G061          | 1   |
|         | B001          | M           | 424341G     |           | 1/2" BSPT Ball Valve M/F (Full Bore)               |                  | 7   |
|         | B002          | M           | 4231622G    |           | Reducing Bush 1/2" x 1/4" BSPT S.S                 |                  | 1   |
|         | B003          | M           | 423813      |           | Nipple Hexagon 1/2" BSP ST.ST.                     |                  | 1   |
|         | B004          | M           | 4232230G    |           | Nipple Crane 2.1/2"BSP SS Hex 144/65               |                  | 1   |
|         | B005          | M           | 423751      |           | Plug Solid 11/2"BSP Crane Type 148/40 SS           |                  | 2   |
|         | B006          | M           | 4231703G    |           | Plug Solid 1/2"BSPT Crane Type 148/15 St Steel     |                  | 10  |
|         | B007          | M           | 4232086G    |           | Blanking Plug 3/4"BSPP VSTI-12-EDA3CSS Viton       |                  | 2   |
|         | B008          | M           | 4231053     |           | Elbow Male/Female 1/2" BSP SS                      |                  | 2   |
|         | B009          | M           | 423750      |           | Socket 1/2" BSP St Steel                           |                  | 1   |
|         | B010          | M           | 4232234G    |           | Union Male/Female 2.1/2" BSP SS 257-65             |                  | 1   |
|         | B011          | M           | 423026      |           | Hose Adaptor 1/2" BSPT x 1/2" Hose                 |                  | 6   |
|         | B012          | M           | 4232253G    |           | Rubber Water Hose 2.1/2" Bore                      |                  | 3   |
|         | B013          | M           | 4232102G    |           | IFM Adaptor G1 - G1 E40217 for SM8000              |                  | 2   |
|         | B014          | M           | 4231922G    |           | IFM SM8004 Magnetic Flow Meter 100 l/min           |                  | 1   |
|         | B015          | M           | 4232291G    |           | Union Elbow 1" BSP M/F SS                          |                  | 2   |
|         | B016          | M           | 4232002G    |           | Adaptor Stem 1"BSPT x 1" HOSE                      |                  | 2   |
|         | B017          | M           | 4232231G    |           | Stem Tailpiece Adaptor 2.1/2" BSPT x 2.1/2" Hose   |                  | 4   |
|         | B018          | M           | 288941G     |           | Asco E290A488 Angle Seat Valve G2.1/2" NC          |                  | 1   |
|         | B019          | M           | 427073      |           | Swagelok Male Hose Adaptor                         |                  | 2   |
|         | B020          | M           | 423690      |           | IFM Efector 500 G1/4" - G 1/2" Adaptor E30000      |                  | 3   |
|         | B021          | M           | 4232229G    |           | IFM TM4411 TEMPERATURE SENSOR                      |                  | 2   |
|         | B022          | M           | 4231592G    |           | High Torque Jubilee Clip 60-80mm                   |                  | 8   |
|         | B023          | M           | 423689      |           | IFM Efector 500 Pressure Transmitter PA3024        |                  | 3   |
|         | B024          | M           | 424246G     |           | Gauge Pressure 65dia 6Bar1/4BSPT                   |                  | 1   |
|         | B025          | M           | 423483      |           | DN100 PN16 IBC asbestos free j/ring gasket 1mm thk |                  | 4   |
|         | B026          | M           | 4232254G    |           | Y-STRAINER_100_PN16_ULVAC                          |                  | 1   |
|         | B027          | M           | 423547      |           | U' Bolt                                            |                  | 4   |
|         | B028          | M           | 427598G     |           | Tube 1" P.O. Multi Purpose Hose Blue               |                  | 1   |
|         | B029          | M           | 427599G     |           | Tube 1" P.O. Multi Purpose Hose Green              |                  | 2   |
|         | B030          | M           | 427265      |           | Tube 3/4" P.O.Multi purpose Hose (Green)           |                  | 2   |
|         | B031          | M           | 427098      |           | Tube 3/4" P.O.Multi purpose Hose (Blue)            |                  | 2   |
|         | B032          | M           | 423487      |           | Nipple Crane 1" BSP S.S. Hex 144/25                |                  | 1   |
|         | B034          | M           | 239216      |           | Seal O Ring RM 0246-24 Nitrile                     |                  | 2   |
|         | B035          | M           | 4231626G    |           | High Torque Jubilee Clip 25-40mm                   |                  | 4   |
|         | B036          | M           | 423734      |           | Plug Solid 3/4"BSPT SS Crane Type 148/20           |                  | 2   |
|         | M001          | M           | 421G062     |           | RETURN WATER MANIFOLD                              |                  | 1   |
|         | M002          | M           | 421G063     |           | FEED WATER MANIFOLD                                |                  | 1   |
|         | M003          | M           | 421G092     |           | INLET WATER PIPE                                   |                  | 1   |
|         | M004          | M           | 426G722     |           | MANIFOLD SUPPORT BRACKET                           |                  | 1   |
|         | M006          | M           | 426G723     |           | MANIFOLD SUPPORT BRACKET (ELEC. ROUTING)           |                  | 1   |
|         | M007          | M           | 426G605     | IT1       | Flow Restrictor 1" BSPT M/F                        |                  | 1   |
|         | M008          | M           | 419G788     | IT2       | Flow Restrictor                                    |                  | 1   |
|         | S001          | M           | 421617      |           | SOCKET CAP SCREW M16 x 65 GRADE 12.9               |                  | 16  |
|         | S002          | M           | 421038      |           | NUT FULL HEXAGON M16                               |                  | 16  |
|         | S003          | M           | 421059      |           | Washer Plain M16                                   |                  | 32  |
|         | S004          | M           | 421042      |           | NUT FULL HEX M12                                   |                  | 8   |
|         | S005          | M           | 421030      |           | PLAIN WASHER M12                                   |                  | 12  |
|         | S006          | M           | 421210      |           | SOCKET CAP SCREW M12 x 35 GRADE 12.9               |                  | 4   |
|         | S007          | M           | 4211905G    |           | Screw Panhead M4 x 35                              |                  | 4   |
|         | S008          | M           | 421227      |           | Washer Spring M4                                   |                  | 4   |
| 01-095  | 908G725       |             |             |           | <b>Water Piping - Cooling Ring Connections</b>     | 421G960          | 1   |
|         | B001          | M           | 427073      |           | Swagelok Male Hose Adaptor                         |                  | 2   |
|         | B002          | M           | 427098      |           | Tube 3/4" P.O.Multi purpose Hose (Blue)            |                  | 2   |
|         | B003          | M           | 427265      |           | Tube 3/4" P.O.Multi purpose Hose (Green)           |                  | 1   |
| 01-100  | 908G650       |             |             |           | <b>Diffusion Pump Water Fittings</b>               | 421G924          | 2   |
|         | B001          | M           | 424341G     |           | 1/2" BSPT Ball Valve M/F (Full Bore)               |                  | 4   |
|         | B002          | M           | 427072      |           | Male Hose Adaptor                                  |                  | 4   |
| 01-110  | 910G552       |             |             |           | <b>Logo &amp; I/D Plate (Dahlia)</b>               | 426G254          | 1   |

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| Section | Bubble Number | Mech / Elec | Part Number | Parameter | Part Description                                          | Assembly Drawing | Qty |
|---------|---------------|-------------|-------------|-----------|-----------------------------------------------------------|------------------|-----|
|         | M001          | M           | 329G422     |           | Logo & I/D Plate (Dahlia)                                 |                  | 1   |
|         | S001          | M           | 421193      |           | SOCKET CAP SCREW M4 x 6 GRADE 12.9                        |                  | 2   |
| 01-120  | 908G866       |             |             |           | <b>Vessel Warning Label Locations</b>                     | 328G488          | 1   |
|         | B001          | M           | 2442392G    |           | 3M VHB TAPE RP25 - 1" x 36 yds.                           |                  | 1   |
|         | M004          | M           | 323G510     | IT11      | Label (Air supply shut off)                               |                  | 1   |
|         | M011          | M           | 323G842     | IT11      | Warning Label                                             |                  | 1   |
|         | M023          | M           | 327G075     | IT11      | warning label combined                                    |                  | 1   |
|         | M025          | M           | 328G869     | IT11      | Pump Group Lubrication Label                              |                  | 1   |
| 01-125  | 909G788       |             |             |           | <b>O2 Vessel Warning Labels</b>                           | 328G970          | 1   |
|         | M020          | M           | 327G054     | IT11      | COMBINED WARNING LABEL                                    |                  | 2   |
|         | M021          | M           | 327G053     | IT11      | WARNING LABEL (CAUTION/PRESSURISED GAS)                   |                  | 1   |
|         | M022          | M           | 327G073     | IT11      | WARNING LABEL (DANGER RISK OF ASPHYXIATION)               |                  | 1   |
| 01-130  | 909G359       |             |             |           | <b>Bobst Logo and Product Name</b>                        | 426G247          | 1   |
|         | B001          | M           | 2442392G    |           | 3M VHB TAPE RP25 - 1" x 36 yds.                           |                  | 1   |
|         | M001          | M           | 425G330     |           | BOBST Logo                                                |                  | 4   |
|         | M002          | M           | 425G043     |           | K5 Expert Product Name                                    |                  | 2   |
| 01-140  | 907G358       |             |             |           | <b>Viewing Window Plasma Treater</b>                      | 417G447          | 3   |
|         | B002          | M           | 239229      |           | Seal O Ring RM0545-30Nitrile                              |                  | 2   |
|         | M001          | M           | 412G554     |           | Window adaptor                                            |                  | 1   |
|         | M002          | M           | 412G555     |           | Polycarbonate Window                                      |                  | 1   |
|         | M003          | M           | 412G732     |           | WINDOW CLAMP                                              |                  | 1   |
|         | S001          | M           | 421145      |           | SOCKET CAP SCREW M6 x 30 GRADE 12.9                       |                  | 4   |
| 01-141  | 905G969       |             |             |           | <b>Viewing Window</b>                                     | 412G471          | 1   |
|         | B001          | M           | 204117      |           | 32774 Nylon Flanged Bush                                  |                  | 8   |
|         | B002          | M           | 239339      |           | Seal O Ring RM1695-30 Nitrile                             |                  | 1   |
|         | M001          | M           | 404G873     |           | Polycarbonate Window                                      |                  | 1   |
|         | S001          | M           | 421122      |           | SOCKET CAP SCREW M10 x 30 GRADE 12.9                      |                  | 8   |
|         | S002          | M           | 421029      |           | PLAIN WASHER M10                                          |                  | 8   |
| 01-150  | 909G546       |             |             |           | <b>Neutral busbar Connections</b>                         | 425G993          | 1   |
|         | M001          | M           | 426G446     |           | Neutral Busbar Angle                                      |                  | 14  |
|         | R0EF          | M           | 426G468     | IT1       | Welding Cable                                             |                  | 28  |
|         | R0EF          | M           | 425G957     | 2900HD    | Transformer Assembly 2 STAGE                              |                  | 1   |
|         | S001          | M           | 421018      |           | SOCKET CAP SCREW M10 x 35 GRADE 12.9                      |                  | 28  |
|         | S002          | M           | 421189      |           | Screw HEXHD Setscrew M10x50 GRADE 8.8                     |                  | 56  |
|         | S003          | M           | 421029      |           | PLAIN WASHER M10                                          |                  | 116 |
|         | S004          | M           | 421027      |           | NUT FULL HEX M10                                          |                  | 32  |
| 01-160  | 909G666       |             |             |           | <b>Cryo Connections - Evap Zone - Maxcool 4000H Ver 2</b> | 426G111          | 1   |
|         | B001          | M           | 423718      |           | Reducer 5/8"-3/8"                                         |                  | 1   |
|         | B002          | M           | 423061      |           | Reducer 5/8"-1/2"                                         |                  | 1   |
|         | B003          | M           | 423062      |           | Reducer 3/8"-1/2"                                         |                  | 1   |
|         | B004          | M           | 423063      |           | Copper Tube Fully Annealed 3/8" O/D                       |                  | 3   |
|         | B005          | M           | 423064      |           | Copper Tube Fully Annealed 5/8" O/D                       |                  | 3   |
|         | B006          | M           | 423065      |           | Armaflex Insulation AF 22mm x 25mm Wall                   |                  | 7   |
|         | B007          | M           | 423066      |           | Armaflex Insulation Tape 50x3x15m Lg                      |                  | 1   |
|         | B008          | M           | 423067      |           | Armaflex Non Flam Adhesive 500ml Tin                      |                  | 1   |
|         | M001          | M           | 426G272     |           | Bracket                                                   |                  | 2   |
|         | M002          | M           | 430G239     |           | Cryo Spill Guard                                          |                  | 1   |
|         | S001          | M           | 421347      |           | SOCKET CAP SCREW M20 x 55 GRADE 12.9                      |                  | 4   |
|         | S003          | M           | 421070      |           | PLAIN WASHER M20                                          |                  | 8   |
|         | S006          | M           | 421072      |           | NUT FULL HEXAGON M20                                      |                  | 4   |
|         | S007          | M           | 421164      |           | WASHER PLAIN M5                                           |                  | 2   |
|         | S008          | M           | 421031      |           | SOCKET CAP SCREW M5 x 20 GRADE 12.9                       |                  | 2   |
| 01-170  | 908G943       |             |             |           | <b>Cryo Coil Connections Winding Zone</b>                 | 422G948          | 1   |
|         | B001          | M           | 423062      |           | Reducer 3/8"-1/2"                                         |                  | 1   |
|         | B002          | M           | 423061      |           | Reducer 5/8"-1/2"                                         |                  | 1   |
|         | B003          | M           | 423718      |           | Reducer 5/8"-3/8"                                         |                  | 1   |
|         | B004          | M           | 4232256G    |           | Armaflex Ins. Class 'O' Self Seal 15 ID x 25 Wall         |                  | 3   |
|         | B005          | M           | 423633      |           | Armaflex Insulation Class 'O' 10 ID x 19 Wall             |                  | 3   |
|         | B006          | M           | 423064      |           | Copper Tube Fully Annealed 5/8" O/D                       |                  | 4   |
|         | B007          | M           | 423063      |           | Copper Tube Fully Annealed 3/8" O/D                       |                  | 4   |
|         | B008          | M           | 423067      |           | Armaflex Non Flam Adhesive 500ml Tin                      |                  | 1   |
|         | B009          | M           | 423066      |           | Armaflex Insulation Tape 50x3x15m Lg                      |                  | 1   |
|         | REF1          | M           | 425177G     |           | Polycold Unit Max Cool 4000H Ver 2 WebGUI                 |                  | 1   |
| 01-200  | 910G238       |             |             |           | <b>Gas Control Panel - O2 Only (Vessel Mounted)</b>       | 429G113          | 1   |
|         | B001          | M           | 213141G     |           | PRESSURE GAUGE 0-10 BAR OXYGEN SERVICE                    |                  | 1   |
|         | B002          | M           | 213095      |           | Pressure Regulator 0.3 - 10 bar                           |                  | 1   |
|         | B003          | M           | 423843      |           | Port Connector 1/2" O/D                                   |                  | 3   |
|         | B004          | M           | 423366      |           | Male Connector 1/4 Tube Stem -1/4 Male NPT                |                  | 2   |

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| Section       | Bubble Number  | Mech / Elec | Part Number | Parameter | Part Description                                        | Assembly Drawing | Qty      |
|---------------|----------------|-------------|-------------|-----------|---------------------------------------------------------|------------------|----------|
|               | B005           | M           | 423365      |           | Reducing Union Tee 1/2-1/4 Swagelok                     |                  | 2        |
|               | B006           | M           | 423512      |           | Union straight 1/2" O/D                                 |                  | 2        |
|               | B007           | M           | 424419G     |           | H6800- High Performance Ball Valve                      |                  | 1        |
|               | B008           | M           | 4231636G    |           | Swagelok Poppet Check Valve                             |                  | 1        |
|               | B009           | M           | 424418G     |           | HAM-LET DIAPHRAGM VALVE,LET-LOK 1/2"                    |                  | 2        |
|               | B010           | E           | 674009G     |           | Pressure Switch 4 Bar Oxygen                            |                  | 2        |
|               | B011           | M           | 423541      |           | Male Connector 1/2"BSP                                  |                  | 2        |
|               | B012           | M           | 423281      |           | Tube 1/2" O/D x 18 SWG St.Steel                         |                  | 1        |
|               | B013           | M           | 423274      |           | Tube 1/4" O/D x 20 SWG S.S.                             |                  | 1        |
|               | B014           | M           | 427103      |           | Mounting bracket for single yoke                        |                  | 1        |
|               | B017           | M           | 427501G     |           | SINGLE YOKE 1/2" BSP CONN. FOR OXYGEN SERVICE           |                  | 1        |
|               | B018           | M           | 4232393G    |           | Hose Assy Braided St.St Teflon 1/2" Tube Adaptors       |                  | 1        |
|               | M001           | M           | 429G185     |           | Gas control Panel Back plate                            |                  | 1        |
|               | M002           | M           | 418G758     |           | 1/2" Pipe Clamp Bracket                                 |                  | 4        |
|               | M003           | M           | 428G507     |           | Spacer - Cable Tray                                     |                  | 4        |
|               | M004           | M           | 428G508     | IT1       | Gas Panel Mounting Bracket                              |                  | 2        |
|               | M005           | M           | 428G508     | IT3       | Gas Panel Mounting Bracket                              |                  | 2        |
|               | M006           | M           | 426G873     |           | O2PS1 SET TO 3 BAR - LABEL                              |                  | 1        |
|               | M007           | M           | 426G874     |           | SET TO 2 BAR - LABEL                                    |                  | 1        |
|               | M008           | M           | 426G875     |           | O2PS2 SET TO 1.5 BAR - LABEL                            |                  | 1        |
|               | M009           | M           | 426G876     |           | GAS SUPPLY LABEL                                        |                  | 1        |
|               | M010           | M           | 426G885     |           | OXYGEN LABEL                                            |                  | 1        |
|               | M015           | M           | 428G664     |           | REGULATOR BRACKET SHIM                                  |                  | 2        |
|               | M016           | M           | 428G654     |           | GAS PANEL SUPPLY TYPE PLATE                             |                  | 1        |
|               | M018           | M           | 428G656     |           | OXYGEN LABEL                                            |                  | 1        |
|               | S001           | M           | 421149      |           | SOCKET CAP SCREW M10 x 20 GRADE 12.9                    |                  | 16       |
|               | S002           | M           | 421121      |           | SOCKET CAP SCREW M6 x 16 GRADE 12.9                     |                  | 16       |
|               | S003           | M           | 421542      |           | SOCKET CAP SCREW M6 x 65 GRADE 12.9                     |                  | 8        |
|               | S004           | M           | 421702      |           | SOCKET CAP SCREW M3 x 16 GRADE 12.9                     |                  | 3        |
|               | S005           | M           | 421029      |           | PLAIN WASHER M10                                        |                  | 16       |
|               | S006           | M           | 421034      |           | WASHER PLAIN M6                                         |                  | 20       |
|               | S007           | M           | 421117      |           | Washer Plain M3                                         |                  | 3        |
|               | S008           | M           | 421019      |           | Socket Head Capscrew M6 x 20 GRADE 12.9                 |                  | 4        |
| <b>01-220</b> | <b>910G454</b> |             |             |           | <b>Vessel Panelling K5-Expert</b>                       | <b>430G000</b>   | <b>1</b> |
|               | B001           | M           | 4232416G    |           | Magnetic Catch                                          |                  | 2        |
|               | M001           | M           | 428G961     |           | 2900 Vessel Panel LHS Top Edge                          |                  | 1        |
|               | M002           | M           | 428G962     |           | 2900 Vessel Panel RHS Top Edge                          |                  | 1        |
|               | M003           | M           | 430G538     |           | Vessel Upper LHS Corner Panel                           |                  | 1        |
|               | M004           | M           | 430G541     |           | Vessel Upper RHS Corner Panel                           |                  | 1        |
|               | M005           | M           | 428G965     |           | 2900 Vessel panel upper centre with attachment for ligh |                  | 1        |
|               | M006           | M           | 428G968     |           | 2900 Vessel Front Lower LH Corner Panel                 |                  | 1        |
|               | M007           | M           | 428G969     |           | VESSEL FRONT LOWER RH COVER PANEL                       |                  | 1        |
|               | M008           | M           | 428G822     |           | 840 VESSEL BOTTOM LH PANEL                              |                  | 1        |
|               | M009           | M           | 428G853     |           | 846 VESSEL BOTTOM RH PANEL                              |                  | 1        |
|               | M010           | M           | 428G817     |           | Spacer                                                  |                  | 10       |
|               | M011           | M           | 429G110     |           | UPPER COVER PANEL FOR VESSEL                            |                  | 1        |
|               | M012           | M           | 429G111     |           | SPACER                                                  |                  | 1        |
|               | M015           | M           | 425G721     | IT1       | VESSEL BOTTOM CENTRE PANEL                              |                  | 1        |
|               | M016           | M           | 425G733     |           | 558 REMOVABLE ACCESS PANEL                              |                  | 1        |
|               | M017           | M           | 425G733     |           | 370 REMOVABLE ACCESS PANEL                              |                  | 1        |
|               | M018           | M           | 425G737     |           | BRACKET                                                 |                  | 2        |
|               | M019           | M           | 430G078     |           | CABINET BANK END PANEL                                  |                  | 2        |
|               | M020           | M           | 424G717     |           | Panel                                                   |                  | 2        |
|               | M021           | M           | 430G034     |           | 1200 Cabinet Top Panel RHS                              |                  | 1        |
|               | M022           | M           | 430G079     |           | PUMP GROUP INTERFACE PANEL                              |                  | 1        |
|               | M023           | M           | 424G769     |           | Panel Bracket                                           |                  | 8        |
|               | M024           | M           | 424G782     |           | Panel Bracket                                           |                  | 6        |
|               | M025           | M           | 426G351     |           | Vessel Door Transit Lock Bracket                        |                  | 2        |
|               | M026           | M           | 426G376     |           | Vessel Door Hinge Bracket                               |                  | 2        |
|               | M027           | M           | 425G553     |           | CARTRIDGE                                               |                  | 4        |
|               | M028           | M           | 430G031     | IT2       | Cabinet Top Panel                                       |                  | 1        |
|               | M029           | M           | 427G357     |           | Hook Pin                                                |                  | 4        |
|               | M031           | M           | 425G121     |           | Camera Television Bracket                               |                  | 1        |
|               | M032           | M           | 425G559     |           | Light diffuser                                          |                  | 4        |
|               | M033           | M           | 430G081     |           | PANEL BRACKET                                           |                  | 1        |
|               | M034           | M           | 430G080     |           | PANEL BRACKET                                           |                  | 1        |
|               | M035           | M           | 430G035     | IT1       | Magnet Packer                                           |                  | 2        |
|               | M036           | M           | 430G035     | IT2       | Magnet Packer                                           |                  | 4        |

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| Section       | Bubble Number  | Mech / Elec | Part Number | Parameter | Part Description                                  | Assembly Drawing | Qty      |
|---------------|----------------|-------------|-------------|-----------|---------------------------------------------------|------------------|----------|
|               | M037           | M           | 430G035     | IT3       | Magnet Packer                                     |                  | 4        |
|               | M038           | M           | 430G055     |           | Door Magnet Bracket                               |                  | 2        |
|               | S001           | M           | 421989      |           | Screw Socket Button Head M5 x 12Lg                |                  | 12       |
|               | S002           | M           | 421872      |           | Screw Socket Buttonhead M5x16                     |                  | 12       |
|               | S003           | M           | 421282      |           | Screw CSK HD M3 x 10                              |                  | 4        |
|               | S004           | M           | 421940      |           | SOCKET CAP SCREW M3 x 10 GRADE 12.9               |                  | 4        |
|               | S005           | M           | 421722      |           | SOCKET CAP SCREW M4 x 20 GRADE 12.9               |                  | 4        |
|               | S006           | M           | 421172      |           | SOCKET CAP SCREW M5 x 12 GRADE 12.9               |                  | 28       |
|               | S007           | M           | 421133      |           | SOCKET CAP SCREW M5 x 16 GRADE 12.9               |                  | 8        |
|               | S008           | M           | 421031      |           | SOCKET CAP SCREW M5 x 20 GRADE 12.9               |                  | 22       |
|               | S010           | M           | 421351      |           | SOCKET CAP SCREW M5 x 50 GRADE 12.9               |                  | 2        |
|               | S011           | M           | 421019      |           | Socket Head Capscrew M6 x 20 GRADE 12.9           |                  | 30       |
|               | S012           | M           | 421182      |           | SOCKET CAP SCREW M8 x 16 GRADE 12.9               |                  | 4        |
|               | S013           | M           | 421049      |           | SOCKET CAP SCREW M8 x 20 GRADE 12.9               |                  | 8        |
|               | S014           | M           | 421210      |           | SOCKET CAP SCREW M12 x 35 GRADE 12.9              |                  | 6        |
|               | S015           | M           | 421288      |           | GRUB SCREW M4x10                                  |                  | 16       |
|               | S016           | M           | 421194      |           | Washer Plain M4                                   |                  | 4        |
|               | S017           | M           | 421164      |           | WASHER PLAIN M5                                   |                  | 94       |
|               | S018           | M           | 421034      |           | WASHER PLAIN M6                                   |                  | 56       |
|               | S019           | M           | 421054      |           | PLAIN WASHER M8                                   |                  | 8        |
|               | S020           | M           | 421030      |           | PLAIN WASHER M12                                  |                  | 6        |
|               | S021           | M           | 421170      |           | Nut Full Hex M5                                   |                  | 14       |
|               | S022           | M           | 421057      |           | NUT FULL HEXAGON M6                               |                  | 26       |
|               | S023           | M           | 421703      |           | SOCKET CAP SCREW M4 x 16 GRADE 12.9               |                  | 4        |
|               | S024           | M           | 421890      |           | SOCKET CAP SCREW M5 x 40 GRADE 12.9               |                  | 4        |
| <b>01-221</b> | <b>910G478</b> |             |             |           | <b>BLANK BOAT BUTTON PANEL K5 EXPERT</b>          | <b>430G088</b>   | <b>1</b> |
|               | M001           | M           | 429G147     | 2900HD    | Boat Button Panel                                 |                  | 1        |
|               | S001           | M           | 421033      |           | SOCKET CAP SCREW M6 x 12 GRADE 12.9               |                  | 4        |
|               | S002           | M           | 421034      |           | WASHER PLAIN M6                                   |                  | 4        |
| <b>01-225</b> | <b>910G455</b> |             |             |           | <b>Additional Pump Group Panelling</b>            | <b>430G003</b>   | <b>1</b> |
|               | M001           | M           | 430G009     | IT1       | Cabinet Top Panel RHS                             |                  | 1        |
|               | S001           | M           | 421210      |           | SOCKET CAP SCREW M12 x 35 GRADE 12.9              |                  | 3        |
|               | S002           | M           | 421172      |           | SOCKET CAP SCREW M5 x 12 GRADE 12.9               |                  | 2        |
|               | S003           | M           | 421170      |           | Nut Full Hex M5                                   |                  | 2        |
|               | S004           | M           | 421030      |           | PLAIN WASHER M12                                  |                  | 3        |
|               | S005           | M           | 421164      |           | WASHER PLAIN M5                                   |                  | 4        |
| <b>01-250</b> | <b>909G719</b> |             |             |           | <b>Static Discharge Wand Assembly</b>             | <b>426G387</b>   | <b>1</b> |
|               | B001           | E           | 600037G     |           | Coiled Static Grounding Cable                     |                  | 1        |
|               | B002           | E           | 600038G     |           | Cable Gland M20 4-9mm Diameter Cable              |                  | 1        |
|               | B003           | E           | 600036G     |           | Central Earthing Point (7829.000)                 |                  | 1        |
|               | B004           | M           | 4211477G    |           | Terry Spring Clip 1"                              |                  | 1        |
|               | M001           | M           | 426G402     |           | Insulated Handle                                  |                  | 1        |
|               | M002           | M           | 426G405     |           | Static Copper Rod                                 |                  | 1        |
|               | S001           | M           | 4211263     |           | SOCKET CAP SCREW M4 x 10 GRADE 12.9               |                  | 1        |
|               | S002           | M           | 421194      |           | Washer Plain M4                                   |                  | 1        |
|               | S003           | M           | 421180      |           | Nut Full Hex M4                                   |                  | 1        |
|               | S004           | M           | 421964      |           | SCREW, SOCKET HEAD CAP, M6x8 GRADE 12.9           |                  | 2        |
|               | S006           | M           | 421054      |           | PLAIN WASHER M8                                   |                  | 1        |
|               | S007           | M           | 421036      |           | NUT FULL HEXAGON M8                               |                  | 3        |
| <b>01-700</b> | <b>909G563</b> |             |             |           | <b>Operator Controls Cubicle</b>                  | <b>430G003</b>   | <b>1</b> |
|               | 1              | M           | 245G169     |           | K5000 - E05_0 Cubicle - Operator Panel            |                  | 1        |
|               | 3              | E           | 618952G     |           | Olfex Robot F1 Highflex Cable 24Core+GN/YL, 0.5mm |                  | 20       |
|               | 4              | E           | 618953G     |           | Olfex Torsion Cable, 4 Twisted Pairs, 0.5mm       |                  | 20       |
|               | 5              | E           | 618974G     |           | Olfex Robot F1 Highflex Cable 3Core+GN/YL, 1mm    |                  | 20       |
| <b>01-702</b> | <b>908G700</b> |             |             |           | <b>K5000 - O2 Gas Control &amp; MFC Cubicles</b>  | <b>426G387</b>   | <b>1</b> |
|               | 1              | M           | 245G177     |           | K5000 - E10_0 Cubicle - Gas Control Cubicle       |                  | 1        |
|               | 2              | E           | 600068G     |           | Hoffman E10_0 Cubicle - 400x200x130mm             |                  | 1        |
|               | 3              | E           | 685005G     |           | E10_0 - Siemens Free Issue Parts To Panel Builder |                  | 1        |
| <b>01-720</b> | <b>908G761</b> |             |             |           | <b>K5000 - Main Water Manifold Electrical</b>     | <b>422G644</b>   | <b>1</b> |
|               | 1              | E           | 651001G     |           | ET200AL 4 Channel Analogue Input U/I/RTD, 4xM12   |                  | 1        |
|               | 2              | E           | 6131343G    |           | 4 Pole M12 Straight Male - M12 90 Deg Female 1.5m |                  | 3        |
|               | 11             | E           | 651001G     |           | ET200AL 4 Channel Analogue Input U/I/RTD, 4xM12   |                  | 1        |
|               | 12             | E           | 6161320G    |           | M12 4 Pin 90Deg Female Cable 1.5m - Open End      |                  | 2        |
|               | 13             | E           | 245G185     | 4.0M      | CABLE ASSEMBLY FLOW-TEMP SENSOR / ET200AL         |                  | 1        |
|               | 14             | E           | 6131619G    |           | M12 Male Rewirable Connector 4 Pin                |                  | 2        |
|               | 21             | E           | 652176G     |           | ET200AL Bus Connection Cable 2xM8, 0.3M           |                  | 1        |
|               | 31             | E           | 652180G     |           | ET200AL Power Connection Cable 2xM8, 0.3M         |                  | 1        |
|               | 40             | M           | 421769      |           | SOCKET CAP SCREW M4x30 GRADE 12.9                 |                  | 4        |



## V18G019 - Bill of Materials



| Section       | Bubble Number  | Mech / Elec | Part Number | Parameter | Part Description                                         | Assembly Drawing | Qty      |
|---------------|----------------|-------------|-------------|-----------|----------------------------------------------------------|------------------|----------|
| <b>01-721</b> | <b>908G762</b> |             |             |           | <b>K5000 - Vessel Electrical Connections</b>             | <b>422G599</b>   | <b>1</b> |
|               | 1              | E           | 651004G     |           | ET200AL PROFINET HEADER MODULE                           |                  | 1        |
|               | 2              | E           | 6131433G    |           | Profinet Connection Cable, M12 To RJ45 - 10M             |                  | 1        |
|               | 11             | E           | 651003G     |           | ET200AL 4DI+4DQx24V0.5A, 8xM8 Module IP67                |                  | 1        |
|               | 12             | E           | 245G184     | 2.0M      | CABLE ASSEMBLY ROTARY ACT'R PROXIMITY / ET200AL          |                  | 1        |
|               | 13             | E           | 245G184     | 1.5M      | CABLE ASSEMBLY ROTARY ACT'R PROXIMITY / ET200AL          |                  | 1        |
|               | 15             | E           | 6131464G    |           | M8 Male - M8 Female 3 Pin Conn. Cable PUR/PVC 2.5m       |                  | 2        |
|               | 16             | E           | 6131458G    |           | Form 'B1' 24V Solenoid Valve Conn. M8 3 Pin 11mm         |                  | 2        |
|               | 17             | E           | 6131432G    |           | M8 Male - M8 Female 3 Pin Conn. Cable PUR/PVC 1M         |                  | 1        |
|               | 21             | E           | 651003G     |           | ET200AL 4DI+4DQx24V0.5A, 8xM8 Module IP67                |                  | 1        |
|               | 22             | E           | 245G184     | 1.5M      | CABLE ASSEMBLY ROTARY ACT'R PROXIMITY / ET200AL          |                  | 1        |
|               | 25             | E           | 6131427G    |           | M8 Male - M8 Female 3 Pin Conn. Cable PUR/PVC 3M         |                  | 1        |
|               | 26             | E           | 6131458G    |           | Form 'B1' 24V Solenoid Valve Conn. M8 3 Pin 11mm         |                  | 1        |
|               | 31             | M           | 245G270     | 2.0M      | CABLE ASSEMBLY MKS BARATRON AA01A/ ET200AL               |                  | 1        |
|               | 41             | E           | 651003G     |           | ET200AL 4DI+4DQx24V0.5A, 8xM8 Module IP67                |                  | 1        |
|               | 43             | E           | 6131426G    |           | M8 Male To Open End Connection Cable, 3 Wire - 3m        |                  | 2        |
|               | 44             | E           | 6131432G    |           | M8 Male - M8 Female 3 Pin Conn. Cable PUR/PVC 1M         |                  | 3        |
|               | 51             | E           | 651001G     |           | ET200AL 4 Channel Analogue Input U/VRTD, 4xM12           |                  | 1        |
|               | 52             | E           | 245G186     | 2.0M      | CABLE ASSEMBLY PIRANI / ET200AL                          |                  | 1        |
|               | 53             | E           | 245G186     | 1.0M      | CABLE ASSEMBLY PIRANI / ET200AL                          |                  | 1        |
|               | 54             | E           | 245G186     | 0.5M      | CABLE ASSEMBLY PIRANI / ET200AL                          |                  | 1        |
|               | 61             | E           | 652176G     |           | ET200AL Bus Connection Cable 2xM8, 0.3M                  |                  | 3        |
|               | 63             | E           | 652189G     |           | ET200AL Bus Connection Cable 2xM8, 4.0m                  |                  | 2        |
|               | 64             | E           | 652179G     |           | ET200AL Bus Connection Cable 2xM8, 10M                   |                  | 2        |
|               | 71             | E           | 652191G     |           | ET200AL Power Connection Cable 2xM8, 0.3m                |                  | 2        |
|               | 73             | E           | 652190G     |           | ET200AL Power Connection Cable 2xM8, 4.0m                |                  | 1        |
|               | 74             | E           | 652183G     |           | ET200AL Power Connection Cable 2xM8, 10.0M               |                  | 3        |
|               | 75             | E           | 652185G     |           | ET200AL Power Conn. Cable M8 Female - Open , 20.0M       |                  | 3        |
|               | 80             | M           | 421769      |           | SOCKET CAP SCREW M4x30 GRADE 12.9                        |                  | 12       |
|               | 81             | M           | 421054      |           | PLAIN WASHER M8                                          |                  | 4        |
|               | 82             | M           | 421138      |           | SOCKET CAP SCREW M8 x 25 GRADE 12.9                      |                  | 4        |
|               | M001           | M           | 429G090     |           | Electrical Connections Panel                             |                  | 1        |
| <b>01-722</b> | <b>908G804</b> |             |             |           | <b>K5000 - VV11 &amp; VV12 ELECTRICAL (8 MIN PMPDWN)</b> | <b>422G628</b>   | <b>1</b> |
|               | 1              | E           | 651003G     |           | ET200AL 4DI+4DQx24V0.5A, 8xM8 Module IP67                |                  | 1        |
|               | 2              | E           | 245G184     | 3.0M      | CABLE ASSEMBLY ROTARY ACT'R PROXIMITY / ET200AL          |                  | 1        |
|               | 3              | E           | 245G184     | 3.5M      | CABLE ASSEMBLY ROTARY ACT'R PROXIMITY / ET200AL          |                  | 1        |
|               | 4              | E           | 6131463G    |           | M8 Male - M8 Female 3 Pin Conn. Cable PUR/PVC 3.5m       |                  | 1        |
|               | 5              | E           | 6131427G    |           | M8 Male - M8 Female 3 Pin Conn. Cable PUR/PVC 3M         |                  | 1        |
|               | 6              | E           | 6131458G    |           | Form 'B1' 24V Solenoid Valve Conn. M8 3 Pin 11mm         |                  | 2        |
|               | 11             | E           | 652186G     |           | ET200AL Bus Connection Cable 2xM8, 0.5M                  |                  | 1        |
|               | 21             | E           | 652187G     |           | ET200AL Power Connection Cable 2xM8, 0.5M                |                  | 1        |
|               | 30             | M           | 421769      |           | SOCKET CAP SCREW M4x30 GRADE 12.9                        |                  | 2        |
| <b>01-725</b> | <b>909G378</b> |             |             |           | <b>K5 Expert - Vessel Panel Lights</b>                   | <b>427G020</b>   | <b>1</b> |
|               | 1              | M           | 245G263     |           | 3 Colour LED Light Assembly                              |                  | 4        |
|               | 2              | E           | 6131498G    |           | T COUPLER M12 MALE TO M12 FEMALE - PIN TO PIN            |                  | 4        |
|               | 3              | E           | 6131499G    |           | 4 PIN CONNECTION CABLE 1M M12 MALE TO FEMALE             |                  | 3        |
| <b>01-730</b> | <b>908G691</b> |             |             |           | <b>Source Leadthrough(s) HD</b>                          | <b>422G644</b>   | <b>1</b> |
|               | 1              | E           | 616598      |           | FC LANE 25 PIN SOCKET CODE 0                             |                  | 1        |
|               | 2              | E           | 616599      |           | FC LANE 25 PIN PLUG CODE 0                               |                  | 1        |
|               | 3              | E           | 616600      |           | FC LANE 25 PIN BULKHEAD CODE 0                           |                  | 1        |
|               | 4              | E           | 616601      |           | FC Lane Cable Outlet Connector                           |                  | 2        |
|               | 5              | E           | 616602      |           | FC Lane Outlet Connector Gasket                          |                  | 2        |
|               | 11             | E           | 616603      |           | FC LANE 25 PIN SOCKET CODE 1                             |                  | 1        |
|               | 12             | E           | 616604      |           | FC LANE 25 PIN PLUG CODE 1                               |                  | 1        |
|               | 13             | E           | 616605      |           | FC LANE 25 PIN BULKHEAD CODE 1                           |                  | 1        |
|               | 14             | E           | 616601      |           | FC Lane Cable Outlet Connector                           |                  | 2        |
|               | 15             | E           | 616602      |           | FC Lane Outlet Connector Gasket                          |                  | 2        |
|               | 21             | E           | 616606      |           | FC LANE 25 PIN SOCKET CODE 2                             |                  | 0        |
|               | 22             | E           | 616607      |           | FC LANE 25 PIN PLUG CODE 2                               |                  | 0        |
|               | 23             | E           | 616608      |           | FC LANE 25 PIN BULKHEAD CODE 2                           |                  | 0        |
|               | 24             | E           | 616601      |           | FC Lane Cable Outlet Connector                           |                  | 0        |
|               | 25             | E           | 616602      |           | FC Lane Outlet Connector Gasket                          |                  | 0        |
|               | 31             | E           | 616598      |           | FC LANE 25 PIN SOCKET CODE 0                             |                  | 1        |
|               | 32             | E           | 616599      |           | FC LANE 25 PIN PLUG CODE 0                               |                  | 1        |
|               | 33             | E           | 616600      |           | FC LANE 25 PIN BULKHEAD CODE 0                           |                  | 1        |
|               | 34             | E           | 616601      |           | FC Lane Cable Outlet Connector                           |                  | 2        |
|               | 35             | E           | 616602      |           | FC Lane Outlet Connector Gasket                          |                  | 2        |
|               | 41             | E           | 616603      |           | FC LANE 25 PIN SOCKET CODE 1                             |                  | 1        |

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| Section       | Bubble Number  | Mech / Elec | Part Number | Parameter | Part Description                                      | Assembly Drawing | Qty      |
|---------------|----------------|-------------|-------------|-----------|-------------------------------------------------------|------------------|----------|
|               | 42             | E           | 616604      |           | FC LANE 25 PIN PLUG CODE 1                            |                  | 1        |
|               | 43             | E           | 616605      |           | FC LANE 25 PIN BULKHEAD CODE 1                        |                  | 1        |
|               | 44             | E           | 616601      |           | FC Lane Cable Outlet Connector                        |                  | 2        |
|               | 45             | E           | 616602      |           | FC Lane Outlet Connector Gasket                       |                  | 2        |
|               | 51             | E           | 616606      |           | FC LANE 25 PIN SOCKET CODE 2                          |                  | 0        |
|               | 52             | E           | 616607      |           | FC LANE 25 PIN PLUG CODE 2                            |                  | 0        |
|               | 53             | E           | 616608      |           | FC LANE 25 PIN BULKHEAD CODE 2                        |                  | 0        |
|               | 54             | E           | 616601      |           | FC Lane Cable Outlet Connector                        |                  | 0        |
|               | 55             | E           | 616602      |           | FC Lane Outlet Connector Gasket                       |                  | 0        |
|               | 61             | E           | 616609      |           | FC LANE 25 PIN SOCKET CODE 3                          |                  | 1        |
|               | 62             | E           | 616610      |           | FC LANE 25 PIN PLUG CODE 3                            |                  | 1        |
|               | 63             | E           | 616611      |           | FC LANE 25 PIN BULKHEAD CODE 3                        |                  | 1        |
|               | 64             | E           | 616601      |           | FC Lane Cable Outlet Connector                        |                  | 2        |
|               | 65             | E           | 616602      |           | FC Lane Outlet Connector Gasket                       |                  | 2        |
| <b>01-731</b> | <b>908G692</b> |             |             |           | <b>Camera Leadthrough(s)</b>                          | <b>422G632</b>   | <b>1</b> |
|               | 1              | E           | 616598      |           | FC LANE 25 PIN SOCKET CODE 0                          |                  | 1        |
|               | 2              | E           | 616599      |           | FC LANE 25 PIN PLUG CODE 0                            |                  | 1        |
|               | 3              | E           | 616600      |           | FC LANE 25 PIN BULKHEAD CODE 0                        |                  | 1        |
|               | 4              | E           | 616601      |           | FC Lane Cable Outlet Connector                        |                  | 2        |
|               | 5              | E           | 616602      |           | FC Lane Outlet Connector Gasket                       |                  | 2        |
|               | 11             | E           | 616598      |           | FC LANE 25 PIN SOCKET CODE 0                          |                  | 1        |
|               | 12             | E           | 616599      |           | FC LANE 25 PIN PLUG CODE 0                            |                  | 1        |
|               | 13             | E           | 616600      |           | FC LANE 25 PIN BULKHEAD CODE 0                        |                  | 1        |
|               | 14             | E           | 616601      |           | FC Lane Cable Outlet Connector                        |                  | 2        |
|               | 15             | E           | 616602      |           | FC Lane Outlet Connector Gasket                       |                  | 2        |
| <b>01-732</b> | <b>908G693</b> |             |             |           | <b>Vessel Lights &amp; Leadthrough(s)</b>             | <b>422G629</b>   | <b>1</b> |
|               | 1              | E           | 616620      |           | FC LANE 6 PIN PLUG CODE 0                             |                  | 1        |
|               | 2              | E           | 616619      |           | FC LANE 6 PIN SOCKET CODE 0                           |                  | 1        |
|               | 3              | E           | 616623      |           | FC LANE 6 PIN BULKHEAD CODE 0                         |                  | 1        |
|               | 4              | E           | 616601      |           | FC Lane Cable Outlet Connector                        |                  | 2        |
|               | 5              | E           | 616602      |           | FC Lane Outlet Connector Gasket                       |                  | 2        |
|               | 11             | E           | 616620      |           | FC LANE 6 PIN PLUG CODE 0                             |                  | 1        |
|               | 12             | E           | 616619      |           | FC LANE 6 PIN SOCKET CODE 0                           |                  | 1        |
|               | 13             | E           | 616623      |           | FC LANE 6 PIN BULKHEAD CODE 0                         |                  | 1        |
|               | 14             | E           | 616601      |           | FC Lane Cable Outlet Connector                        |                  | 2        |
|               | 15             | E           | 616602      |           | FC Lane Outlet Connector Gasket                       |                  | 2        |
|               | 21             | M           | 559897G     |           | FLOURESCENT TUBE T8 4FT (1200mm) DAYLIGHT             |                  | 4        |
|               | 22             | M           | 4211477G    |           | Terry Spring Clip 1"                                  |                  | 8        |
|               | 23             | E           | 603104      |           | FLUORESCENT LIGHT FITTING T8, PUSH ON                 |                  | 8        |
|               | 24             | E           | 603019G     |           | M4 Plastic Saddle Clip                                |                  | 12       |
| <b>01-733</b> | <b>910G520</b> |             |             |           | <b>K5000 - O2 Wiring</b>                              | <b>430G657</b>   | <b>1</b> |
|               | 14             | E           | 6131432G    |           | M8 Male - M8 Female 3 Pin Conn. Cable PUR/PVC 1M      |                  | 1        |
| <b>01-735</b> | <b>908G817</b> |             |             |           | <b>Vessel Standard Electrical Leadthrough Plates</b>  | <b>422G641</b>   | <b>1</b> |
|               | 2              | M           | 418G295     |           | Electrical Leadthrough Flange (3 x FC Lane)           |                  | 3        |
|               | 3              | M           | 422G468     |           | Electrical Leadthrough Flange (1 x FC Lane)           |                  | 5        |
|               | 4              | M           | 239373      |           | Seal O Ring RM1245-30 Nitrile                         |                  | 8        |
|               | 5              | M           | 421049      |           | SOCKET CAP SCREW M8 x 20 GRADE 12.9                   |                  | 24       |
|               | 6              | M           | 421054      |           | PLAIN WASHER M8                                       |                  | 24       |
| <b>01-736</b> | <b>909G565</b> |             |             |           | <b>K5-Expert Lighting Junction Box</b>                | <b>422G641</b>   | <b>1</b> |
|               | 1              | M           | 245G173     |           | T03_7 Cubicle - Vessel Lighting Cubicle               |                  | 1        |
|               | 2              | E           | 600048G     |           | Hoffman T03_7 Cubicle - 400x400x120mm                 |                  | 1        |
| <b>01-745</b> | <b>908G887</b> |             |             |           | <b>LEADTHROUGH BLANK</b>                              | <b>422G641</b>   | <b>4</b> |
|               | 1              | M           | 301G086     |           | LEADTHROUGH BLANK                                     |                  | 1        |
|               | 2              | M           | 421101      |           | Nut Full Hex M24                                      |                  | 1        |
|               | 3              | M           | 421069      |           | PLAIN WASHER M24                                      |                  | 1        |
|               | 4              | M           | 239671G     |           | Seal O Ring RM0355-30                                 |                  | 1        |
| <b>01-750</b> | <b>909G488</b> |             |             |           | <b>K5000-2900 40 Boat, 75mm Pitch, 2-Stage, 15kVA</b> | <b>430G657</b>   | <b>1</b> |
|               | 1              | E           | 709418G     | 400V      | K5000-2900 40 Boat 75mm Pitch, 2-Stage Trans Assem    |                  | 1        |
|               | 3              | E           | 616719      |           | HARTING HAN 24B HIGH HOOD M40 SIDE ENTRY              |                  | 13       |
|               | 4              | E           | 6161120G    |           | HARTING HAN 16B HIGH HOOD M40 SIDE ENTRY              |                  | 14       |
|               | 5              | E           | 616859      |           | HAN-MODULAR 6 WAY FRAME HOUSING -FEMALE (HINGED FR    |                  | 13       |
|               | 6              | E           | 616306      |           | HAN-MODULAR 4 WAY FRAME HOUSING -FEMALE (HINGED FR    |                  | 14       |
|               | 7              | E           | 616858      |           | HAN-MODULAR 2 WAY MODULE - FEMALE - 100A              |                  | 67       |
|               | 8              | E           | 616302      |           | HAN-MODULAR DUMMY MODULE                              |                  | 0        |
| <b>01-751</b> | <b>910G123</b> |             |             |           | <b>Source Interconnect Cables</b>                     | <b>428G478</b>   | <b>1</b> |
|               | 1              | M           | 428G473     | IT1       | Transformer Secondary Cable (Live)                    |                  | 8        |
|               | 2              | M           | 428G473     | IT2       | Transformer Secondary Cable (Live)                    |                  | 12       |
|               | 3              | M           | 428G473     | IT3       | Transformer Secondary Cable (Live)                    |                  | 14       |

## V18G019 - Bill of Materials



| Section       | Bubble Number  | Mech / Elec | Part Number | Parameter | Part Description                                          | Assembly Drawing | Qty      |
|---------------|----------------|-------------|-------------|-----------|-----------------------------------------------------------|------------------|----------|
|               | 4              | M           | 428G473     | IT5       | Transformer Secondary Cable (Live)                        |                  | 8        |
|               | 5              | M           | 428G473     | IT7       | Transformer Secondary Cable (Live)                        |                  | 8        |
|               | 6              | M           | 426G468     | IT1       | Welding Cable                                             |                  | 28       |
|               | 7              | M           | 428G473     | IT4       | Transformer Secondary Cable (Live)                        |                  | 12       |
|               | 8              | M           | 428G473     | IT6       | Transformer Secondary Cable (Live)                        |                  | 14       |
|               | 9              | M           | 428G473     | IT8       | Transformer Secondary Cable (Live)                        |                  | 4        |
| <b>02</b>     |                |             |             |           | <b>Vessel Internal</b>                                    |                  |          |
| <b>02-010</b> | <b>909G178</b> |             |             |           | <b>MESH SCREENS</b>                                       | <b>424G087</b>   | <b>1</b> |
|               | M001           | M           | 323G059     |           | Mesh Screen 8" Port                                       |                  | 1        |
|               | S001           | M           | 421172      |           | SOCKET CAP SCREW M5 x 12 GRADE 12.9                       |                  | 4        |
|               | S002           | M           | 421164      |           | WASHER PLAIN M5                                           |                  | 4        |
|               | S003           | M           | 423458      |           | Pipe Clip 6dia Enots 34-0218-04                           |                  | 4        |
| <b>02-015</b> | <b>909G178</b> |             |             |           | <b>MESH SCREENS</b>                                       | <b>424G087</b>   | <b>2</b> |
|               | M001           | M           | 403G003     |           | Mesh Screen 12" Port                                      |                  | 1        |
|               | S001           | M           | 421172      |           | SOCKET CAP SCREW M5 x 12 GRADE 12.9                       |                  | 4        |
|               | S002           | M           | 421164      |           | WASHER PLAIN M5                                           |                  | 4        |
|               | S003           | M           | 423458      |           | Pipe Clip 6dia Enots 34-0218-04                           |                  | 4        |
| <b>02-020</b> | <b>906G696</b> |             |             |           | <b>Vessel Light Assembly K5000</b>                        | <b>414G763</b>   | <b>2</b> |
|               | B002           | M           | 421078      |           | Terry Spring Clip 1.1/2"                                  |                  | 7        |
|               | B003           | M           | 422089G     |           | Thumb Screw M6 x 15                                       |                  | 12       |
|               | M001           | M           | 414G772     | 2900      | VESEL LIGHT COVER                                         |                  | 2        |
|               | S001           | M           | 421364      |           | SOCKET CAP SCREW M4 x 8 GRADE 12.9                        |                  | 7        |
| <b>02-030</b> | <b>908G634</b> |             |             |           | <b>Rotary Viewing Window Assembly</b>                     | <b>421G808</b>   | <b>1</b> |
|               | B003           | M           | 232312      |           | LENZE HUB U4-14 710                                       |                  | 1        |
|               | B004           | M           | 232313      |           | LENZE HUB U4-75 766                                       |                  | 1        |
|               | B005           | M           | 232311      |           | SPIDER U4-26 752                                          |                  | 1        |
|               | B007           | M           | 202236      |           | BEARING PILLOW BLOCK MP 25                                |                  | 2        |
|               | M001           | M           | 418G492     | 2900      | ROTOR                                                     |                  | 1        |
|               | M002           | M           | 418G393     |           | Rotor End Stub Shaft                                      |                  | 1        |
|               | M003           | M           | 418G394     |           | Rotor End Stub Shaft                                      |                  | 1        |
|               | M004           | M           | 418G299     | 2900      | Shroud                                                    |                  | 1        |
|               | M005           | M           | 418G311     |           | Bracket motor mtg                                         |                  | 1        |
|               | M006           | M           | 418G303     |           | Bracket                                                   |                  | 1        |
|               | M007           | M           | 405G813     |           | Support Bracket - Rotary Window Guard                     |                  | 1        |
|               | M010           | M           | 418G325     |           | Heat Sink                                                 |                  | 1        |
|               | M011           | M           | 418G392     |           | stud                                                      |                  | 2        |
|               | M012           | M           | 418G357     |           | Motor Clamp                                               |                  | 1        |
|               | M013           | M           | 418G381     |           | Thermal Sheet Pad                                         |                  | 1        |
|               | M014           | M           | 418G382     |           | Thermal Sheet Pad                                         |                  | 1        |
|               | M015           | M           | 418G249     | IT14      | Key                                                       |                  | 1        |
|               | S001           | M           | 421149      |           | SOCKET CAP SCREW M10 x 20 GRADE 12.9                      |                  | 4        |
|               | S002           | M           | 421210      |           | SOCKET CAP SCREW M12 x 35 GRADE 12.9                      |                  | 4        |
|               | S003           | M           | 4211388     |           | SOCKET CAP SCREW M10 x 160 GRADE 12.9                     |                  | 2        |
|               | S005           | M           | 421019      |           | Socket Head Capscrew M6 x 20 GRADE 12.9                   |                  | 5        |
|               | S006           | M           | 421418      |           | Screw Socket Countersunk Head M6 x 16                     |                  | 8        |
|               | S007           | M           | 2441506     |           | DOWEL PIN DIA.6x25                                        |                  | 3        |
|               | S008           | M           | 421029      |           | PLAIN WASHER M10                                          |                  | 6        |
|               | S009           | M           | 421030      |           | PLAIN WASHER M12                                          |                  | 4        |
|               | S010           | M           | 421034      |           | WASHER PLAIN M6                                           |                  | 20       |
|               | S012           | M           | 421054      |           | PLAIN WASHER M8                                           |                  | 3        |
|               | S013           | M           | 421052      |           | Nut Wing M8 Steel                                         |                  | 3        |
|               | S014           | M           | 421049      |           | SOCKET CAP SCREW M8 x 20 GRADE 12.9                       |                  | 2        |
|               | S015           | M           | 421826      |           | Screw Caphead M6x12 Fine Thread (0.75)                    |                  | 1        |
|               | S017           | M           | 421475      |           | Washer Plain Large Dia. M8                                |                  | 2        |
| <b>02-050</b> | <b>908G671</b> |             |             |           | <b>Vessel Collector Plate Assy</b>                        | <b>422G040</b>   | <b>1</b> |
|               | M001           | M           | 422G041     |           | Collector Plate                                           |                  | 1        |
|               | M002           | M           | 422G042     |           | Collector Plate                                           |                  | 1        |
|               | M003           | M           | 404G444     |           | Spacer                                                    |                  | 5        |
|               | M004           | M           | 422G043     |           | Collector Plate                                           |                  | 1        |
|               | S001           | M           | 421208      |           | SCREW PANHEAD M6x10                                       |                  | 5        |
| <b>02-070</b> | <b>908G508</b> |             |             |           | <b>Mechanism Traverse Limit Switch Positions (Vessel)</b> | <b>421G171</b>   | <b>1</b> |
|               | M001           | M           | 425G177     |           | Flag Closed Proxi Switch                                  |                  | 1        |
|               | M002           | M           | 421G175     |           | Flag Open Proxi Switch                                    |                  | 1        |
|               | REF1           | E           | 698471G     |           | SENSOR MAGNETICALLY ACT SWITCH CAT 4 8PIN MICRO           |                  | 1        |
|               | S001           | M           | 421033      |           | SOCKET CAP SCREW M6 x 12 GRADE 12.9                       |                  | 5        |
|               | S002           | M           | 421147      |           | SOCKET CAP SCREW M4 x 12 GRADE 12.9                       |                  | 2        |
|               | S003           | M           | 421034      |           | WASHER PLAIN M6                                           |                  | 5        |
|               | S004           | M           | 421194      |           | Washer Plain M4                                           |                  | 2        |

## V18G019 - Bill of Materials



| Section       | Bubble Number  | Mech / Elec | Part Number | Parameter | Part Description                                   | Assembly Drawing | Qty      |
|---------------|----------------|-------------|-------------|-----------|----------------------------------------------------|------------------|----------|
|               | S006           | M           | 421227      |           | Washer Spring M4                                   |                  | 2        |
| <b>02-090</b> | <b>910G261</b> |             |             |           | <b>Evap Cryo Coil Assy</b>                         | <b>429G232</b>   | <b>1</b> |
|               | B001           | M           | 239218      |           | O-Ring Seal RM0995-30 (Nitrile)                    |                  | 2        |
|               | B002           | M           | 4232534G    |           | 15mm Pipe Clip                                     |                  | 43       |
|               | M001           | M           | 429G236     |           | Mounting Bracket                                   |                  | 17       |
|               | M002           | M           | 409G166     | 2900      | CRYO COIL                                          |                  | 1        |
|               | M003           | M           | 429G237     |           | Mounting Bracket                                   |                  | 4        |
|               | M005           | M           | 426G494     |           | Cryo Coil Evap Zone                                |                  | 1        |
|               | M006           | M           | 401G592     |           | Cryocoil Feedthrough Flange                        |                  | 2        |
|               | M007           | M           | 429G244     |           | Mounting Bracket                                   |                  | 3        |
|               | S001           | M           | 421121      |           | SOCKET CAP SCREW M6 x 16 GRADE 12.9                |                  | 21       |
|               | S002           | M           | 421019      |           | Socket Head Capscrew M6 x 20 GRADE 12.9            |                  | 6        |
|               | S003           | M           | 421049      |           | SOCKET CAP SCREW M8 x 20 GRADE 12.9                |                  | 8        |
|               | S004           | M           | 421543      |           | Screw Countersunk Head Setscrew M4 x 16            |                  | 43       |
|               | S005           | M           | 421034      |           | WASHER PLAIN M6                                    |                  | 27       |
|               | S006           | M           | 421054      |           | PLAIN WASHER M8                                    |                  | 8        |
| <b>02-100</b> | <b>910G265</b> |             |             |           | <b>Winding Zone Over Pump Ports cryo-Coil Assy</b> | <b>429G227</b>   | <b>1</b> |
|               | B001           | M           | 239218      |           | O-Ring Seal RM0995-30 (Nitrile)                    |                  | 1        |
|               | B002           | M           | 424342G     |           | 3/4" BSP Ball Valve M/F (Full Bore)                |                  | 1        |
|               | B003           | M           | 4232353G    |           | Brass 3/4"BSPT Male/3/4" Hose Tail                 |                  | 1        |
|               | B004           | M           | 4232534G    |           | 15mm Pipe Clip                                     |                  | 32       |
|               | M001           | M           | 425G977     |           | Back Wall Cryo Coil                                |                  | 1        |
|               | M002           | M           | 425G997     |           | Cryo Coil Drip Tray                                |                  | 1        |
|               | M003           | M           | 401G592     |           | Cryocoil Feedthrough Flange                        |                  | 1        |
|               | M004           | M           | 429G245     |           | Cryo-Coil Bracket (2 Coils)                        |                  | 16       |
|               | M005           | M           | 425G996     |           | Cryo Coil Baton Long                               |                  | 2        |
|               | M006           | M           | 425G989     |           | Cryo Coil Baton Short                              |                  | 1        |
|               | S001           | M           | 421543      |           | Screw Countersunk Head Setscrew M4 x 16            |                  | 32       |
|               | S002           | M           | 421033      |           | SOCKET CAP SCREW M6 x 12 GRADE 12.9                |                  | 35       |
|               | S003           | M           | 421138      |           | SOCKET CAP SCREW M8 x 25 GRADE 12.9                |                  | 10       |
|               | S004           | M           | 421034      |           | WASHER PLAIN M6                                    |                  | 35       |
|               | S005           | M           | 421054      |           | PLAIN WASHER M8                                    |                  | 10       |
| <b>02-120</b> | <b>905G175</b> |             |             |           | <b>Vessel Cryo Blank</b>                           | <b>427G709</b>   | <b>1</b> |
|               | B001           | M           | 239193      |           | Seal O Ring RM0945-30 Nitrile                      |                  | 1        |
|               | M001           | M           | 308G972     | IT.28     | Special Flange                                     |                  | 1        |
|               | S001           | M           | 421138      |           | SOCKET CAP SCREW M8 x 25 GRADE 12.9                |                  | 4        |
|               | S002           | M           | 421054      |           | PLAIN WASHER M8                                    |                  | 4        |
| <b>02-130</b> | <b>909G558</b> |             |             |           | <b>Diff Lid Covers</b>                             | <b>425G979</b>   | <b>1</b> |
|               | M001           | M           | 425G980     |           | Diffusion Pump Shield Mounting Bar                 |                  | 2        |
|               | M005           | M           | 425G981     | IT4       | Diffusion Pump Shield Mounting Bar                 |                  | 1        |
|               | M008           | M           | 422G076     | IT1       | Diffusion Pump Guard                               |                  | 5        |
|               | M012           | M           | 422G076     | IT5       | Diffusion Pump Guard                               |                  | 1        |
|               | S001           | M           | 421049      |           | SOCKET CAP SCREW M8 x 20 GRADE 12.9                |                  | 6        |
|               | S002           | M           | 421054      |           | PLAIN WASHER M8                                    |                  | 6        |
|               | S003           | M           | 421170      |           | Nut Full Hex M5                                    |                  | 6        |
|               | S004           | M           | 421023      |           | Screw Countersunk Head Socket M5x16                |                  | 6        |
|               | S005           | M           | 421164      |           | WASHER PLAIN M5                                    |                  | 6        |
| <b>02-135</b> | <b>910G579</b> |             |             |           | <b>Mesh Cryo Coil Guards</b>                       | <b>430G706</b>   | <b>1</b> |
|               | B001           | M           | 4212017G    |           | Detent Pin                                         |                  | 8        |
|               | B002           | M           | 240056G     |           | Key Ring                                           |                  | 16       |
|               | B003           | M           | 240055G     |           | Chain 510-201X150LG                                |                  | 8        |
|               | M001           | M           | 430G709     | IT9       | Mesh Guard - Faceplate Side                        |                  | 1        |
|               | M002           | M           | 430G709     | IT5       | Mesh Guard - Faceplate Side                        |                  | 1        |
|               | M003           | M           | 430G709     | IT8       | Mesh Guard - Faceplate Side                        |                  | 1        |
|               | M004           | M           | 430G712     |           | Mesh Guard - Sideplate Floor                       |                  | 1        |
|               | M005           | M           | 430G707     |           | Guard Latch Support                                |                  | 6        |
|               | M006           | M           | 430G708     | IT1       | Guard Foot Support                                 |                  | 3        |
|               | M007           | M           | 430G708     | IT2       | Guard Foot Support                                 |                  | 3        |
|               | M008           | M           | 430G710     |           | Guard Top Support                                  |                  | 3        |
|               | M009           | M           | 430G711     |           | Guard Foot Support                                 |                  | 2        |
|               | S001           | M           | 421033      |           | SOCKET CAP SCREW M6 x 12 GRADE 12.9                |                  | 16       |
|               | S002           | M           | 421121      |           | SOCKET CAP SCREW M6 x 16 GRADE 12.9                |                  | 18       |
|               | S003           | M           | 421034      |           | WASHER PLAIN M6                                    |                  | 34       |
| <b>02-734</b> | <b>908G696</b> |             |             |           | <b>Strobe Motor &amp; Leadthrough</b>              | <b>422G636</b>   | <b>1</b> |
|               | 1              | E           | 606065      |           | Stepper Motor SM87-3-18-M-V-6                      |                  | 1        |
|               | 11             | E           | 616620      |           | FC LANE 6 PIN PLUG CODE 0                          |                  | 1        |
|               | 12             | E           | 616619      |           | FC LANE 6 PIN SOCKET CODE 0                        |                  | 1        |
|               | 13             | E           | 616623      |           | FC LANE 6 PIN BULKHEAD CODE 0                      |                  | 1        |

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|---------|---------------|-------------|-------------|-----------|--------------------------------------------------|------------------|-----|
|         | 14            | E           | 616601      |           | FC Lane Cable Outlet Connector                   |                  | 2   |
|         | 15            | E           | 616602      |           | FC Lane Outlet Connector Gasket                  |                  | 2   |
| 02-750  | 908G935       |             |             |           | K5-EXPERT CAMERA SYSTEM                          | 422G636          | 1   |
|         | 1             | E           | 618619      |           | CAMERA QUAD UNIT (No-HDD)                        |                  | 1   |
|         | 2             | E           | 618749G     |           | 2M SHOTGUN CABLE WITH BNC CONN. TO WIRE END      |                  | 4   |
|         | 3             | E           | 618905G     |           | 7M SHOTGUN CABLE WITH BNC CONN. TO WIRE END      |                  | 4   |
|         | 4             | M           | 430348G     |           | Dome Camera 2MP 1080P HD-TVI, PAL, 1/3" Sensor   |                  | 4   |
|         | 5             | E           | 618906G     |           | 18.5" CCTV Monitor With HDMI, VGA Input          |                  | 1   |
|         | 6             | E           | 670292G     |           | 12V DC 9 Way Surge Protected Enclosed PSU        |                  | 1   |
|         | 7             | E           | 6131451G    |           | HDMI Connection Lead 1.5m                        |                  | 1   |
|         | 8             | E           | 6161327G    |           | 6 Way 13A, 240V Rewirable Extension              |                  | 1   |
|         | M001          | M           | 429G103     |           | CCTV Cable Cover Long                            |                  | 1   |
|         | M002          | M           | 429G104     |           | CCTV Cable Cover Short                           |                  | 1   |
|         | S001          | M           | 421147      |           | SOCKET CAP SCREW M4 x 12 GRADE 12.9              |                  | 8   |
|         | S002          | M           | 421194      |           | Washer Plain M4                                  |                  | 8   |
| 02-751  | 909G163       |             |             |           | HD SECURITY CAMERA ADAPTER PLATE                 | 426G654          | 4   |
|         | 1             | M           | 426G406     |           | Camera Adaptor Plate - HIKVISION DS-2CE56D7T     |                  | 1   |
|         | 2             | M           | 421418      |           | Screw Socket Countersunk Head M6 x 16            |                  | 4   |
|         | 3             | M           | 4211312     |           | Screw Countersunk Head Slotted M4 x 12 ST.ST. A4 |                  | 3   |
| 03      |               |             |             |           | Source                                           |                  |     |
| 03-010  | 910G074       |             |             |           | Source Box Assembly                              | 428G229          | 1   |
|         | B001          | M           | 423904      |           | Blanking Plug 3/8"BSPP ST.ST. VSTI-3/8ED71       |                  | 2   |
|         | B002          | M           | 4231447G    |           | RS FITTING STEEL GASKET SS-8-RS-2V               |                  | 4   |
|         | B003          | M           | 4231544G    |           | Blanking Plug 1/2"BSPP ST.ST. VSTI-1/2           |                  | 6   |
|         | B004          | M           | 427144      |           | MALE CONNECTOR 3/4" TO 1/2" BSPP                 |                  | 4   |
|         | B005          | M           | 427147      |           | TUBE ADAPTOR 3/4" TUBE TO 3/4" HOSE              |                  | 4   |
|         | B006          | M           | 203338G     |           | Bush Flanged Glacier DU 30 id x 34 od x 16 lg    |                  | 4   |
|         | B007          | M           | 239404      |           | Seal, Viton Face Seal                            |                  | 6   |
|         | B008          | M           | 427633G     |           | SWAGelok SEALING WASHER SS-16-RS-2V              |                  | 2   |
|         | B009          | M           | 427632G     |           | MALE ADAPTOR 1" T.A TO 1" BSPP                   |                  | 2   |
|         | B010          | M           | 4232239G    |           | Swagelok Tube Fitting, Union Elbow, 1"           |                  | 2   |
|         | B011          | M           | 4232240G    |           | Tube Adaptor 1" Tube to 1" Hose                  |                  | 2   |
|         | B012          | M           | 427098      |           | Tube 3/4" P.O.Multi purpose Hose (Blue)          |                  | 1   |
|         | M001          | M           | 421G247     |           | 2900 Source Trough (Reel Side)                   |                  | 1   |
|         | M002          | M           | 421G248     |           | 2900 Source Trough (OP SIDE)                     |                  | 1   |
|         | M003          | M           | 421G249     |           | Source End Plate (Front)                         |                  | 1   |
|         | M004          | M           | 421G250     |           | Source End Plate (Rear)                          |                  | 1   |
|         | M005          | M           | 421G251     | IT11      | Trough Spacer                                    |                  | 1   |
|         | M006          | M           | 421G015     |           | 2900 Source Side                                 |                  | 2   |
|         | M007          | M           | 414G313     | IT1       | Slide Screw HD Source Covers                     |                  | 12  |
|         | M008          | M           | 421G263     |           | Stop Plate                                       |                  | 2   |
|         | M009          | M           | 425G641     | 2900HD    | Water Cooled Side Bar                            |                  | 1   |
|         | M010          | M           | 428G178     | 2900HD    | Source Side Guard Operator Side                  |                  | 1   |
|         | M011          | M           | 425G640     | 2900HD    | Source Side Guard Reel Side                      |                  | 1   |
|         | M012          | M           | 421G256     | IT11      | Source Spacer Block (Short)                      |                  | 38  |
|         | M013          | M           | 421G256     | IT12      | Source Spacer Block (Short)                      |                  | 1   |
|         | M014          | M           | 421G257     | IT10      | Source Spacer Block (Long)                       |                  | 1   |
|         | M015          | M           | 421G257     | IT11      | Source Spacer Block (Long)                       |                  | 1   |
|         | M016          | M           | 421G257     | IT12      | Source Spacer Block (Long)                       |                  | 1   |
|         | M017          | M           | 425G646     | 2900HD    | Source Bolting Strip Reel Side                   |                  | 2   |
|         | M018          | M           | 425G645     | 2900HD    | Source Bolting Strip Operator Side               |                  | 2   |
|         | M019          | M           | 421G326     | IT2       | CLAMPING STRIP                                   |                  | 4   |
|         | M020          | M           | 425G642     | 2900HD    | Source Bolting Strip Operator Side - Blocks      |                  | 1   |
|         | M021          | M           | 425G643     | 2900HD    | Source Bolting Strip Reel Side - Blocks          |                  | 1   |
|         | M022          | M           | 421G433     |           | 2900 Source Bolting Strip-TROUGH                 |                  | 2   |
|         | M023          | M           | 425G644     | 2900HD    | Source Clamping Strip (CENTRE BAR BRACKET)       |                  | 4   |
|         | M024          | M           | 421G251     | IT12      | Trough Spacer                                    |                  | 1   |
|         | M025          | M           | 420G763     |           | JACKING BLOCK                                    |                  | 1   |
|         | M026          | M           | 421G018     |           | 3324 Source Extrusion                            |                  | 2   |
|         | M027          | M           | 421G018     |           | 3228 Source Extrusion                            |                  | 2   |
|         | M028          | M           | 421G222     |           | 3324 Side Bar Extrusion                          |                  | 1   |
|         | M029          | M           | 428G221     | 2900HD    | Source Guard Packer                              |                  | 1   |
|         | M030          | M           | 428G159     |           | Guard Support                                    |                  | 6   |
|         | M031          | M           | 428G162     |           | Slide Nut                                        |                  | 6   |
|         | M032          | M           | 428G193     | IT2       | Source Guard Upper (Reel Side)                   |                  | 1   |
|         | M033          | M           | 428G193     | IT3       | Source Guard Upper (Reel Side)                   |                  | 1   |
|         | M034          | M           | 428G193     | IT8       | Source Guard Upper (Reel Side)                   |                  | 1   |
|         | M036          | M           | 429G336     |           | Guard support extension                          |                  | 6   |

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|---------------|----------------|-------------|-------------|-----------|--------------------------------------------------|------------------|-----------|
|               | S001           | M           | 421611      |           | Screw Countersunk Head Slotted M8 x 16           |                  | 55        |
|               | S002           | M           | 421138      |           | SOCKET CAP SCREW M8 x 25 GRADE 12.9              |                  | 4         |
|               | S003           | M           | 421146      |           | SOCKET CAP SCREW M8 x 30 GRADE 12.9              |                  | 32        |
|               | S004           | M           | 421237      |           | SOCKET CAP SCREW M8 x 35 GRADE 12.9              |                  | 4         |
|               | S005           | M           | 421278      |           | SOCKET CAP SCREW M8 x 45 GRADE 12.9              |                  | 10        |
|               | S006           | M           | 421524      |           | SOCKET CAP SCREW M8 x 55 GRADE 12.9              |                  | 45        |
|               | S007           | M           | 421040      |           | Screw HEXHD Setscrew M8 x 25 GRADE 8.8           |                  | 8         |
|               | S008           | M           | 421285      |           | Screw HEXHD Setscrew M10x60 GRADE 8.8            |                  | 1         |
|               | S009           | M           | 421054      |           | PLAIN WASHER M8                                  |                  | 10        |
|               | S010           | M           | 421029      |           | PLAIN WASHER M10                                 |                  | 18        |
|               | S011           | M           | 421027      |           | NUT FULL HEX M10                                 |                  | 13        |
|               | S012           | M           | 421149      |           | SOCKET CAP SCREW M10 x 20 GRADE 12.9             |                  | 12        |
| <b>03-020</b> | <b>909G516</b> |             |             |           | <b>Source Stand &amp; Wire Reel Assy</b>         | <b>425G637</b>   | <b>1</b>  |
|               | B001           | M           | 268254G     |           | Wire Reel 14.6kg 2.0mm Dia                       |                  | 40        |
|               | B002           | M           | 427260      |           | Swagelok Tube Adaptor 3/8"                       |                  | 4         |
|               | B003           | M           | 423555      |           | Positionable Male Elbow 3/8" o/d x 3/8"          |                  | 4         |
|               | B004           | M           | 427088      |           | Tube 3/8" P.O. Multi purpose hose Blue           |                  | 1         |
|               | M001           | M           | 425G651     | 2900HD    | Support Beam (Reel Side)                         |                  | 1         |
|               | M002           | M           | 425G650     | 2900HD    | Support Beam (Operator Side)                     |                  | 1         |
|               | M003           | M           | 426G078     |           | Source Mounting Bracket (Lower)                  |                  | 2         |
|               | M004           | M           | 421G524     |           | Source Mounting Bracket (Upper)                  |                  | 2         |
|               | M005           | M           | 421G543     |           | Reel Hanger(Reel Side)                           |                  | 20        |
|               | M006           | M           | 429G303     |           | Wire Reel Axle                                   |                  | 40        |
|               | M007           | M           | 422G046     |           | Source mounting bracket                          |                  | 1         |
|               | M008           | M           | 421G844     |           | Insulation Bush                                  |                  | 8         |
|               | M009           | M           | 422G012     | IT4       | Cover (Source Support Beam)                      |                  | 6         |
|               | M010           | M           | 422G012     | IT3       | Cover (Source Support Beam)                      |                  | 2         |
|               | M011           | M           | 421G906     |           | Reel Hanger(Operator Side)                       |                  | 20        |
|               | M012           | M           | 421G786     |           | Insulation Plate                                 |                  | 2         |
|               | M013           | M           | 408G502     |           | Angle bracket                                    |                  | 2         |
|               | M014           | M           | 425G983     |           | Wire feed bracket                                |                  | 40        |
|               | M015           | M           | 421G859     | 3290      | Support Beam Extrusion                           |                  | 2         |
|               | M016           | M           | 429G430     |           | Lower Guide                                      |                  | 40        |
|               | S001           | M           | 4211902G    |           | Screw HEXHD Setscrew M12x150 GRADE 8.8           |                  | 4         |
|               | S002           | M           | 421042      |           | NUT FULL HEX M12                                 |                  | 4         |
|               | S003           | M           | 421034      |           | WASHER PLAIN M6                                  |                  | 40        |
|               | S004           | M           | 421248      |           | DOWEL 6 dia x 20 mm LONG                         |                  | 16        |
|               | S005           | M           | 421054      |           | PLAIN WASHER M8                                  |                  | 8         |
|               | S006           | M           | 421049      |           | SOCKET CAP SCREW M8 x 20 GRADE 12.9              |                  | 2         |
|               | S007           | M           | 421122      |           | SOCKET CAP SCREW M10 x 30 GRADE 12.9             |                  | 8         |
|               | S008           | M           | 421147      |           | SOCKET CAP SCREW M4 x 12 GRADE 12.9              |                  | 40        |
|               | S009           | M           | 421237      |           | SOCKET CAP SCREW M8 x 35 GRADE 12.9              |                  | 8         |
|               | S010           | M           | 421210      |           | SOCKET CAP SCREW M12 x 35 GRADE 12.9             |                  | 16        |
|               | S011           | M           | 421030      |           | PLAIN WASHER M12                                 |                  | 16        |
|               | S012           | M           | 4211484G    |           | SCREW SOCKET BUTTON HEAD M4 x 10                 |                  | 16        |
|               | S013           | M           | 421091      |           | SOCKET CAP SCREW M6 x 25 GRADE 12.9              |                  | 40        |
|               | S014           | M           | 421029      |           | PLAIN WASHER M10                                 |                  | 8         |
|               | S015           | M           | 421191      |           | Locknut M10                                      |                  | 40        |
| <b>03-030</b> | <b>908G530</b> |             |             |           | <b>Wire Feeder Assy</b>                          | <b>421G268</b>   | <b>40</b> |
|               | B001           | M           | 200909G     |           | DEEP GROOVE BALL BEARING 6000-2RSH               |                  | 1         |
|               | B003           | M           | 204135      |           | Insulating Sleeve                                |                  | 2         |
|               | B005           | M           | 240017G     |           | Thumb Nut                                        |                  | 1         |
|               | B006           | M           | 224421G     |           | LOCKING ASSEMBLY TLK350 12mm                     |                  | 1         |
|               | B007           | M           | 240016G     |           | Swing Bolt                                       |                  | 1         |
|               | B008           | M           | 235263G     |           | Spring LHL 500A 01                               |                  | 1         |
|               | B009           | M           | 234005G     |           | CIRCLIP EXTERNAL 1400-0010                       |                  | 1         |
|               | B010           | M           | 4211892G    |           | Shoulder Screw - Dia 6mm x (M5 Thread) x 6mm Lg. |                  | 1         |
|               | B011           | M           | 4211893G    |           | NYLOC NUT M6                                     |                  | 1         |
|               | B012           | M           | 204168G     |           | Bush Igus ZFM-0810-09 Iglidur 'Z'                |                  | 1         |
|               | M001           | M           | 420G974     |           | Wire Feeder Drive Body                           |                  | 1         |
|               | M002           | M           | 429G181     |           | Drive Wheel                                      |                  | 1         |
|               | M003           | M           | 421G087     |           | Wire Feeder Pivot Arm                            |                  | 1         |
|               | M004           | M           | 428G927     |           | Insulator                                        |                  | 1         |
|               | M005           | M           | 421G289     |           | Spout Clamp                                      |                  | 1         |
|               | M006           | M           | 426G479     |           | Spout                                            |                  | 1         |
|               | M007           | M           | 415G026     |           | Idler shaft                                      |                  | 1         |
|               | M008           | M           | 427G486     |           | Pivot Pin                                        |                  | 1         |
|               | M009           | M           | 415G027     |           | Spacer                                           |                  | 1         |

# V18G019 - Bill of Materials



| Section       | Bubble Number  | Mech / Elec | Part Number | Parameter | Part Description                                  | Assembly Drawing | Qty       |
|---------------|----------------|-------------|-------------|-----------|---------------------------------------------------|------------------|-----------|
|               | M010           | M           | 429G306     |           | Lead-in bush                                      |                  | 1         |
|               | M011           | M           | 427G487     |           | Spacer                                            |                  | 1         |
|               | REF1           | E           | 606175G     |           | Stepper Motor SM88-1-18-ML4.Z352 (8 Wire Stepper) |                  | 1         |
|               | S001           | M           | 421133      |           | SOCKET CAP SCREW M5 x 16 GRADE 12.9               |                  | 2         |
|               | S002           | M           | 421368      |           | SOCKET CAP SCREW M5 x 25 GRADE 12.9               |                  | 2         |
|               | S003           | M           | 421019      |           | Socket Head Capscrew M6 x 20 GRADE 12.9           |                  | 2         |
|               | S004           | M           | 421449      |           | SOCKET C'SK SCREW M4 x 10                         |                  | 1         |
|               | S005           | M           | 421164      |           | WASHER PLAIN M5                                   |                  | 4         |
|               | S006           | M           | 421034      |           | WASHER PLAIN M6                                   |                  | 2         |
|               | S007           | M           | 421655      |           | Washer Heavy Duty M6                              |                  | 2         |
| <b>03-035</b> | <b>908G758</b> |             |             |           | <b>Source Tools</b>                               | <b>426G255</b>   | <b>1</b>  |
|               | M002           | M           | 426G235     |           | Live Post lever                                   |                  | 1         |
|               | M003           | M           | 422G248     |           | Wire Feeder Hand Tool                             |                  | 1         |
|               | M004           | M           | 426G270     |           | Source Tools Hanger                               |                  | 1         |
|               | M005           | M           | 428G119     |           | Cleaning Tray                                     |                  | 3         |
|               | S001           | M           | 421444      |           | SOCKET C'SK SCREW M5 x 10                         |                  | 2         |
| <b>03-040</b> | <b>910G165</b> |             |             |           | <b>Live Post Assembly</b>                         | <b>428G698</b>   | <b>40</b> |
|               | B001           | M           | 427078      |           | Tube 1/2" P.O. Multi purpose Hose Blue            |                  | 1         |
|               | B003           | M           | 427305      |           | Union Elbow 1/2" OD Brass                         |                  | 1         |
|               | B004           | M           | 427152      |           | Swagelok Tube Fitting B-PB8-SL8                   |                  | 1         |
|               | B005           | M           | 203261      |           | Plain Clamping Ring GR10-19.05 Plain Bore 19.05mm |                  | 1         |
|               | B006           | M           | 235259G     |           | Spring Compression LC 135M 01 M                   |                  | 1         |
|               | B007           | M           | 239244      |           | Seal O Ring BS210 Nitrile                         |                  | 2         |
|               | B008           | M           | 240045G     |           | Lobe Knob With Solid Knob                         |                  | 1         |
|               | B009           | M           | 234155G     |           | CIRCLIP EXTERNAL 1400-0060                        |                  | 2         |
|               | B010           | M           | 421639      |           | Dowel Pin 5 x 20 mm Long                          |                  | 4         |
|               | B011           | M           | 239822G     |           | Seal O Ring RM 0161-16 VITON                      |                  | 2         |
|               | B012           | M           | 4232058G    |           | High Torque Jubilee Clip 20-35mm                  |                  | 8         |
|               | M001           | M           | 421G014     |           | Live Post Channel                                 |                  | 1         |
|               | M002           | M           | 423G664     |           | Pivot Block                                       |                  | 1         |
|               | M003           | M           | 428G702     |           | Live Post                                         |                  | 1         |
|               | M004           | M           | 428G705     |           | Live Extension Block                              |                  | 1         |
|               | M005           | M           | 428G731     |           | Live End Block                                    |                  | 1         |
|               | M006           | M           | 421G272     |           | Deflector Plate                                   |                  | 1         |
|               | M007           | M           | 412G735     |           | Live Busbar lead through HD source                |                  | 1         |
|               | M008           | M           | 428G730     |           | Busbar Connector                                  |                  | 1         |
|               | M009           | M           | 421G274     |           | Spring Clamp                                      |                  | 1         |
|               | M010           | M           | 421G275     |           | Spring Holder                                     |                  | 1         |
|               | M011           | M           | 421G276     |           | Jacking Stud                                      |                  | 1         |
|               | M012           | M           | 428G710     |           | Live Post Guard                                   |                  | 1         |
|               | M013           | M           | 428G748     |           | Busbar Plate                                      |                  | 1         |
|               | M014           | M           | 411G732     |           | Lead Through Plate                                |                  | 1         |
|               | M015           | M           | 320G733     |           | Lead Through Bush                                 |                  | 2         |
|               | M016           | M           | 406G275     |           | Kant Seal 12.7 x 2 Viton                          |                  | 2         |
|               | M017           | M           | 421G273     |           | Live End Pivot                                    |                  | 1         |
|               | M018           | M           | 426G097     |           | Bush                                              |                  | 2         |
|               | M019           | M           | 404G209     | IT3       | Braided Cable                                     |                  | 2         |
|               | M020           | M           | 300G113     | IT.1      | Busbar Clamp                                      |                  | 2         |
|               | M021           | M           | 300G113     | IT.2      | Busbar Clamp                                      |                  | 2         |
|               | M022           | M           | 417G013     |           | Hose Tail                                         |                  | 2         |
|               | M023           | M           | 421G016     |           | 170 Wire Feeder Channel Extrusion                 |                  | 1         |
|               | S001           | M           | 421364      |           | SOCKET CAP SCREW M4 x 8 GRADE 12.9                |                  | 2         |
|               | S002           | M           | 421121      |           | SOCKET CAP SCREW M6 x 16 GRADE 12.9               |                  | 2         |
|               | S003           | M           | 421019      |           | Socket Head Capscrew M6 x 20 GRADE 12.9           |                  | 2         |
|               | S004           | M           | 421091      |           | SOCKET CAP SCREW M6 x 25 GRADE 12.9               |                  | 2         |
|               | S005           | M           | 421145      |           | SOCKET CAP SCREW M6 x 30 GRADE 12.9               |                  | 2         |
|               | S006           | M           | 421219      |           | SOCKET CAP SCREW M6 x 40 GRADE 12.9               |                  | 8         |
|               | S007           | M           | 4211899G    |           | SOCKET CAP SCREW M6 x 120 GRADE 12.9              |                  | 2         |
|               | S008           | M           | 421146      |           | SOCKET CAP SCREW M8 x 30 GRADE 12.9               |                  | 2         |
|               | S009           | M           | 421156      |           | SOCKET CAP SCREW M8 x 40 GRADE 12.9               |                  | 1         |
|               | S010           | M           | 421569      |           | SOCKET CAP SCREW M8 x 100 GRADE 12.9              |                  | 2         |
|               | S011           | M           | 421034      |           | WASHER PLAIN M6                                   |                  | 8         |
|               | S012           | M           | 421022      |           | Washer Spring M6                                  |                  | 2         |
|               | S013           | M           | 421054      |           | PLAIN WASHER M8                                   |                  | 6         |
|               | S014           | M           | 421044      |           | Screw HEXHD Setscrew M8x30 GRADE 8.8              |                  | 2         |
|               | S015           | M           | 421036      |           | NUT FULL HEXAGON M8                               |                  | 2         |
| <b>03-051</b> | <b>909G711</b> |             |             |           | <b>Neutral Centre Bar Assy - HD</b>               | <b>426G417</b>   | <b>1</b>  |
|               | B001           | M           | 268259G     |           | Intermetallic Evaporation Boat T-VAP 125x40x10    |                  | 4         |



**V18G019 - Bill of Materials**


| Section       | Bubble Number  | Mech / Elec | Part Number | Parameter | Part Description                               | Assembly Drawing | Qty      |
|---------------|----------------|-------------|-------------|-----------|------------------------------------------------|------------------|----------|
|               | B002           | M           | 423904      |           | Blanking Plug 3/8"BSPP ST.ST. VSTI-3/8ED71     |                  | 4        |
|               | B003           | M           | 4211468G    |           | DOWEL PIN 8x 20 mm LONG                        |                  | 2        |
|               | B004           | M           | 268275G     |           | 3M Evaporation Boat 125 x 40 x 9               |                  | 4        |
|               | M001           | M           | 426G418     | 4B        | Neutral Centre Bar-HD                          |                  | 1        |
|               | M002           | M           | 428G139     |           | Neutral End Block                              |                  | 4        |
|               | M003           | M           | 426G427     | 4B        | Fixing Plate - HD                              |                  | 1        |
|               | M004           | M           | 426G433     |           | Mounting Block                                 |                  | 2        |
|               | M005           | M           | 426G431     |           | Trough Brace                                   |                  | 2        |
|               | M006           | M           | 406G166     |           | Kant Seal 16 x 2 Viton                         |                  | 8        |
|               | S001           | M           | 421237      |           | SOCKET CAP SCREW M8 x 35 GRADE 12.9            |                  | 4        |
|               | S002           | M           | 421049      |           | SOCKET CAP SCREW M8 x 20 GRADE 12.9            |                  | 8        |
|               | S003           | M           | 4211870G    |           | Bolt HEXHD M6x30 St/St                         |                  | 8        |
|               | S004           | M           | 421209      |           | Washer Plain M6 st,st                          |                  | 8        |
|               | S005           | M           | 421103      |           | Washer Spring M8                               |                  | 4        |
|               | S006           | M           | 421054      |           | PLAIN WASHER M8                                |                  | 4        |
|               | S007           | M           | 4211828G    |           | DOWEL PIN DIA 6 X 20 LONG                      |                  | 8        |
|               | S008           | M           | 421035      |           | SOCKET CAP SCREW M10 x 25 GRADE 12.9           |                  | 4        |
|               | S009           | M           | 421029      |           | PLAIN WASHER M10                               |                  | 4        |
| <b>03-053</b> | <b>909G711</b> |             |             |           | <b>Neutral Centre Bar Assy - HD</b>            | <b>426G417</b>   | <b>6</b> |
|               | B001           | M           | 268259G     |           | Intermetallic Evaporation Boat T-VAP 125x40x10 |                  | 6        |
|               | B002           | M           | 423904      |           | Blanking Plug 3/8"BSPP ST.ST. VSTI-3/8ED71     |                  | 4        |
|               | B003           | M           | 4211468G    |           | DOWEL PIN 8x 20 mm LONG                        |                  | 2        |
|               | B004           | M           | 268275G     |           | 3M Evaporation Boat 125 x 40 x 9               |                  | 6        |
|               | M001           | M           | 426G418     | 6B        | Neutral Centre Bar-HD                          |                  | 1        |
|               | M002           | M           | 428G139     |           | Neutral End Block                              |                  | 6        |
|               | M003           | M           | 426G427     | 6B        | Fixing Plate - HD                              |                  | 1        |
|               | M004           | M           | 426G433     |           | Mounting Block                                 |                  | 2        |
|               | M005           | M           | 426G431     |           | Trough Brace                                   |                  | 2        |
|               | M006           | M           | 406G166     |           | Kant Seal 16 x 2 Viton                         |                  | 12       |
|               | S001           | M           | 421237      |           | SOCKET CAP SCREW M8 x 35 GRADE 12.9            |                  | 4        |
|               | S002           | M           | 421049      |           | SOCKET CAP SCREW M8 x 20 GRADE 12.9            |                  | 10       |
|               | S003           | M           | 4211870G    |           | Bolt HEXHD M6x30 St/St                         |                  | 12       |
|               | S004           | M           | 421209      |           | Washer Plain M6 st,st                          |                  | 12       |
|               | S005           | M           | 421103      |           | Washer Spring M8                               |                  | 4        |
|               | S006           | M           | 421054      |           | PLAIN WASHER M8                                |                  | 4        |
|               | S007           | M           | 4211828G    |           | DOWEL PIN DIA 6 X 20 LONG                      |                  | 12       |
|               | S008           | M           | 421035      |           | SOCKET CAP SCREW M10 x 25 GRADE 12.9           |                  | 4        |
|               | S009           | M           | 421029      |           | PLAIN WASHER M10                               |                  | 4        |
| <b>03-060</b> | <b>910G187</b> |             |             |           | <b>K5 Expert Neutral Post A HD &amp; STD</b>   | <b>428G860</b>   | <b>7</b> |
|               | B001           | M           | 427078      |           | Tube 1/2" P.O. Multi purpose Hose Blue         |                  | 1        |
|               | B002           | M           | 239873G     |           | Seal, Viton Face Seal                          |                  | 1        |
|               | B003           | M           | 239243      |           | Seal O Ring RM0265-30 Nitrile                  |                  | 2        |
|               | B004           | M           | 427305      |           | Union Elbow 1/2" OD Brass                      |                  | 1        |
|               | B005           | M           | 427152      |           | Swagelok Tube Fitting B-PB8-SL8                |                  | 2        |
|               | B007           | M           | 4232058G    |           | High Torque Jubilee Clip 20-35mm               |                  | 4        |
|               | M001           | M           | 430G018     |           | Neutral Busbar                                 |                  | 1        |
|               | M002           | M           | 428G791     |           | Neutral Connection Block                       |                  | 1        |
|               | M003           | M           | 428G794     |           | Water Cooled Neutral Lead Through Tube         |                  | 1        |
|               | M004           | M           | 428G852     |           | Neutral Clamp Block                            |                  | 2        |
|               | M005           | M           | 428G873     | IT1       | Neutral Busbar Connection                      |                  | 1        |
|               | M006           | M           | 428G874     | IT1       | Neutral Busbar Adaptor                         |                  | 1        |
|               | M007           | M           | 428G875     |           | Neutral Busbar Adaptor                         |                  | 1        |
|               | M008           | M           | 300G114     | IT.1      | CLAMPING BLOCK                                 |                  | 1        |
|               | M009           | M           | 300G114     | IT.2      | CLAMPING BLOCK                                 |                  | 1        |
|               | S001           | M           | 421368      |           | SOCKET CAP SCREW M5 x 25 GRADE 12.9            |                  | 2        |
|               | S002           | M           | 421121      |           | SOCKET CAP SCREW M6 x 16 GRADE 12.9            |                  | 2        |
|               | S003           | M           | 421415      |           | SOCKET CAP SCREW M6 x 55 GRADE 12.9            |                  | 4        |
|               | S004           | M           | 421040      |           | Screw HEXHD Setscrew M8 x 25 GRADE 8.8         |                  | 16       |
|               | S005           | M           | 421036      |           | NUT FULL HEXAGON M8                            |                  | 4        |
|               | S006           | M           | 421034      |           | WASHER PLAIN M6                                |                  | 4        |
|               | S007           | M           | 421054      |           | PLAIN WASHER M8                                |                  | 20       |
| <b>03-061</b> | <b>910G188</b> |             |             |           | <b>K5 Expert Neutral Post B HD &amp; STD</b>   | <b>428G861</b>   | <b>7</b> |
|               | B001           | M           | 427078      |           | Tube 1/2" P.O. Multi purpose Hose Blue         |                  | 1        |
|               | B002           | M           | 239873G     |           | Seal, Viton Face Seal                          |                  | 1        |
|               | B003           | M           | 239243      |           | Seal O Ring RM0265-30 Nitrile                  |                  | 2        |
|               | B004           | M           | 427305      |           | Union Elbow 1/2" OD Brass                      |                  | 1        |
|               | B005           | M           | 427152      |           | Swagelok Tube Fitting B-PB8-SL8                |                  | 2        |
|               | B007           | M           | 4232058G    |           | High Torque Jubilee Clip 20-35mm               |                  | 4        |



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| Section       | Bubble Number  | Mech / Elec | Part Number | Parameter | Part Description                               | Assembly Drawing | Qty       |
|---------------|----------------|-------------|-------------|-----------|------------------------------------------------|------------------|-----------|
|               | M001           | M           | 430G018     |           | Neutral Busbar                                 |                  | 1         |
|               | M002           | M           | 428G791     |           | Neutral Connection Block                       |                  | 1         |
|               | M003           | M           | 428G794     |           | Water Cooled Neutral Lead Through Tube         |                  | 1         |
|               | M004           | M           | 428G852     |           | Neutral Clamp Block                            |                  | 2         |
|               | M005           | M           | 428G876     | IT2       | Neutral Busbar Connection                      |                  | 1         |
|               | M006           | M           | 428G877     | IT3       | Neutral Busbar Adaptor                         |                  | 1         |
|               | M007           | M           | 428G878     |           | Neutral Busbar Adaptor                         |                  | 1         |
|               | M008           | M           | 300G114     | IT.1      | CLAMPING BLOCK                                 |                  | 1         |
|               | M009           | M           | 300G114     | IT.2      | CLAMPING BLOCK                                 |                  | 1         |
|               | S001           | M           | 421368      |           | SOCKET CAP SCREW M5 x 25 GRADE 12.9            |                  | 2         |
|               | S002           | M           | 421121      |           | SOCKET CAP SCREW M6 x 16 GRADE 12.9            |                  | 2         |
|               | S003           | M           | 421415      |           | SOCKET CAP SCREW M6 x 55 GRADE 12.9            |                  | 4         |
|               | S004           | M           | 421040      |           | Screw HEXHD Setscrew M8 x 25 GRADE 8.8         |                  | 16        |
|               | S005           | M           | 421036      |           | NUT FULL HEXAGON M8                            |                  | 4         |
|               | S006           | M           | 421034      |           | WASHER PLAIN M6                                |                  | 4         |
|               | S007           | M           | 421054      |           | PLAIN WASHER M8                                |                  | 20        |
| <b>03-070</b> | <b>909G515</b> |             |             |           | <b>Viewing Flap Assy</b>                       | <b>425G636</b>   | <b>1</b>  |
|               | B001           | M           | 427525G     |           | ISO/BSP PARALLEL GASKET                        |                  | 2         |
|               | B002           | M           | 4231544G    |           | Blanking Plug 1/2"BSPP ST.ST. VSTI-1/2         |                  | 2         |
|               | B003           | M           | 234126      |           | CIRCLIP EXTERNAL 1400-0300                     |                  | 2         |
|               | B004           | M           | 200077      |           | BEARING NO SEALS 6008                          |                  | 1         |
|               | B005           | M           | 200253      |           | BEARING 61908                                  |                  | 1         |
|               | B006           | M           | 233127      |           | CIRCLIP INTERNAL 1300-52                       |                  | 1         |
|               | B007           | M           | 234005      |           | CIRCLIP EXTERNAL 1400-0400                     |                  | 2         |
|               | B008           | M           | 239198      |           | Seal O Ring RM0745-30 Nitrile                  |                  | 1         |
|               | B009           | M           | 239356      |           | Seal O Ring RM0895-30Nitrile                   |                  | 2         |
|               | B010           | M           | 423112      |           | Seal 1/4"bsp Enots 34-0156-02                  |                  | 2         |
|               | B011           | M           | 423113      |           | Coned Locknut 1/4"BSP Enots 36-0540-02         |                  | 2         |
|               | B012           | M           | 423111      |           | Elbow Adaptor Hobbs 8 O/D - 1/4" BSP           |                  | 2         |
|               | B013           | M           | 423038      |           | Tubing Nut 8 OD                                |                  | 2         |
|               | B014           | M           | 423039      |           | Tubing Sleeve 8 OD                             |                  | 2         |
|               | B015           | M           | 427113      |           | Pneumatic Actuator Kinetrol 070-104B002000     |                  | 1         |
|               | B016           | M           | 239347      |           | Seal Simmerring BA40-52-7-72NBR                |                  | 2         |
|               | B017           | M           | 4232239G    |           | Swagelok Tube Fitting, Union Elbow, 1"         |                  | 2         |
|               | B018           | M           | 4232240G    |           | Tube Adaptor 1" Tube to 1" Hose                |                  | 2         |
|               | B019           | M           | 423044      |           | Elbow Stem 8 OD                                |                  | 1         |
|               | B020           | M           | 425003      |           | Oil Reservoir 1/4"BSP (Ref OR 7730)            |                  | 1         |
|               | B021           | M           | 423476      |           | 8mm O/D Half Hard Copper Tube X 3000mm lg      |                  | 1         |
|               | B022           | M           | 423045      |           | Adaptor Nippled 8mm-1/4"BSP Enots 36-0557-11   |                  | 1         |
|               | M001           | M           | 427G605     |           | Flap Pivot Arm                                 |                  | 1         |
|               | M002           | M           | 427G606     |           | Flap Pivot Arm                                 |                  | 1         |
|               | M004           | M           | 428G197     | 2900HD    | Viewing Flap                                   |                  | 1         |
|               | M005           | M           | 421G400     |           | FLAP STOP BRACKET                              |                  | 1         |
|               | M006           | M           | 421G691     |           | Seal Retaining Cap                             |                  | 1         |
|               | M007           | M           | 402G339     |           | Bearing spacer                                 |                  | 1         |
|               | M008           | M           | 402G340     |           | Seal Dust shield                               |                  | 1         |
|               | M009           | M           | 421G692     |           | Kinetrol Adaptor                               |                  | 1         |
|               | M010           | M           | 421G576     |           | Spacer Ring Insulator                          |                  | 1         |
|               | M011           | M           | 403G793     |           | Flanged Bush Insulator                         |                  | 4         |
|               | M012           | M           | 421G575     |           | Bearing Housing                                |                  | 1         |
|               | M013           | M           | 404G053     |           | LEAD THROUGH SHAFT                             |                  | 1         |
|               | M014           | M           | 408G512     |           | Flap Telescopic Universal Joint                |                  | 1         |
|               | M015           | M           | 417G799     | IT3       | Standpipe Adaptor 1" BSPP x 25                 |                  | 2         |
|               | M016           | M           | 418G248     | IT11      | Key                                            |                  | 2         |
|               | M017           | M           | 418G248     | IT24      | Key                                            |                  | 1         |
|               | M018           | M           | 421G018     | 3220      | Source Extrusion                               |                  | 1         |
|               | S001           | M           | 421138      |           | SOCKET CAP SCREW M8 x 25 GRADE 12.9            |                  | 4         |
|               | S002           | M           | 421278      |           | SOCKET CAP SCREW M8 x 45 GRADE 12.9            |                  | 8         |
|               | S003           | M           | 421033      |           | SOCKET CAP SCREW M6 x 12 GRADE 12.9            |                  | 4         |
|               | S004           | M           | 421019      |           | Socket Head Capscrew M6 x 20 GRADE 12.9        |                  | 5         |
|               | S005           | M           | 421289      |           | GRUB SCREW M5x8                                |                  | 2         |
| <b>03-080</b> | <b>910G423</b> |             |             |           | <b>Source Leadthrough Blanking Plug Assy</b>   | <b>429G888</b>   | <b>8</b>  |
|               | B001           | M           | 239243      |           | Seal O Ring RM0265-30 Nitrile                  |                  | 2         |
|               | M001           | M           | 417G325     |           | Source Lead Through Blanking Plate             |                  | 1         |
|               | S001           | M           | 421121      |           | SOCKET CAP SCREW M6 x 16 GRADE 12.9            |                  | 2         |
| <b>03-085</b> | <b>908G749</b> |             |             |           | <b>1/2" BSPT Blanking Plug</b>                 | <b>429G888</b>   | <b>18</b> |
|               | B001           | M           | 4231703G    |           | Plug Solid 1/2"BSPT Crane Type 148/15 St Steel |                  | 1         |
| <b>03-086</b> | <b>905G694</b> |             |             |           | <b>Faceplate 3/4" BSP Blank</b>                | <b>411G305</b>   | <b>1</b>  |

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| Section | Bubble Number | Mech / Elec | Part Number | Parameter | Part Description                               | Assembly Drawing | Qty |
|---------|---------------|-------------|-------------|-----------|------------------------------------------------|------------------|-----|
|         | B001          | M           | 423671      |           | Blanking Plug 3/4"BSPP                         |                  | 1   |
|         | B002          | M           | 239216      |           | Seal O Ring RM 0246-24 Nitrile                 |                  | 1   |
| 03-090  | 910G244       |             |             |           | <b>Source Water Assembly</b>                   | 428G948          | 1   |
|         | B001          | M           | 4232263G    |           | Swagelok Reducing Union Tee                    |                  | 2   |
|         | B002          | M           | 4232264G    |           | Swagelok Tube Fitting Union 1" Brass           |                  | 2   |
|         | B003          | M           | 4231034     |           | TUBE ADAPTOR 3/4" TUBE TO 3/4" HOSE BRASS      |                  | 2   |
|         | B004          | M           | 4232240G    |           | Tube Adaptor 1" Tube to 1" Hose                |                  | 2   |
|         | B005          | M           | 427260      |           | Swagelok Tube Adaptor 3/8"                     |                  | 2   |
|         | B006          | M           | 427072      |           | Male Hose Adaptor                              |                  | 54  |
|         | B007          | M           | 427075      |           | Tube Adaptor 1/2" OD x 1/2" Hose               |                  | 54  |
|         | B008          | M           | 239060      |           | O' RING RM 0315.30                             |                  | 2   |
|         | B009          | M           | 239450      |           | Seal O Ring RM0375-30                          |                  | 2   |
|         | B010          | M           | 427598G     |           | Tube 1" P.O. Multi Purpose Hose Blue           |                  | 2   |
|         | B011          | M           | 427599G     |           | Tube 1" P.O. Multi Purpose Hose Green          |                  | 1   |
|         | B012          | M           | 427098      |           | Tube 3/4" P.O.Multi purpose Hose (Blue)        |                  | 1   |
|         | B013          | M           | 427265      |           | Tube 3/4" P.O.Multi purpose Hose (Green)       |                  | 1   |
|         | B014          | M           | 427088      |           | Tube 3/8" P.O. Multi purpose hose Blue         |                  | 1   |
|         | B015          | M           | 427636G     |           | Tube 3/8" P.O Multi purpose Hose Green         |                  | 1   |
|         | B016          | M           | 427078      |           | Tube 1/2" P.O. Multi purpose Hose Blue         |                  | 27  |
|         | B017          | M           | 427263      |           | Tube 1/2" P.O. Multi purpose hose Green        |                  | 27  |
|         | B018          | M           | 427262      |           | Goodridge Fire Proof Sleeve 3/4" ID            |                  | 2   |
|         | B019          | M           | 427282      |           | Goodridge Fire Proof Sleeve 1" ID              |                  | 2   |
|         | B020          | M           | 427616G     |           | Pyro jacket (41mm inside diameter)             |                  | 3   |
|         | B023          | M           | 4231703G    |           | Plug Solid 1/2"BSPT Crane Type 148/15 St Steel |                  | 15  |
|         | M001          | M           | 417G924     |           | 3/4" Water Leadthrough                         |                  | 2   |
|         | M002          | M           | 420G174     |           | Hose Adaptor 1" BSPP to 1" Hose                |                  | 2   |
| 03-095  | 908G759       |             |             |           | <b>Cooling Ring Water Leadthrough</b>          | 422G213          | 1   |
|         | B001          | M           | 4231032     |           | Reducing Union 3/4o/d-1/2o/d BRASS             |                  | 2   |
|         | B002          | M           | 427066      |           | Swagelok Tube Adaptor 1/2"                     |                  | 2   |
|         | B003          | M           | 239060      |           | O' RING RM 0315.30                             |                  | 2   |
|         | B004          | M           | 427078      |           | Tube 1/2" P.O. Multi purpose Hose Blue         |                  | 1   |
|         | B005          | M           | 427263      |           | Tube 1/2" P.O. Multi purpose hose Green        |                  | 1   |
|         | B006          | M           | 427262      |           | Goodridge Fire Proof Sleeve 3/4" ID            |                  | 2   |
|         | B007          | M           | 4232058G    |           | High Torque Jubilee Clip 20-35mm               |                  | 8   |
|         | M001          | M           | 417G924     |           | 3/4" Water Leadthrough                         |                  | 2   |
| 03-100  | 909G517       |             |             |           | <b>Wirefeed Covers HD/STD</b>                  | 425G638          | 1   |
|         | M002          | M           | 427G406     | IT2       | Source Guard (Operator Side)                   |                  | 1   |
|         | M007          | M           | 427G406     | IT7       | Source Guard (Operator Side)                   |                  | 1   |
|         | M008          | M           | 427G406     | IT8       | Source Guard (Operator Side)                   |                  | 1   |
|         | M011          | M           | 425G653     | IT2       | Source Guard (Reel Side)                       |                  | 1   |
|         | M012          | M           | 425G653     | IT3       | Source Guard (Reel Side)                       |                  | 1   |
|         | M017          | M           | 425G653     | IT8       | Source Guard (Reel Side)                       |                  | 1   |
| 03-110  | 908G635       |             |             |           | <b>Source Cooling Ring</b>                     | 421G788          | 1   |
|         | B001          | M           | 423904      |           | Blanking Plug 3/8"BSPP ST.ST. VSTI-3/8ED71     |                  | 4   |
|         | B002          | M           | 239854G     |           | Seal, Viton Face Seal                          |                  | 2   |
|         | B003          | M           | 427072      |           | Male Hose Adaptor                              |                  | 1   |
|         | B004          | M           | 423817      |           | 3/8" BSP X 1/2" TUBE MALE ADAPTOR              |                  | 2   |
|         | B005          | M           | 423528      |           | 1/2" Union Elbow St.St.                        |                  | 3   |
|         | B006          | M           | 427066      |           | Swagelok Tube Adaptor 1/2"                     |                  | 1   |
|         | B007          | M           | 427524G     |           | Male Connector                                 |                  | 1   |
|         | B008          | M           | 427540G     |           | Stainless Steel Tube 1/2" OD X 0.049 Wall      |                  | 1   |
|         | M001          | M           | 422G033     | 2900      | O2 Source Piece (Reel Side)                    |                  | 1   |
|         | M002          | M           | 422G034     |           | O2 Source Piece (Front)                        |                  | 1   |
|         | M003          | M           | 422G035     | 2900      | O2 Source Piece (Operator Side)                |                  | 1   |
|         | M004          | M           | 422G036     |           | O2 Source Packing Piece                        |                  | 2   |
|         | M005          | M           | 422G038     |           | O2 Source Piece (Rear)                         |                  | 1   |
|         | M006          | M           | 427G658     |           | Centre Support Bracket                         |                  | 1   |
|         | M007          | M           | 421G222     | 3324      | Side Bar Extrusion                             |                  | 2   |
|         | M008          | M           | 421G222     | 327       | Side Bar Extrusion                             |                  | 2   |
|         | M009          | M           | 429G065     |           | O2 Source Packing Plate                        |                  | 2   |
|         | S001          | M           | 421019      |           | Socket Head Capscrew M6 x 20 GRADE 12.9        |                  | 24  |
|         | S002          | M           | 4211956G    |           | SOCKET CAP SCREW M4X40 GRADE 12.9              |                  | 4   |
|         | S003          | M           | 421524      |           | SOCKET CAP SCREW M8 x 55 GRADE 12.9            |                  | 4   |
|         | S004          | M           | 4211957G    |           | SOCKET CAP SCREW M4x55 GRADE 12.9              |                  | 4   |
|         | S005          | M           | 421227      |           | Washer Spring M4                               |                  | 8   |
|         | S006          | M           | 245032      |           | M8 x 20 SOCKET CSK SELF/COLOUR                 |                  | 2   |
|         | S007          | M           | 4211012     |           | Screw Countersunk Head Socket M8x16            |                  | 1   |
| 03-750  | 909G496       |             |             |           | <b>Source Wire Feed - HD</b>                   | 422G672          | 1   |

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| Section       | Bubble Number  | Mech / Elec | Part Number | Parameter | Part Description                                   | Assembly Drawing | Qty      |
|---------------|----------------|-------------|-------------|-----------|----------------------------------------------------|------------------|----------|
|               | 1              | E           | 606183G     |           | Stepper Motor SM88-1-18-ML4.Z352 - M12 Connection  |                  | 40       |
|               | 11             | E           | 606091      |           | B&R X67SM2436 STEPPER MTR MODULE, 2 x 3AMP         |                  | 20       |
|               | 12             | E           | 606119      |           | B&R X67 THREADED CAP M8 X67AC0M08                  |                  | 20       |
|               | 13             | E           | 606120      |           | B&R X67 THREADED CAP M12 X67AC0M12                 |                  | 40       |
|               | 14             | M           | 421147      |           | SOCKET CAP SCREW M4 x 12 GRADE 12.9                |                  | 40       |
|               | 21             | E           | 606110      |           | B&R X67CA0P20.0050, POWER CABLE 5m                 |                  | 21       |
|               | 31             | E           | 606118      |           | B&R X67CA0X21.0050, X2X CABLE 5m                   |                  | 1        |
|               | 41             | E           | 606177G     |           | B&R X67CA0X01.0005, X2X LINK CABLE 0.5m            |                  | 18       |
|               | 51             | E           | 606113      |           | B&R X67CA0X01.0050, X2X LINK CABLE 5.0m            |                  | 1        |
|               | 51             | E           | 606185G     |           | B&R X67CA0X01.0100, X2X LINK CABLE 10.0m           |                  | 0        |
|               | 61             | E           | 606090      |           | X2X Link Supply 24Vdc                              |                  | 1        |
|               | 71             | E           | 606113      |           | B&R X67CA0X01.0050, X2X LINK CABLE 5.0m            |                  | 1        |
|               | 72             | E           | 606185G     |           | B&R X67CA0X01.0100, X2X LINK CABLE 10.0m           |                  | 0        |
|               | 80             | M           | 421147      |           | SOCKET CAP SCREW M4 x 12 GRADE 12.9                |                  | 2        |
| <b>04</b>     |                |             |             |           | <b>Mech External</b>                               |                  |          |
| <b>04-010</b> | <b>910G264</b> |             |             |           | <b>Faceplate Sideplate &amp; Tie Bar Assy</b>      | <b>429G207</b>   | <b>1</b> |
|               | B001           | M           | 239216      |           | Seal O Ring RM 0246-24 Nitrile                     |                  | 2        |
|               | B002           | M           | 239374      |           | O Ring Cord 16.0mm c/s Nitrile                     |                  | 14       |
|               | B003           | M           | 239324      |           | Nitrile O-Ring RM2093-57                           |                  | 2        |
|               | B004           | M           | 239288      |           | NITRILE O-RING RM4193-57                           |                  | 2        |
|               | B005           |             | 242A080     |           | LOCKNUT KM30                                       |                  | 1        |
|               | B006           | M           | 268121      |           | 3M Safety - Walk General Purpose 51 wide Black     |                  | 1        |
|               | M001           | M           | 429G208     |           | Mechanism Faceplate                                |                  | 1        |
|               | M002           | M           | 429G209     |           | Mechanism Sideplate                                |                  | 1        |
|               | M003           | M           | 303G508     |           | 1/2" BSP PLUG                                      |                  | 2        |
|               | M004           | M           | 419G467     |           | 2900 Tie Bar                                       |                  | 2        |
|               | M005           | M           | 421G232     |           | Top Hat                                            |                  | 2        |
|               | M006           | M           | 417G382     |           | 2900 Tie Bar                                       |                  | 1        |
|               | M007           | M           | 404G158     |           | Lifting Bracket                                    |                  | 2        |
|               | M008           | M           | 420G960     |           | Adjustable Stiffener Mounting                      |                  | 1        |
|               | M009           | M           | 420G961     |           | Adjustable Stiffener                               |                  | 1        |
|               | M010           | M           | 421G986     |           | 2900 Tie Bar                                       |                  | 1        |
|               | M012           | M           | 421G996     | IT1       | Transport Bracket                                  |                  | 3        |
|               | M013           | M           | 421G996     | IT2       | Transport Bracket                                  |                  | 1        |
|               | M014           | M           | 424G076     |           | Trunking Support Bracket                           |                  | 2        |
|               | S001           | M           | 421115      |           | SOCKET CAP SCREW M16 x 45 GRADE 12.9               |                  | 28       |
|               | S002           | M           | 421310      |           | SOCKET CAP SCREW M12 x 65 GRADE 12.9               |                  | 12       |
|               | S003           | M           | 421104      |           | SOCKET CAP SCREW M16 x 50 GRADE 12.9               |                  | 46       |
|               | S004           | M           | 421059      |           | Washer Plain M16                                   |                  | 22       |
|               | S005           | M           | 421116      |           | Washer Spring M16                                  |                  | 46       |
| <b>04-040</b> | <b>910G274</b> |             |             |           | <b>Roller Drive - Draw Roller</b>                  | <b>429G292</b>   | <b>1</b> |
|               | B001           | M           | 232462G     |           | KTR RADEX-N NN Size 60 (48-38 H7 Bore)             |                  | 1        |
|               | B002           | M           | 664483G     |           | SIMOTICS M MOTOR 1PH8107-1DF02-2GA1, 23.5A         |                  | 1        |
|               | M001           | M           | 429G167     |           | Bell Housing                                       |                  | 1        |
|               | M002           | M           | 425G902     | IT1       | Shaft Extension                                    |                  | 1        |
|               | M003           | M           | 418G248     | IT13      | Key                                                |                  | 1        |
|               | S001           | M           | 421086      |           | Screw HEXHD Setscrew M12x40 GRADE 8.8              |                  | 4        |
|               | S002           | M           | 421030      |           | PLAIN WASHER M12                                   |                  | 7        |
|               | S003           | M           | 421210      |           | SOCKET CAP SCREW M12 x 35 GRADE 12.9               |                  | 3        |
|               | S006           | M           | 421969      |           | GRUB SCREW M8x10                                   |                  | 2        |
| <b>04-050</b> | <b>908G728</b> |             |             |           | <b>WATER COOLED DRUM DRIVE ASSY</b>                | <b>420G964</b>   | <b>1</b> |
|               | B001           | M           | 221789      |           | Belt Polychain 14M-1568-37                         |                  | 1        |
|               | B002           | M           | 225338      |           | Sprocket Polychain 14M-28S-37                      |                  | 1        |
|               | B003           | M           | 220663      |           | Pulley Polychain 14M-80S-90                        |                  | 1        |
|               | B004           | M           | 224174      |           | TAPERLOCK BUSH 2012-48                             |                  | 1        |
|               | B005           | M           | 224285      |           | Bush Taperlock 4030 Bore 115 W/- Std KeyTL4030/115 |                  | 1        |
|               | B006           | M           | 664504G     |           | SIMOTICS M MOTOR 1PH8133-1DF00-2GA1 17.5kW 34A     |                  | 1        |
|               | M001           | M           | 425G844     |           | Motor Support Bracket                              |                  | 1        |
|               | S001           | M           | 421706      |           | SOCKET CAP SCREW M10 x 60 GRADE 12.9               |                  | 4        |
|               | S002           | M           | 421104      |           | SOCKET CAP SCREW M16 x 50 GRADE 12.9               |                  | 4        |
|               | S003           | M           | 421027      |           | NUT FULL HEX M10                                   |                  | 4        |
|               | S004           | M           | 421042      |           | NUT FULL HEX M12                                   |                  | 3        |
|               | S005           | M           | 421455      |           | Screw HEXHD Setscrew M12x70 GRADE 8.8              |                  | 3        |
|               | S006           | M           | 421029      |           | PLAIN WASHER M10                                   |                  | 8        |
|               | S007           | M           | 421059      |           | Washer Plain M16                                   |                  | 4        |
| <b>04-060</b> | <b>908G673</b> |             |             |           | <b>ROLLER DRIVE ASSY</b>                           | <b>421G938</b>   | <b>1</b> |
|               | B001           | M           | 232463G     |           | KTR Radex Coupling NN Size 60 (48-48 H7 BORE)      |                  | 1        |
|               | B002           | M           | 664500G     |           | SIMOTICS M MOTOR 1PH8133-1DD02-2GA1 13.5kW, 29A    |                  | 1        |

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| Section       | Bubble Number  | Mech / Elec | Part Number | Parameter | Part Description                                  | Assembly Drawing | Qty      |
|---------------|----------------|-------------|-------------|-----------|---------------------------------------------------|------------------|----------|
|               | M001           | M           | 421G935     |           | BELL HOUSING                                      |                  | 1        |
|               | S001           | M           | 421231      |           | SOCKET CAP SCREW M16 x 70 GRADE 12.9              |                  | 4        |
|               | S002           | M           | 421030      |           | PLAIN WASHER M12                                  |                  | 3        |
|               | S003           | M           | 421210      |           | SOCKET CAP SCREW M12 x 35 GRADE 12.9              |                  | 3        |
|               | S004           | M           | 421038      |           | NUT FULL HEXAGON M16                              |                  | 4        |
|               | S005           | M           | 421059      |           | Washer Plain M16                                  |                  | 8        |
| <b>04-070</b> | <b>908G676</b> |             |             |           | <b>WCR DRIVE ASSY</b>                             | <b>421G968</b>   | <b>1</b> |
|               | B001           | M           | 664480G     |           | SINAMICS VECTOR MOTOR 6.25KW 34NM 1750RPM 13A     |                  | 1        |
|               | B002           | M           | 225295      |           | Sprocket Polychain 14M-40S-20                     |                  | 1        |
|               | B003           | M           | 225296      |           | Sprocket Polychain 14M-44S-20                     |                  | 1        |
|               | B004           | M           | 221784      |           | Belt Polychain 14M-1400-20                        |                  | 1        |
|               | B005           | M           | 224271      |           | Bush Taperlock 2517 Bore 38 W/- Std Key TL2517/38 |                  | 1        |
|               | B006           | M           | 224375G     |           | Bush Taperlock 3020 Bore 75 W/- Std Key TL3020/75 |                  | 1        |
|               | M001           | M           | 421G973     |           | MOTOR SUPPORT BRACKET                             |                  | 1        |
|               | M002           | M           | 422G064     |           | JACKING BLOCK                                     |                  | 1        |
|               | S001           | M           | 421029      |           | PLAIN WASHER M10                                  |                  | 8        |
|               | S002           | M           | 421021      |           | Washer Spring M10                                 |                  | 4        |
|               | S003           | M           | 421059      |           | Washer Plain M16                                  |                  | 4        |
|               | S004           | M           | 421161      |           | SOCKET CAP SCREW M10 x 50 GRADE 12.9              |                  | 4        |
|               | S005           | M           | 421303      |           | SOCKET CAP SCREW M16 x 40 GRADE 12.9              |                  | 4        |
|               | S006           | M           | 421210      |           | SOCKET CAP SCREW M12 x 35 GRADE 12.9              |                  | 2        |
|               | S007           | M           | 421027      |           | NUT FULL HEX M10                                  |                  | 4        |
|               | S008           | M           | 421358      |           | Screw HEXHD Setscrew M12x80 GRADE 8.8             |                  | 2        |
|               | S009           | M           | 421042      |           | NUT FULL HEX M12                                  |                  | 2        |
| <b>04-080</b> | <b>908G624</b> |             |             |           | <b>Rewind Layon Drive Assy</b>                    | <b>425G936</b>   | <b>1</b> |
|               | B001           | M           | 250480G     |           | SIMOGear HELICAL GEARBOX                          |                  | 1        |
|               | B002           | M           | 232406G     |           | Nomar MCLC Rigid Coupling Size 25                 |                  | 1        |
|               | B003           | M           | 664526G     |           | AC MOTOR 0.82KW 3NM 3000RPM DRIVE-CLIQ            |                  | 1        |
|               | M001           | M           | 421G321     |           | Bell Housing Rewind Layon                         |                  | 1        |
|               | S001           | M           | 421122      |           | SOCKET CAP SCREW M10 x 30 GRADE 12.9              |                  | 3        |
|               | S003           | M           | 421114      |           | SOCKET CAP SCREW M10 x 40 GRADE 12.9              |                  | 4        |
|               | S004           | M           | 421019      |           | Socket Head Capscrew M6 x 20 GRADE 12.9           |                  | 4        |
|               | S005           | M           | 421027      |           | NUT FULL HEX M10                                  |                  | 4        |
|               | S006           | M           | 421029      |           | PLAIN WASHER M10                                  |                  | 11       |
|               | S008           | M           | 421021      |           | Washer Spring M10                                 |                  | 7        |
| <b>04-090</b> | <b>908G627</b> |             |             |           | <b>Unwind Layon Drive Assy</b>                    | <b>425G938</b>   | <b>1</b> |
|               | B001           | M           | 225297      |           | Sprocket Polychain 8M-34S-36                      |                  | 2        |
|               | B002           | M           | 221768      |           | Belt Polychain 8M-1000-36                         |                  | 1        |
|               | B003           | M           | 250482G     |           | SIMOGear HELICAL GEARBOX                          |                  | 1        |
|               | B004           | M           | 224286      |           | Bush Taperlock 1610 Bore 25W/- Std Key TL1610/25  |                  | 1        |
|               | B005           | M           | 224284      |           | Bush Taperlock 1610 Bore 30 W/- Std Key TL1610/30 |                  | 1        |
|               | B006           | M           | 664526G     |           | AC MOTOR 0.82KW 3NM 3000RPM DRIVE-CLIQ            |                  | 1        |
|               | M001           | M           | 421G111     |           | Unwind Layon Motor Bracket                        |                  | 1        |
|               | M002           | M           | 418G248     | IT20      | Key                                               |                  | 1        |
|               | S001           | M           | 421122      |           | SOCKET CAP SCREW M10 x 30 GRADE 12.9              |                  | 6        |
|               | S002           | M           | 421402      |           | SOCKET CAP SCREW M12 x 60 GRADE 12.9              |                  | 4        |
|               | S003           | M           | 421036      |           | NUT FULL HEXAGON M8                               |                  | 2        |
|               | S004           | M           | 421019      |           | Socket Head Capscrew M6 x 20 GRADE 12.9           |                  | 4        |
|               | S005           | M           | 421042      |           | NUT FULL HEX M12                                  |                  | 4        |
|               | S006           | M           | 421047      |           | Screw HEXHD Setscrew M8x60 GRADE 8.8              |                  | 2        |
|               | S008           | M           | 421029      |           | PLAIN WASHER M10                                  |                  | 6        |
|               | S009           | M           | 421030      |           | PLAIN WASHER M12                                  |                  | 8        |
|               | S012           | M           | 421021      |           | Washer Spring M10                                 |                  | 6        |
|               | S013           | M           | 421073      |           | Washer Spring M12                                 |                  | 4        |
| <b>04-100</b> | <b>908G176</b> |             |             |           | <b>Flanged Castor Drive Assembly</b>              | <b>420G072</b>   | <b>1</b> |
|               | B001           | M           | 225346      |           | SPROCKET 214100 (1" PITCH 17 TEETH Duplex)        |                  | 1        |
|               | B002           | M           | 225347      |           | SPROCKET 214114-75 (1" PITCH 30 TEETH Duplex)     |                  | 1        |
|               | B003           | M           | 225348      |           | Chain 1"P Duplex                                  |                  | 1        |
|               | B005           | M           | 202046      |           | FLANGED CARTRIDGE MFC75                           |                  | 4        |
|               | B006           | M           | 662192G     |           | MOTOR VARIO BRAKE MOTOR/GEAR BOX                  |                  | 1        |
|               | M001           | M           | 418G181     |           | Flanged Castor - K5000                            |                  | 1        |
|               | M002           | M           | 414G962     |           | Plain Wheel                                       |                  | 1        |
|               | M003           | M           | 419G680     |           | Castor Positioning Block                          |                  | 1        |
|               | M004           | M           | 420G073     |           | CASTOR BRACKET NON DRIVEN                         |                  | 1        |
|               | M005           | M           | 420G074     |           | CASTOR BRACKET DRIVEN                             |                  | 1        |
|               | M006           | M           | 420G081     |           | JACKING BLOCK DRIVEN CASTOR                       |                  | 2        |
|               | M007           | M           | 420G129     |           | COMPRESSION PLATE                                 |                  | 2        |
|               | M008           | M           | 420G132     |           | DRIVE COLLAR K5000 DRIVEN CASTOR                  |                  | 2        |

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| Section       | Bubble Number  | Mech / Elec | Part Number | Parameter | Part Description                                 | Assembly Drawing | Qty      |
|---------------|----------------|-------------|-------------|-----------|--------------------------------------------------|------------------|----------|
|               | M009           | M           | 420G138     |           | CASTOR SHAFT (LONG)                              |                  | 1        |
|               | M010           | M           | 418G249     | IT3       | Key                                              |                  | 4        |
|               | M011           | M           | 418G249     | IT4       | Key                                              |                  | 1        |
|               | S001           | M           | 421616      |           | GRUB SCREW M12x10                                |                  | 10       |
|               | S002           | M           | 421302      |           | Bolt HEXHD M16X40 GRADE 8.8                      |                  | 16       |
|               | S003           | M           | 421251      |           | Screw HEXHD Setscrew M16x50 GRADE 8.8            |                  | 4        |
|               | S004           | M           | 421601      |           | Bolt HEXHD M16x75 GRADE 8.8                      |                  | 4        |
|               | S005           | M           | 4211878G    |           | Bolt HEXHD Setscrew M24x50 GRADE 8.8             |                  | 16       |
|               | S006           | M           | 421068      |           | Screw HEXHD Setscrew M12x30 GRADE 8.8            |                  | 2        |
|               | S007           | M           | 421038      |           | NUT FULL HEXAGON M16                             |                  | 12       |
|               | S008           | M           | 421059      |           | Washer Plain M16                                 |                  | 24       |
|               | S009           | M           | 421030      |           | PLAIN WASHER M12                                 |                  | 2        |
|               | S010           | M           | 421069      |           | PLAIN WASHER M24                                 |                  | 22       |
|               | S011           | M           | 4211879G    |           | Bolt HEXHD M24x90 GRADE 8.8                      |                  | 6        |
|               | S012           | M           | 421583      |           | GRUB SCREW M12x12                                |                  | 4        |
|               | S013           | M           | 421026      |           | Screw Socket Countersunk Head M6 x 25            |                  | 4        |
|               | S014           | M           | 421298      |           | Screw HEXHD Setscrew M16X120 GRADE 8.8           |                  | 4        |
| <b>04-130</b> | <b>908G808</b> |             |             |           | <b>Mech Pneumatic Items</b>                      | <b>422G184</b>   | <b>1</b> |
|               | B001           | M           | 2121520G    |           | VTUG Series VG VALVE TERMINAL                    |                  | 1        |
|               | B002           | M           | 2121507G    |           | Pilot Operated Check valve                       |                  | 2        |
|               | B003           | M           | 427461G     |           | 6mm union tee                                    |                  | 4        |
|               | B006           | M           | 423265      |           | Nylon Tube Natural 6mm O/D                       |                  | 15       |
|               | B007           | M           | 4232181G    |           | Male Elbow 6mm OD to 1/4" BSP                    |                  | 4        |
|               | B008           | M           | 427639G     |           | Swagelok Reducing Union 6mm to 1/4"              |                  | 5        |
|               | B009           | M           | 239060      |           | O' RING RM 0315.30                               |                  | 4        |
|               | B010           | M           | 427641G     |           | Swagelok Reducing Union 4mm to 1/4"              |                  | 1        |
|               | B011           | M           | 211373      |           | REED SWITCH & CONNECTOR                          |                  | 2        |
|               | B012           | M           | 2121487G    |           | Sintered Bronze Silencer 1/8"                    |                  | 1        |
|               | B013           | M           | 427656G     |           | 1/4" tube - 1/8" Female Connector                |                  | 1        |
|               | B014           | M           | 427704G     |           | 6mm to 1/8BSP Male Connector                     |                  | 3        |
|               | B015           | M           | 427709G     |           | Reducing Union Elbow Stainless Steel 6mm to 1/8" |                  | 1        |
|               | M001           | M           | 418G015     |           | 1/4" Leadthrough                                 |                  | 4        |
|               | REF1           | M           | 419G411     |           | Faceplate/Sideplate & Tie Bar Assy               |                  | 1        |
|               | REF2           | M           | 421G980     |           | MECH END WATER PIPING ASSEMBLY                   |                  | 1        |
|               | REF3           | M           | 421G631     |           | Rotary Union Pipework                            |                  | 1        |
|               | REF4           | M           | 422G149     |           | PRE PLASMA PSU WATER                             |                  | 1        |
|               | REF5           | M           | 422G150     |           | POST PLASMA PSU WATER                            |                  | 1        |
|               | REF6           | M           | 419G095     |           | LATCH ASSY                                       |                  | 1        |
|               | S001           | M           | 423520      |           | POLYURETHANE TUBE 20MTR COIL 6MM O/D             |                  | 1        |
|               | S002           | M           | 423313      |           | Tube Polyurethane 100mt Coil of 12mm TU1208C-100 |                  | 1        |
|               | S003           | M           | 427021      |           | SMC' KQR08-12 REDUCER PUSH IN 8mm - 12mm         |                  | 1        |
|               | S004           | M           | 421890      |           | SOCKET CAP SCREW M5 x 40 GRADE 12.9              |                  | 4        |
|               | S005           | M           | 427642G     |           | Swagelok Reducing Union 8mm to 1/4"              |                  | 1        |
|               | S006           | M           | 423034      |           | Tube 8mm O/D Polyurethane                        |                  | 4        |
|               | S007           | M           | 423265      |           | Nylon Tube Natural 6mm O/D                       |                  | 5        |
| <b>04-140</b> | <b>908G868</b> |             |             |           | <b>Mech Warning Label Locations</b>              | <b>328G489</b>   | <b>1</b> |
|               | M007           | M           | 323G716     | IT11      | Warning Label                                    |                  | 1        |
|               | M013           | M           | 323G844     | IT11      | Label (Grease castors)                           |                  | 6        |
|               | M024           | M           | 328G090     | IT11      | Warning Label - O2 Gas Nozzle Cleaning           |                  | 1        |
|               | M026           | M           | 328G761     | IT11      | Combined Warning Labels                          |                  | 1        |
|               | M027           | M           | 328G832     | IT11      | Combined Warning Labels                          |                  | 1        |
|               | S001           | M           | 421364      |           | SOCKET CAP SCREW M4 x 8 GRADE 12.9               |                  | 12       |
| <b>04-150</b> | <b>905G841</b> |             |             |           | <b>Rotary union blanks</b>                       | <b>422G184</b>   | <b>1</b> |
|               | B001           | M           | 4231537G    |           | 2" BSPP FEMALE CAP 60 CONE AF0132                |                  | 2        |
|               | B002           | M           | 4231536G    |           | 1-1/2"BSPP FEMALE CAP 60 CONE AF0124             |                  | 2        |
|               | B003           | M           | 4231637G    |           | 2" BSPP MALE CAP - 60 DEGREE CONE                |                  | 2        |
|               | B004           | M           | 4231638G    |           | 1 1/2" BSPP MALE CAP - 60 DEGREE CONE            |                  | 2        |
| <b>04-160</b> | <b>908G754</b> |             |             |           | <b>Oil Bottle Assy</b>                           | <b>422G185</b>   | <b>4</b> |
|               | B001           | M           | 425003      |           | Oil Reservoir 1/4"BSP (Ref OR 7730)              |                  | 1        |
|               | B002           | M           | 423038      |           | Tubing Nut 8 OD                                  |                  | 1        |
|               | B003           | M           | 423036      |           | Adaptor Male 8mm-1/4"BSPT 36-0530-11             |                  | 1        |
|               | B004           | M           | 423039      |           | Tubing Sleeve 8 OD                               |                  | 1        |
|               | B005           | M           | 423049      |           | Nylon 12 BS5409 Normal Duty 8mm O/D              |                  | 1        |
|               | M001           | M           | 422G186     |           | Oil Bottle Bracket                               |                  | 1        |
|               | S001           | M           | 421049      |           | SOCKET CAP SCREW M8 x 20 GRADE 12.9              |                  | 2        |
|               | S002           | M           | 421054      |           | PLAIN WASHER M8                                  |                  | 2        |
| <b>04-170</b> | <b>910G465</b> |             |             |           | <b>O2 MFC and Pipework Assembly</b>              | <b>430G036</b>   | <b>1</b> |
|               | B001           | M           | 4231657G    |           | Tube Adaptor 1/4" OD x 7/16-20 SAE FKM O-seal    |                  | 40       |

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| Section       | Bubble Number  | Mech / Elec | Part Number | Parameter | Part Description                                         | Assembly Drawing | Qty      |
|---------------|----------------|-------------|-------------|-----------|----------------------------------------------------------|------------------|----------|
|               | B002           | M           | 423844      |           | Union Elbow 1/4" O/D.(Brass)                             |                  | 80       |
|               | B003           | M           | 427158      |           | Stainless Steel Tube 1/4" OD                             |                  | 2        |
|               | B004           | M           | 4231659G    |           | Male Conn 1/4" x 7/16-20 SAE/MS O-seal bored thro'       |                  | 40       |
|               | B005           | M           | 427696G     |           | Male Connector                                           |                  | 2        |
|               | B006           | M           | 4232542G    |           | Plug 1/4" Brass                                          |                  | 2        |
|               | B007           | M           | 4232014G    |           | PORT CONNECTOR 1/2" OD BRASS                             |                  | 2        |
|               | B008           | M           | 427368      |           | Union Tee 1/2" OD                                        |                  | 1        |
|               | B009           | M           | 427305      |           | Union Elbow 1/2" OD Brass                                |                  | 1        |
|               | B010           | M           | 426225G     |           | MFC GE50A no shut-off valve 3000scm O2 1/4"swag          |                  | 40       |
|               | B011           | M           | 239386      |           | Seal O Ring RM3593-57 Nitrile                            |                  | 1        |
|               | B012           | M           | 4211908G    |           | Pin-Lock - Stainless Steel Tumbler Lk - 20mm grip        |                  | 6        |
|               | B013           | M           | 427548G     |           | Swagelok PFA Tubing 1/4" x 0.062 wall                    |                  | 3        |
|               | B018           | M           | 427497G     |           | Male Connector                                           |                  | 2        |
|               | B019           | M           | 4231645G    |           | Quick Connecting Flexible Hose 1/2"                      |                  | 1        |
|               | B020           | E           | 650689      |           | ET200B L2 PROFIBUS CONNECTOR 6ES7-972-0BA42-0XA0         |                  | 40       |
|               | B023           | M           | 268071      |           | Grease UT 18 (100g tube) -40/+250                        |                  | 1        |
|               | B024           | M           | 424418G     |           | HAM-LET DIAPHRAGM VALVE,LET-LOK 1/2"                     |                  | 1        |
|               | B025           | M           | 2121467G    |           | Solenoid Valve & Plug Socket Kit                         |                  | 1        |
|               | B026           | M           | 427176      |           | Straight connector 1/8" BSPT male to 6mm push in         |                  | 1        |
|               | B027           | M           | 427353      |           | Plastic tubing PUN-6x1-SI                                |                  | 1        |
|               | B028           | M           | 423843      |           | Port Connector 1/2" O/D                                  |                  | 1        |
|               | M001           | M           | 425G811     | IT4       | MFC Shelf                                                |                  | 1        |
|               | M002           | M           | 428G869     | IT4       | Guard SIDE                                               |                  | 2        |
|               | M003           | M           | 428G872     | IT9       | Leadthrough Plate                                        |                  | 1        |
|               | M004           | M           | 425G850     | IT4       | Guard                                                    |                  | 1        |
|               | M005           | M           | 428G868     | IT4       | Guard BASE                                               |                  | 1        |
|               | M006           | M           | 412G896     | IT16      | Alox Manifold                                            |                  | 2        |
|               | M007           | M           | 430G045     | IT1       | Hose Cover                                               |                  | 1        |
|               | M008           | M           | 430G089     |           | Hose Cover                                               |                  | 1        |
|               | M012           | M           | 430G052     |           | Guard Edge                                               |                  | 1        |
|               | M013           | M           | 430G053     |           | Guard Edge                                               |                  | 1        |
|               | REF2           | M           | 425G813     |           | O2 Faceplate Adaptor Plate                               |                  | 1        |
|               | S001           | M           | 421030      |           | PLAIN WASHER M12                                         |                  | 4        |
|               | S002           | M           | 421245      |           | SOCKET CAP SCREW M5 x 10 GRADE 12.9                      |                  | 8        |
|               | S003           | M           | 421526      |           | SOCKET CAP SCREW M12 x 45 GRADE 12.9                     |                  | 4        |
|               | S004           | M           | 421146      |           | SOCKET CAP SCREW M8 x 30 GRADE 12.9                      |                  | 10       |
|               | S005           | M           | 421861      |           | Screw Caphead 8-32 UNC-2B x1/2"                          |                  | 80       |
|               | S006           | M           | 421164      |           | WASHER PLAIN M5                                          |                  | 4        |
|               | S007           | M           | 421034      |           | WASHER PLAIN M6                                          |                  | 7        |
|               | S008           | M           | 421033      |           | SOCKET CAP SCREW M6 x 12 GRADE 12.9                      |                  | 7        |
|               | S009           | M           | 421182      |           | SOCKET CAP SCREW M8 x 16 GRADE 12.9                      |                  | 4        |
|               | S012           | M           | 421054      |           | PLAIN WASHER M8                                          |                  | 4        |
|               | S016           | M           | 421658      |           | SOCKET CAP SCREW M4 x 25 GRADE 12.9                      |                  | 2        |
|               | S017           | M           | 421194      |           | Washer Plain M4                                          |                  | 2        |
| <b>04-180</b> | <b>909G544</b> |             |             |           | <b>M.E. Pre Plasma Treater Dual Circuit Water Piping</b> | <b>426G019</b>   | <b>1</b> |
|               | B001           | M           | 424341G     |           | 1/2" BSPT Ball Valve M/F (Full Bore)                     |                  | 4        |
|               | B002           | M           | 423750      |           | Socket 1/2" BSP St Steel                                 |                  | 2        |
|               | B003           | M           | 427078      |           | Tube 1/2" P.O. Multi purpose Hose Blue                   |                  | 10       |
|               | B004           | M           | 427263      |           | Tube 1/2" P.O. Multi purpose hose Green                  |                  | 8        |
|               | B005           | M           | 427072      |           | Male Hose Adaptor                                        |                  | 6        |
|               | B006           | M           | 4231910G    |           | IFM Adaptor G1/2 - R1/2 E40199 for SM6000                |                  | 4        |
|               | B007           | M           | 4231917G    |           | IFM SM6004 Magnetic Flow Meter 25 l/min                  |                  | 2        |
|               | B008           | M           | 423813      |           | Nipple Hexagon 1/2" BSP ST.ST.                           |                  | 2        |
|               | B009           | M           | 4231053     |           | Elbow Male/Female 1/2" BSP SS                            |                  | 4        |
|               | B010           | M           | 4231018     |           | Branch Tee 1/2" Type 161 Stainless Steel                 |                  | 2        |
|               | M001           | M           | 426G020     |           | WATER HOSE BRACKET                                       |                  | 2        |
|               | M002           | M           | 414G407     | IT3       | Flow Restrictor                                          |                  | 2        |
|               | S001           | M           | 421019      |           | Socket Head Capscrew M6 x 20 GRADE 12.9                  |                  | 4        |
|               | S002           | M           | 421769      |           | SOCKET CAP SCREW M4x30 GRADE 12.9                        |                  | 2        |
|               | S003           | M           | 421034      |           | WASHER PLAIN M6                                          |                  | 4        |
| <b>04-190</b> | <b>910G255</b> |             |             |           | <b>Auto Gas Control Pre Plasma Treater</b>               | <b>429G161</b>   | <b>1</b> |
|               | B001           | M           | 426223G     |           | MFC GE50A with shut off valve 2000scm O2 1/4"swa         |                  | 1        |
|               | B002           | M           | 426222G     |           | MFC GE50A with shut off valve 2000scm Ar 1/4"swa         |                  | 1        |
|               | B004           | M           | 423191      |           | Union Tee 1/4" O/D                                       |                  | 1        |
|               | B005           | M           | 423211      |           | Union Elbow 1/4" O/D                                     |                  | 1        |
|               | B006           | M           | 427158      |           | Stainless Steel Tube 1/4" OD                             |                  | 1        |
|               | B007           | M           | 4232296G    |           | Hose Assy Braided St.St. Ends 1/4" tube adapters         |                  | 1        |
|               | B008           | M           | 421861      |           | Screw Caphead 8-32 UNC-2B x1/2"                          |                  | 4        |

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| Section       | Bubble Number  | Mech / Elec | Part Number | Parameter | Part Description                                             | Assembly Drawing | Qty      |
|---------------|----------------|-------------|-------------|-----------|--------------------------------------------------------------|------------------|----------|
|               | B011           | M           | 424032      |           | Centering Ring DN 16 ISO-KF (OBSOLETE)                       |                  | 1        |
|               | B012           | M           | 424033      |           | Rapid Fastening Clamp Ring DN 10-16                          |                  | 1        |
|               | B013           | E           | 650689      |           | ET200B L2 PROFIBUS CONNECTOR 6ES7-972-0BA42-0XA0             |                  | 2        |
|               | M001           | M           | 406G639     |           | Spacer                                                       |                  | 2        |
|               | S001           | M           | 421703      |           | SOCKET CAP SCREW M4 x 16 GRADE 12.9                          |                  | 4        |
|               | S002           | M           | 421194      |           | Washer Plain M4                                              |                  | 8        |
| <b>04-195</b> | <b>908G485</b> |             |             |           | <b>Baratron Assy</b>                                         | <b>421G202</b>   | <b>1</b> |
|               | B001           | M           | 426158G     |           | Baratron Type 722B 1 Torr range KF16                         |                  | 1        |
|               | B002           | M           | 424032      |           | Centering Ring DN 16 ISO-KF (OBSOLETE)                       |                  | 2        |
|               | B003           | M           | 424033      |           | Rapid Fastening Clamp Ring DN 10-16                          |                  | 1        |
|               | B004           | M           | 424008      |           | Elbow 90deg NW16KF                                           |                  | 1        |
| <b>04-250</b> | <b>908G799</b> |             |             |           | <b>Electrical Mounting Brackets (Mech External)</b>          | <b>422G053</b>   | <b>1</b> |
|               | M001           | M           | 422G092     |           | Remote I/O Bracket                                           |                  | 1        |
|               | S001           | M           | 421769      |           | SOCKET CAP SCREW M4x30 GRADE 12.9                            |                  | 20       |
|               | S002           | M           | 421133      |           | SOCKET CAP SCREW M5 x 16 GRADE 12.9                          |                  | 4        |
|               | S003           | M           | 421049      |           | SOCKET CAP SCREW M8 x 20 GRADE 12.9                          |                  | 2        |
|               | S004           | M           | 421170      |           | Nut Full Hex M5                                              |                  | 4        |
|               | S005           | M           | 421164      |           | WASHER PLAIN M5                                              |                  | 8        |
|               | S006           | M           | 421054      |           | PLAIN WASHER M8                                              |                  | 2        |
| <b>04-260</b> | <b>908G838</b> |             |             |           | <b>GAS WEDGE MFC ASSEMBLY</b>                                | <b>422G377</b>   | <b>1</b> |
|               | B001           | M           | 426221G     |           | MFC GE50A w/shut off valve 20000scm Air 1/4" swa             |                  | 2        |
|               | B002           | M           | 427427G     |           | Union 1/4" x 1/4" BRASS B-400-6                              |                  | 2        |
|               | B003           | M           | 4232430G    |           | Female Adaptor 1/4" OD x 1/4" BSPP                           |                  | 2        |
|               | B004           | M           | 212068      |           | BREATHER                                                     |                  | 2        |
|               | B005           | M           | 423212      |           | Nylon Tube Natural 1/4" O/D                                  |                  | 1        |
|               | B008           | M           | 421861      |           | Screw Caphead 8-32 UNC-2B x1/2"                              |                  | 4        |
|               | B013           | E           | 650689      |           | ET200B L2 PROFIBUS CONNECTOR 6ES7-972-0BA42-0XA0             |                  | 2        |
|               | M001           | M           | 422G329     |           | BRACKET-MFC                                                  |                  | 1        |
|               | M002           | M           | 406G639     |           | Spacer                                                       |                  | 2        |
|               | S001           | M           | 421049      |           | SOCKET CAP SCREW M8 x 20 GRADE 12.9                          |                  | 2        |
|               | S002           | M           | 421703      |           | SOCKET CAP SCREW M4 x 16 GRADE 12.9                          |                  | 4        |
|               | S003           | M           | 421054      |           | PLAIN WASHER M8                                              |                  | 2        |
|               | S004           | M           | 421194      |           | Washer Plain M4                                              |                  | 8        |
| <b>04-270</b> | <b>908G860</b> |             |             |           | <b>Rewind Layon Gas Wedge MFCs</b>                           | <b>422G642</b>   | <b>1</b> |
|               | B001           | M           | 423265      |           | Nylon Tube Natural 6mm O/D                                   |                  | 2        |
|               | B002           | M           | 4232430G    |           | Female Adaptor 1/4" OD x 1/4" BSPP                           |                  | 2        |
|               | B003           | M           | 421861      |           | Screw Caphead 8-32 UNC-2B x1/2"                              |                  | 4        |
|               | B004           | M           | 426221G     |           | MFC GE50A w/shut off valve 20000scm Air 1/4" swa             |                  | 2        |
|               | B005           | E           | 650689      |           | ET200B L2 PROFIBUS CONNECTOR 6ES7-972-0BA42-0XA0             |                  | 2        |
|               | B006           | M           | 212068      |           | BREATHER                                                     |                  | 2        |
|               | M001           | M           | 406G639     |           | Spacer                                                       |                  | 2        |
|               | M002           | M           | 422G330     |           | BRACKET-MFC                                                  |                  | 1        |
|               | M003           | M           | 422G639     |           | Support Bracket                                              |                  | 2        |
|               | M004           | M           | 422G640     |           | Packing Strip                                                |                  | 2        |
|               | S001           | M           | 421703      |           | SOCKET CAP SCREW M4 x 16 GRADE 12.9                          |                  | 4        |
|               | S002           | M           | 421049      |           | SOCKET CAP SCREW M8 x 20 GRADE 12.9                          |                  | 12       |
|               | S003           | M           | 421036      |           | NUT FULL HEXAGON M8                                          |                  | 8        |
|               | S004           | M           | 421194      |           | Washer Plain M4                                              |                  | 8        |
|               | S005           | M           | 421054      |           | PLAIN WASHER M8                                              |                  | 16       |
| <b>04-280</b> | <b>909G833</b> |             |             |           | <b>Oil Bottle Labels</b>                                     | <b>427G017</b>   | <b>1</b> |
|               | M001           | M           | 424G519     |           | Lubricant Label - Draw Roller - Oil AWS68                    |                  | 1        |
|               | M002           | M           | 424G520     |           | Lubricant Label - DTR2 Roller - Oil AWS68                    |                  | 1        |
|               | M003           | M           | 424G521     |           | Lubricant Label - Source Flap - Oil AWS68                    |                  | 1        |
|               | M004           | M           | 424G522     |           | Lubricant Label - Drum - Oil AWS10                           |                  | 1        |
|               | M005           | M           | 424G523     |           | Lubricant Label - DTR1 Roller - Oil AWS10                    |                  | 1        |
| <b>04-700</b> | <b>908G700</b> |             |             |           | <b>K5000 - O2 Gas Control &amp; MFC Cubicles</b>             | <b>429G161</b>   | <b>1</b> |
|               | 11             | M           | 245G175     |           | K5000 - T06_21 Cubicle - O2 Control Cubicle                  |                  | 1        |
|               | 12             | E           | 600050G     |           | Hoffman T03_8/T06_21 Cubicle - 600x600x210mm                 |                  | 1        |
|               | 13             | E           | 6161338G    |           | ELST 4012 M12 Straight Cable Plug 4 Pin                      |                  | 1        |
|               | 21             | E           | 619802G     |           | SKINTOP CABLE GLAND M16                                      |                  | 5        |
| <b>04-720</b> | <b>908G772</b> |             |             |           | <b>Face/Sideplate Electrical (Atmos) - I/O &amp; Leadthr</b> | <b>424G139</b>   | <b>1</b> |
|               | 2              | E           | 651002G     |           | ET200AL 8DIx24V, 8xM8 Module IP67                            |                  | 5        |
|               | 71             | E           | 616598      |           | FC LANE 25 PIN SOCKET CODE 0                                 |                  | 1        |
|               | 72             | E           | 616599      |           | FC LANE 25 PIN PLUG CODE 0                                   |                  | 1        |
|               | 73             | E           | 616600      |           | FC LANE 25 PIN BULKHEAD CODE 0                               |                  | 1        |
|               | 74             | E           | 616601      |           | FC Lane Cable Outlet Connector                               |                  | 2        |
|               | 75             | E           | 616602      |           | FC Lane Outlet Connector Gasket                              |                  | 2        |
|               | 81             | E           | 616603      |           | FC LANE 25 PIN SOCKET CODE 1                                 |                  | 0        |



## V18G019 - Bill of Materials



| Section       | Bubble Number  | Mech / Elec | Part Number | Parameter  | Part Description                                          | Assembly Drawing | Qty      |
|---------------|----------------|-------------|-------------|------------|-----------------------------------------------------------|------------------|----------|
|               | 82             | E           | 616604      |            | FC LANE 25 PIN PLUG CODE 1                                |                  | 1        |
|               | 83             | E           | 616605      |            | FC LANE 25 PIN BULKHEAD CODE 1                            |                  | 1        |
|               | 84             | E           | 616601      |            | FC Lane Cable Outlet Connector                            |                  | 2        |
|               | 85             | E           | 616602      |            | FC Lane Outlet Connector Gasket                           |                  | 2        |
|               | 86             | E           | 245G188     | 2.5M.TYPE1 | CABLE ASSEMBLY FC-LANE / ET200AL - 8 CHANNEL              |                  | 1        |
|               | 91             | E           | 616603      |            | FC LANE 25 PIN SOCKET CODE 1                              |                  | 0        |
|               | 92             | E           | 616604      |            | FC LANE 25 PIN PLUG CODE 1                                |                  | 1        |
|               | 93             | E           | 616605      |            | FC LANE 25 PIN BULKHEAD CODE 1                            |                  | 1        |
|               | 94             | E           | 616601      |            | FC Lane Cable Outlet Connector                            |                  | 2        |
|               | 95             | E           | 616602      |            | FC Lane Outlet Connector Gasket                           |                  | 2        |
|               | 96             | E           | 245G189     | 1.0M.TYPE1 | CABLE ASSEMBLY FC-LANE / ET200AL - 16 CHANNEL             |                  | 1        |
|               | 101            | E           | 616598      |            | FC LANE 25 PIN SOCKET CODE 0                              |                  | 0        |
|               | 102            | E           | 616599      |            | FC LANE 25 PIN PLUG CODE 0                                |                  | 1        |
|               | 103            | E           | 616600      |            | FC LANE 25 PIN BULKHEAD CODE 0                            |                  | 1        |
|               | 104            | E           | 616601      |            | FC Lane Cable Outlet Connector                            |                  | 2        |
|               | 105            | E           | 616602      |            | FC Lane Outlet Connector Gasket                           |                  | 2        |
|               | 106            | E           | 245G189     | 1.0M.TYPE0 | CABLE ASSEMBLY FC-LANE / ET200AL - 16 CHANNEL             |                  | 1        |
|               | 121            | E           | 652176G     |            | ET200AL Bus Connection Cable 2xM8, 0.3M                   |                  | 4        |
|               | 122            | E           | 652195G     |            | ET200AL Bus Connection Cable 2xM8, 3.0m                   |                  | 1        |
|               | 123            | E           | 652196G     |            | ET200AL Bus Connection Cable 2xM8, 2.0m                   |                  | 1        |
|               | 131            | E           | 652180G     |            | ET200AL Power Connection Cable 2xM8, 0.3M                 |                  | 4        |
|               | 132            | E           | 652197G     |            | ET200AL Power Connection Cable 2xM8, 3.0m                 |                  | 1        |
|               | 133            | E           | 652198G     |            | ET200AL Power Connection Cable 2xM8, 2.0m                 |                  | 1        |
|               | 134            | E           | 652184G     |            | ET200AL Power Conn. Cable 1xM8, 10.0M Open Ended          |                  | 1        |
|               | 140            | M           | 421769      |            | SOCKET CAP SCREW M4x30 GRADE 12.9                         |                  | 10       |
|               | 141            | M           | 421049      |            | SOCKET CAP SCREW M8 x 20 GRADE 12.9                       |                  | 2        |
|               | 142            | M           | 421054      |            | PLAIN WASHER M8                                           |                  | 2        |
|               | 143            | M           | 422G092     |            | Remote I/O Bracket                                        |                  | 1        |
|               | 144            | M           | 421245      |            | SOCKET CAP SCREW M5 x 10 GRADE 12.9                       |                  | 1        |
| <b>04-721</b> | <b>908G771</b> |             |             |            | <b>Faceplate Chuck Proxy Electrical (Sidelay/CraneEx)</b> | <b>422G569</b>   | <b>1</b> |
|               | 1              | E           | 651004G     |            | ET200AL PROFINET HEADER MODULE                            |                  | 1        |
|               | 2              | E           | 6131433G    |            | Profinet Connection Cable, M12 To RJ45 - 10M              |                  | 1        |
|               | 11             | E           | 651002G     |            | ET200AL 8Dlx24V, 8xM8 Module IP67                         |                  | 1        |
|               | 12             | E           | 6131447G    |            | Conn. Cable M8 Male - M12 Female, Straight, 3m            |                  | 1        |
|               | 13             | E           | 6131543G    |            | M8 Male - M12 Female Special Connection - 2M              |                  | 3        |
|               | 21             | E           | 651002G     |            | ET200AL 8Dlx24V, 8xM8 Module IP67                         |                  | 1        |
|               | 22             | E           | 6131434G    |            | Conn. Cable M8 Male - M12 Female, Straight, 1.5m          |                  | 1        |
|               | 23             | E           | 6131543G    |            | M8 Male - M12 Female Special Connection - 2M              |                  | 3        |
|               | 41             | E           | 652176G     |            | ET200AL Bus Connection Cable 2xM8, 0.3M                   |                  | 1        |
|               | 42             | E           | 652177G     |            | ET200AL Bus Connection Cable 2xM8, 1.0M                   |                  | 0        |
|               | 43             | E           | 652176G     |            | ET200AL Bus Connection Cable 2xM8, 0.3M                   |                  | 1        |
|               | 44             | E           | 652179G     |            | ET200AL Bus Connection Cable 2xM8, 10M                    |                  | 0        |
|               | 51             | E           | 652180G     |            | ET200AL Power Connection Cable 2xM8, 0.3M                 |                  | 1        |
|               | 52             | E           | 652181G     |            | ET200AL Power Connection Cable 2xM8, 1.0M                 |                  | 0        |
|               | 53             | E           | 652182G     |            | ET200AL Power Connection Cable 2xM8, 5.0M                 |                  | 0        |
|               | 54             | E           | 652183G     |            | ET200AL Power Connection Cable 2xM8, 10.0M                |                  | 1        |
|               | 60             | M           | 421769      |            | SOCKET CAP SCREW M4x30 GRADE 12.9                         |                  | 6        |
| <b>04-722</b> | <b>908G701</b> |             |             |            | <b>Faceplate Pneumatic &amp; Solenoids I/O Electrical</b> | <b>424G112</b>   | <b>1</b> |
|               | 1              | E           | 651012G     |            | SIMATIC ET 200AL, DQ 8X24VDC/2A, 8XM12, IP67              |                  | 1        |
|               | 2              | E           | 6131473G    |            | M12 Male - M8 Female 3 pin Cable 0.5m                     |                  | 4        |
|               | 3              | E           | 6131411G    |            | FORM 'A' SOLENOID VALVE CONNECTION CABLE 4M, M12          |                  | 2        |
|               | 11             | E           | 651012G     |            | SIMATIC ET 200AL, DQ 8X24VDC/2A, 8XM12, IP67              |                  | 1        |
|               | 12             | E           | 6131473G    |            | M12 Male - M8 Female 3 pin Cable 0.5m                     |                  | 2        |
|               | 13             | E           | 6131437G    |            | FORM 'A' SOLENOID VALVE CONNECTION CABLE 3M, M12          |                  | 2        |
|               | 21             | E           | 652195G     |            | ET200AL Bus Connection Cable 2xM8, 3.0m                   |                  | 1        |
|               | 22             | E           | 652176G     |            | ET200AL Bus Connection Cable 2xM8, 0.3M                   |                  | 1        |
|               | 31             | E           | 652184G     |            | ET200AL Power Conn. Cable 1xM8, 10.0M Open Ended          |                  | 1        |
|               | 32             | E           | 652180G     |            | ET200AL Power Connection Cable 2xM8, 0.3M                 |                  | 1        |
|               | 40             | M           | 421770      |            | SOCKET CAP SCREW M4x35 GRADE 12.9                         |                  | 4        |
|               | 41             | M           | 421049      |            | SOCKET CAP SCREW M8 x 20 GRADE 12.9                       |                  | 2        |
|               | 42             | M           | 421054      |            | PLAIN WASHER M8                                           |                  | 2        |
|               | 50             | M           | 427G600     |            | Mounting Plate                                            |                  | 1        |
| <b>04-723</b> | <b>908G934</b> |             |             |            | <b>K5000 MECH CLOSED SAFETY SWITCH</b>                    | <b>422G931</b>   | <b>1</b> |
|               | 1              | E           | 698472G     |            | 8PIN MICRO CONNECT CABLE 10M                              |                  | 1        |
|               | 2              | E           | 698471G     |            | SENSOR MAGNETICALLY ACT SWITCH CAT 4 8PIN MICRO           |                  | 1        |
|               | 10             | M           | 421147      |            | SOCKET CAP SCREW M4 x 12 GRADE 12.9                       |                  | 4        |
|               | 11             | M           | 421227      |            | Washer Spring M4                                          |                  | 4        |
|               | 12             | M           | 421194      |            | Washer Plain M4                                           |                  | 4        |



# V18G019 - Bill of Materials



| Section | Bubble Number | Mech / Elec | Part Number | Parameter | Part Description                                   | Assembly Drawing | Qty |
|---------|---------------|-------------|-------------|-----------|----------------------------------------------------|------------------|-----|
| 04-730  | 908G702       |             |             |           | Faceplate Standard Leadthrough(s)                  | 422G568          | 1   |
|         | 1             | E           | 616598      |           | FC LANE 25 PIN SOCKET CODE 0                       |                  | 1   |
|         | 2             | E           | 616599      |           | FC LANE 25 PIN PLUG CODE 0                         |                  | 1   |
|         | 3             | E           | 616600      |           | FC LANE 25 PIN BULKHEAD CODE 0                     |                  | 1   |
|         | 4             | E           | 616601      |           | FC Lane Cable Outlet Connector                     |                  | 2   |
|         | 5             | E           | 616602      |           | FC Lane Outlet Connector Gasket                    |                  | 2   |
|         | 11            | E           | 616603      |           | FC LANE 25 PIN SOCKET CODE 1                       |                  | 1   |
|         | 12            | E           | 616604      |           | FC LANE 25 PIN PLUG CODE 1                         |                  | 1   |
|         | 13            | E           | 616605      |           | FC LANE 25 PIN BULKHEAD CODE 1                     |                  | 1   |
|         | 14            | E           | 616601      |           | FC Lane Cable Outlet Connector                     |                  | 2   |
|         | 15            | E           | 616602      |           | FC Lane Outlet Connector Gasket                    |                  | 2   |
|         | 21            | E           | 616624      |           | FC LANE 6 PIN SOCKET CODE 1                        |                  | 1   |
|         | 22            | E           | 616625      |           | FC LANE 6 PIN PLUG CODE 1                          |                  | 1   |
|         | 23            | E           | 616626      |           | FC LANE 6 PIN BULKHEAD CODE 1                      |                  | 1   |
|         | 24            | E           | 616601      |           | FC Lane Cable Outlet Connector                     |                  | 2   |
|         | 25            | E           | 616602      |           | FC Lane Outlet Connector Gasket                    |                  | 2   |
|         | 31            | E           | 616627      |           | FC LANE 6 PIN SOCKET CODE 2                        |                  | 1   |
|         | 32            | E           | 616628      |           | FC LANE 6 PIN PLUG CODE 2                          |                  | 1   |
|         | 33            | E           | 616629      |           | FC LANE 6 PIN BULKHEAD CODE 2                      |                  | 1   |
|         | 34            | E           | 616601      |           | FC Lane Cable Outlet Connector                     |                  | 2   |
|         | 35            | E           | 616602      |           | FC Lane Outlet Connector Gasket                    |                  | 2   |
| 04-731  | 908G812       |             |             |           | Standard bowed spreader - electrical - I/O, Leadth | 422G567          | 1   |
|         | 11            | E           | 651001G     |           | ET200AL 4 Channel Analogue Input U/I/RTD, 4xM12    |                  | 1   |
|         | 12            | E           | 6131619G    |           | M12 Male Rewirable Connector 4 Pin                 |                  | 3   |
|         | 13            | E           | 6161296G    |           | M12 4 Pin Straight Socket 4-6mm Cable Entry        |                  | 1   |
|         | 14            | M           | 4232361G    |           | DC/DC Converter, 24V/10V, 25mA, M12                |                  | 1   |
|         | 21            | E           | 652176G     |           | ET200AL Bus Connection Cable 2xM8, 0.3M            |                  | 1   |
|         | 23            | E           | 652180G     |           | ET200AL Power Connection Cable 2xM8, 0.3M          |                  | 1   |
|         | 51            | E           | 616630      |           | FC LANE 6 PIN SOCKET CODE 3                        |                  | 1   |
|         | 52            | E           | 616631      |           | FC LANE 6 PIN PLUG CODE 3                          |                  | 1   |
|         | 53            | E           | 616632      |           | FC LANE 6 PIN BULKHEAD CODE 3                      |                  | 1   |
|         | 54            | E           | 616601      |           | FC Lane Cable Outlet Connector                     |                  | 1   |
|         | 55            | E           | 616602      |           | FC Lane Outlet Connector Gasket                    |                  | 2   |
|         | 61            | E           | 616633      |           | FC LANE 6 PIN SOCKET CODE 4                        |                  | 1   |
|         | 62            | E           | 616634      |           | FC LANE 6 PIN PLUG CODE 4                          |                  | 1   |
|         | 63            | E           | 616635      |           | FC LANE 6 PIN BULKHEAD CODE 4                      |                  | 1   |
|         | 64            | E           | 616601      |           | FC Lane Cable Outlet Connector                     |                  | 2   |
|         | 65            | E           | 616602      |           | FC Lane Outlet Connector Gasket                    |                  | 2   |
|         | 71            | E           | 616615      |           | FC LANE 25 PIN SOCKET CODE 5                       |                  | 1   |
|         | 72            | E           | 616642      |           | FC LANE 25 PIN PLUG CODE 5                         |                  | 1   |
|         | 73            | E           | 616617      |           | FC LANE 25 PIN BULKHEAD CODE 5                     |                  | 1   |
|         | 74            | E           | 616601      |           | FC Lane Cable Outlet Connector                     |                  | 2   |
|         | 75            | E           | 616602      |           | FC Lane Outlet Connector Gasket                    |                  | 2   |
|         | 80            | M           | 421769      |           | SOCKET CAP SCREW M4x30 GRADE 12.9                  |                  | 2   |
| 04-732  | 908G704       |             |             |           | K5000 - Web Light Leadthrough(s)                   | 422G626          | 1   |
|         | 1             | E           | 616619      |           | FC LANE 6 PIN SOCKET CODE 0                        |                  | 1   |
|         | 2             | E           | 616620      |           | FC LANE 6 PIN PLUG CODE 0                          |                  | 1   |
|         | 3             | E           | 616623      |           | FC LANE 6 PIN BULKHEAD CODE 0                      |                  | 1   |
|         | 4             | E           | 616601      |           | FC Lane Cable Outlet Connector                     |                  | 2   |
|         | 5             | E           | 616602      |           | FC Lane Outlet Connector Gasket                    |                  | 2   |
| 04-733  | 908G807       |             |             |           | Faceplate Standard Electrical Leadthrough Plates   | 422G571          | 1   |
|         | 2             | M           | 418G295     |           | Electrical Leadthrough Flange (3 x FC Lane)        |                  | 8   |
|         | 3             | M           | 422G464     |           | ELECTRICAL LEADTHROUGH BLANKING FLANGE             |                  | 2   |
|         | 4             | M           | 239373      |           | Seal O Ring RM1245-30 Nitrile                      |                  | 10  |
|         | 5             | M           | 421049      |           | SOCKET CAP SCREW M8 x 20 GRADE 12.9                |                  | 30  |
|         | 6             | M           | 421054      |           | PLAIN WASHER M8                                    |                  | 30  |
| 04-735  | 908G695       |             |             |           | Monitor Beam Leadthroughs                          | 422G617          | 1   |
|         | 1             | E           | 616598      |           | FC LANE 25 PIN SOCKET CODE 0                       |                  | 1   |
|         | 2             | E           | 616599      |           | FC LANE 25 PIN PLUG CODE 0                         |                  | 1   |
|         | 3             | E           | 616600      |           | FC LANE 25 PIN BULKHEAD CODE 0                     |                  | 1   |
|         | 4             | E           | 616601      |           | FC Lane Cable Outlet Connector                     |                  | 2   |
|         | 5             | E           | 616602      |           | FC Lane Outlet Connector Gasket                    |                  | 2   |
|         | 11            | E           | 616606      |           | FC LANE 25 PIN SOCKET CODE 2                       |                  | 0   |
|         | 12            | E           | 616607      |           | FC LANE 25 PIN PLUG CODE 2                         |                  | 0   |
|         | 13            | E           | 616608      |           | FC LANE 25 PIN BULKHEAD CODE 2                     |                  | 0   |
|         | 14            | E           | 616601      |           | FC Lane Cable Outlet Connector                     |                  | 0   |
|         | 15            | E           | 616602      |           | FC Lane Outlet Connector Gasket                    |                  | 0   |
| 04-736  | 909G565       |             |             |           | K5-Expert Lighting Junction Box                    | 422G571          | 1   |

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| Section | Bubble Number | Mech / Elec | Part Number | Parameter | Part Description                                           | Assembly Drawing | Qty      |
|---------|---------------|-------------|-------------|-----------|------------------------------------------------------------|------------------|----------|
|         | 1             | M           | 245G174     |           | T06_7 Cubicle - Web Lighting Cubicle                       |                  | 1        |
|         | 2             | E           | 600049G     |           | Hoffman T06_7 Cubicle - 400x200x120mm                      |                  | 1        |
| 04-738  | 903G140       |             |             |           | <b>OD Monitor Buffer Box</b>                               | <b>422G617</b>   | <b>1</b> |
|         | 1             | E           | 673132      |           | Optical Buffer Box Assembly Kit                            |                  | 1        |
|         | 2             | E           | 673135      |           | Beam Interface (Buffer) Board Assembly                     |                  | 1        |
|         | 3             | E           | 673177G     |           | OD Detector Board Assembly - Rev 5                         |                  | 1        |
|         | 4             | E           | 673134      |           | Modulator Board Assembly                                   |                  | 1        |
| 04-740  | 908G705       |             |             |           | <b>Adjustb Rewind Layarm Spreader Elect - Leadthr</b>      | <b>422G621</b>   | <b>1</b> |
|         | 1             | E           | 616630      |           | FC LANE 6 PIN SOCKET CODE 3                                |                  | 1        |
|         | 2             | E           | 616631      |           | FC LANE 6 PIN PLUG CODE 3                                  |                  | 1        |
|         | 3             | E           | 616632      |           | FC LANE 6 PIN BULKHEAD CODE 3                              |                  | 1        |
|         | 4             | E           | 616601      |           | FC Lane Cable Outlet Connector                             |                  | 2        |
|         | 5             | E           | 616602      |           | FC Lane Outlet Connector Gasket                            |                  | 2        |
|         | 11            | E           | 616612      |           | FC LANE 25 PIN SOCKET CODE 4                               |                  | 1        |
|         | 12            | E           | 616613      |           | FC LANE 25 PIN PLUG CODE 4                                 |                  | 1        |
|         | 13            | E           | 616614      |           | FC LANE 25 PIN BULKHEAD CODE 4                             |                  | 1        |
|         | 14            | E           | 616601      |           | FC Lane Cable Outlet Connector                             |                  | 2        |
|         | 15            | E           | 616602      |           | FC Lane Outlet Connector Gasket                            |                  | 2        |
|         | 16            | E           | 6131619G    |           | M12 Male Rewirable Connector 4 Pin                         |                  | 2        |
|         | 17            | E           | 6161296G    |           | M12 4 Pin Straight Socket 4-6mm Cable Entry                |                  | 1        |
|         | 18            | M           | 4232361G    |           | DC/DC Converter, 24V/10V, 25mA, M12                        |                  | 1        |
|         | 19            | M           | 418G295     |           | Electrical Leadthrough Flange (3 x FC Lane)                |                  | 1        |
| 04-741  | 908G235       |             |             |           | <b>Edge Guide and/or oscillation - Leadthr &amp; cable</b> | <b>422G601</b>   | <b>1</b> |
|         | 21            | E           | 651001G     |           | ET200AL 4 Channel Analogue Input U//RTD, 4xM12             |                  | 1        |
|         | 22            | M           | 421769      |           | SOCKET CAP SCREW M4x30 GRADE 12.9                          |                  | 2        |
|         | 31            | E           | 6161319G    |           | M12 4 Pin Straight Male 8-10mm Cable Entry                 |                  | 2        |
|         | 41            | E           | 652176G     |           | ET200AL Bus Connection Cable 2xM8, 0.3M                    |                  | 1        |
|         | 51            | E           | 652180G     |           | ET200AL Power Connection Cable 2xM8, 0.3M                  |                  | 1        |
|         | 61            | M           | 4232361G    |           | DC/DC Converter, 24V/10V, 25mA, M12                        |                  | 0        |
|         | 62            | E           | 6131619G    |           | M12 Male Rewirable Connector Male 4 Pin                    |                  | 0        |
|         | 63            | E           | 6161296G    |           | M12 4 Pin Straight Socket 4-6mm Cable Entry                |                  | 0        |
| 04-742  | 908G814       |             |             |           | <b>Web threader - Electrical Leadthru &amp; Cables</b>     | <b>422G622</b>   | <b>1</b> |
|         | 1             | E           | 616619      |           | FC LANE 6 PIN SOCKET CODE 0                                |                  | 1        |
|         | 2             | E           | 616620      |           | FC LANE 6 PIN PLUG CODE 0                                  |                  | 1        |
|         | 3             | E           | 616623      |           | FC LANE 6 PIN BULKHEAD CODE 0                              |                  | 1        |
|         | 4             | E           | 616601      |           | FC Lane Cable Outlet Connector                             |                  | 2        |
|         | 5             | E           | 616602      |           | FC Lane Outlet Connector Gasket                            |                  | 2        |
|         | 11            | E           | 636001G     |           | MALE M12 4PIN PVC/PUR CABLE 3M OPEN END                    |                  | 1        |
| 04-743  | 908G806       |             |             |           | <b>Plasma Treater Leadth &amp; Electrical (Faceplate)</b>  | <b>424G151</b>   | <b>1</b> |
|         | 2             | E           | 616612      |           | FC LANE 25 PIN SOCKET CODE 4                               |                  | 0        |
|         | 3             | E           | 616613      |           | FC LANE 25 PIN PLUG CODE 4                                 |                  | 0        |
|         | 4             | E           | 616614      |           | FC LANE 25 PIN BULKHEAD CODE 4                             |                  | 0        |
|         | 5             | E           | 616601      |           | FC Lane Cable Outlet Connector                             |                  | 0        |
|         | 6             | E           | 616602      |           | FC Lane Outlet Connector Gasket                            |                  | 0        |
|         | 11            | E           | 616817      |           | HAN HARTING PG11 CABLE SEAL 6.5/12                         |                  | 0        |
|         | 12            | E           | 616644      |           | HAN HARTING HA4 SOCKET (FEMALE)                            |                  | 0        |
|         | 13            | E           | 616645      |           | HAN HARTING HA3 HOOD TOP ENTRY                             |                  | 0        |
|         | 14            | E           | 616643      |           | HAN HARTING HA4 PLUG (MALE)                                |                  | 0        |
|         | 15            | E           | 6161151G    |           | HAN HARTING HA3 COUPLING HOOD SURFACE MOUNT                |                  | 0        |
|         | 16            | E           | 6161204G    |           | HARTING COVER, SIZE 3A, FEMALE INSERT                      |                  | 0        |
|         | 21            | E           | 651001G     |           | ET200AL 4 Channel Analogue Input U//RTD, 4xM12             |                  | 1        |
|         | 22            | E           | 245G187     | 3.0M      | CABLE ASSEMBLY BARATRON / ET200AL                          |                  | 1        |
|         | 31            | E           | 651012G     |           | SIMATIC ET 200AL, DQ 8X24VDC/2A, 8XM12, IP67               |                  | 1        |
|         | 32            | E           | 6131437G    |           | FORM 'A' SOLENOID VALVE CONNECTION CABLE 3M, M12           |                  | 3        |
|         | 36            | E           | 651012G     |           | SIMATIC ET 200AL, DQ 8X24VDC/2A, 8XM12, IP67               |                  | 0        |
|         | 37            | E           | 6131437G    |           | FORM 'A' SOLENOID VALVE CONNECTION CABLE 3M, M12           |                  | 0        |
|         | 41            | E           | 6131441G    |           | 4 Pole M12 Straight Male - M12 90 Deg Female 5m            |                  | 0        |
|         | 52            | E           | 6131473G    |           | M12 Male - M8 Female 3 pin Cable 0.5m                      |                  | 1        |
|         | 56            | E           | 6131473G    |           | M12 Male - M8 Female 3 pin Cable 0.5m                      |                  | 0        |
|         | 61            | E           | 652176G     |           | ET200AL Bus Connection Cable 2xM8, 0.3M                    |                  | 2        |
|         | 71            | E           | 652180G     |           | ET200AL Power Connection Cable 2xM8, 0.3M                  |                  | 2        |
|         | 100           | M           | 421769      |           | SOCKET CAP SCREW M4x30 GRADE 12.9                          |                  | 4        |
|         | 101           | M           | 421770      |           | SOCKET CAP SCREW M4x35 GRADE 12.9                          |                  | 4        |
| 04-744  | 908G815       |             |             |           | <b>Faceplate - Rewind Gas Wedge</b>                        | <b>422G627</b>   | <b>1</b> |
|         | 1             | E           | 6131437G    |           | FORM 'A' SOLENOID VALVE CONNECTION CABLE 3M, M12           |                  | 2        |
| 04-745  | 908G887       |             |             |           | <b>LEADTHROUGH BLANK</b>                                   | <b>422G627</b>   | <b>6</b> |
|         | 1             | M           | 301G086     |           | LEADTHROUGH BLANK                                          |                  | 1        |
|         | 2             | M           | 421101      |           | Nut Full Hex M24                                           |                  | 1        |

## V18G019 - Bill of Materials



| Section       | Bubble Number  | Mech / Elec | Part Number | Parameter | Part Description                                   | Assembly Drawing | Qty      |
|---------------|----------------|-------------|-------------|-----------|----------------------------------------------------|------------------|----------|
|               | 3              | M           | 421069      |           | PLAIN WASHER M24                                   |                  | 1        |
|               | 4              | M           | 239671G     |           | Seal O Ring RM0355-30                              |                  | 1        |
| <b>05</b>     |                |             |             |           | <b>Mech Internal</b>                               |                  |          |
| <b>05-010</b> | <b>910G547</b> |             |             |           | <b>Splined Chuck With Sidelay (KTR Coupling)</b>   | <b>430G562</b>   | <b>1</b> |
|               | B001           | M           | 235266G     |           | Compression Spring LHC 162N OBS                    |                  | 2        |
|               | B003           | M           | 422008      |           | Toggle Clamp DE-STA-CO Model 341                   |                  | 2        |
|               | B004           | E           | 675226G     |           | Proximity Switch 12mm IFS205                       |                  | 6        |
|               | B005           | M           | 203355G     |           | PLAIN GLACIER BEARING MB2010DU                     |                  | 2        |
|               | B006           | M           | 203233      |           | BUSH GLACIER MB2230DU                              |                  | 2        |
|               | B007           | M           | 242005      |           | LOCKNUT KM15                                       |                  | 1        |
|               | B008           | M           | 242006      |           | LOCKWASHER MB15                                    |                  | 1        |
|               | B010           | M           | 239135      |           | O' RING RM 0176-24                                 |                  | 2        |
|               | B011           | M           | 239661G     |           | Seal O Ring RM0345-30                              |                  | 2        |
|               | B013           | M           | 239255      |           | Seal O Ring RM0795-30 Nitrile                      |                  | 1        |
|               | B015           | M           | 239351      |           | Seal O Ring RM1645-30 Nitrile                      |                  | 1        |
|               | B016           | M           | 239340      |           | Seal O Ring RM1745-30 Nitrile                      |                  | 1        |
|               | B017           | M           | 239386      |           | Seal O Ring RM3593-57 Nitrile                      |                  | 1        |
|               | B018           | M           | 4231561G    |           | 1/2" BSP Plug C/w O-ring seal                      |                  | 2        |
|               | B020           | M           | 2061182G    |           | Linear Guide Block                                 |                  | 4        |
|               | B021           | M           | 2061250G    |           | Linear Rail Hiwin RGR35-R-200-H                    |                  | 4        |
|               | B022           | M           | 239380      |           | SEAL SIMRIT ROD NI300 20-30-10                     |                  | 2        |
|               | B023           | E           | 658001G     |           | Linear Transducer (Balluff)BIW1-A310-M0075-P1-S115 |                  | 1        |
|               | B024           | M           | 2442367G    |           | Precision Screwjack, 25kN, 60mm stroke             |                  | 1        |
|               | B027           | M           | 423155      |           | Nipple 45 DEG Grease 1/8"BSP GN1/8BSP45            |                  | 1        |
|               | B029           | M           | 239381      |           | Simmerring Wiper Seal AUAS 20-30-10                |                  | 2        |
|               | B030           | M           | 232455G     |           | KTR Radex Coupling NN Size 70 (65-70 BORE H7)      |                  | 1        |
|               | B032           | M           | 664514G     |           | SIEMENS 1PH8186-1DB03-2BD1 - 500RPM - 26.5Kw - 65A |                  | 1        |
|               | B034           | M           | 240020G     |           | Spring Plunger WDS 606-5712                        |                  | 1        |
|               | B036           | M           | 200M570     |           | BEARING 61828-2RS1                                 |                  | 1        |
|               | B037           | M           | 206T094     |           | KB 2045                                            |                  | 1        |
|               | B038           | M           | 204M090     |           | FLANGE BUSH GLACIER FMB 3016DU                     |                  | 2        |
|               | B042           | M           | 2061225G    |           | THK-LBF85-UUCL+350L                                |                  | 1        |
|               | B043           | M           | 232474G     |           | Gearx 'SD' 20 Hub                                  |                  | 1        |
|               | B044           | M           | 232487G     |           | Gearx 'SD' 20 sleeve                               |                  | 1        |
|               | B045           | M           | 232476G     |           | KTR Shiftable Linkage Size 3                       |                  | 1        |
|               | B046           | M           | 234010G     |           | External Circlip D1400-1400 (140mm Dia)            |                  | 1        |
|               | B047           | M           | 239926G     |           | Seal Simmerring BABSL90-110-7-5-72 Viton           |                  | 2        |
|               | B048           | M           | 234050      |           | CIRCLIP EXTERNAL 1400-0800                         |                  | 1        |
|               | B049           | M           | 200632      |           | CYLINDRICAL ROLLER BEARING N216 ECP                |                  | 1        |
|               | B050           | M           | 239198      |           | Seal O Ring RM0745-30 Nitrile                      |                  | 2        |
|               | B051           | M           | 239373      |           | Seal O Ring RM1245-30 Nitrile                      |                  | 1        |
|               | B052           | M           | 239368      |           | O' RING RM 1495-30                                 |                  | 1        |
|               | B053           | M           | 200704      |           | BEARING 3216A                                      |                  | 1        |
|               | B054           | M           | 423036      |           | Adaptor Male 8mm-1/4"BSPT 36-0530-11               |                  | 3        |
|               | B055           | M           | 423039      |           | Tubing Sleeve 8 OD                                 |                  | 5        |
|               | B056           | M           | 423038      |           | Tubing Nut 8 OD                                    |                  | 5        |
|               | B057           | M           | 201006G     |           | Bearing Inner Race IR80 x 90 x 25 - EGS            |                  | 1        |
|               | B058           | M           | 201007G     |           | Bearing Inner Race IR80 x 90 x 54 - EGS            |                  | 1        |
|               | B059           | M           | 423044      |           | Elbow Stem 8 OD                                    |                  | 2        |
|               | B060           | M           | 425003      |           | Oil Reservoir 1/4"BSP (Ref OR 7730)                |                  | 1        |
|               | B061           | M           | 423046      |           | Tube Half Hard Copper 8mm O/D CS7008003            |                  | 1        |
|               | M001           | M           | 424G339     |           | Bearing Support IB-Splined Chuck                   |                  | 1        |
|               | M002           | M           | 424G220     |           | Bearing Clamp IB-Splined Chuck                     |                  | 1        |
|               | M003           | M           | 424G420     |           | Spring Pin                                         |                  | 2        |
|               | M004           | M           | 427G719     |           | Splined Chuck Mounting Block                       |                  | 2        |
|               | M005           | M           | 428G777     |           | Bracket                                            |                  | 1        |
|               | M006           | M           | 425G045     |           | Pivot Pin                                          |                  | 2        |
|               | M007           | M           | 426G250     |           | SideShift Drive Block                              |                  | 1        |
|               | M008           | M           | 424G437     |           | Proximity Switch Bracket                           |                  | 2        |
|               | M009           | M           | 421G534     |           | Stop Block                                         |                  | 3        |
|               | M010           | M           | 427G214     |           | Stop Block                                         |                  | 1        |
|               | M014           | M           | 426G122     |           | Latch                                              |                  | 2        |
|               | M015           | M           | 418G653     |           | Spacer - Clamp                                     |                  | 2        |
|               | M017           | M           | 421G446     |           | Ferrofluidic Seal Housing                          |                  | 1        |
|               | M021           | M           | 424G407     |           | Bearing Block OB                                   |                  | 1        |
|               | M024           | M           | 428G949     |           | Chuck Support Bracket (Non Drive Side)             |                  | 1        |
|               | M025           | M           | 424G222     |           | Splined Chuck Bearing Clamp OB                     |                  | 1        |
|               | M026           | M           | 424G100     |           | Chuck Guard                                        |                  | 2        |

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| Section | Bubble Number | Mech / Elec | Part Number | Parameter | Part Description                        | Assembly Drawing | Qty |
|---------|---------------|-------------|-------------|-----------|-----------------------------------------|------------------|-----|
|         | M027          | M           | 424G106     |           | Nylon Wear Plate                        |                  | 2   |
|         | M030          | M           | 427G707     |           | Push Rod                                |                  | 1   |
|         | M031          | M           | 422G706     |           | Push Rod Housing                        |                  | 1   |
|         | M032          | M           | 403G630     |           | Seal Ring                               |                  | 1   |
|         | M033          | M           | 403G364     |           | Seal Housing                            |                  | 1   |
|         | M034          | M           | 417G144     |           | Bearing Retainer                        |                  | 1   |
|         | M035          | M           | 420G837     |           | Linear Transducer Bracket               |                  | 1   |
|         | M036          | M           | 420G838     |           | Linear Potentiometer Square Pin         |                  | 1   |
|         | M037          | M           | 430G612     |           | Splined Chuck Bell Housing              |                  | 1   |
|         | M039          | M           | 421G456     |           | Ball Spline Shaft                       |                  | 1   |
|         | M047          | M           | 424G252     |           | Ballspline Adaptor Housing              |                  | 1   |
|         | M056          | M           | 422G699     |           | Proximity Bracket                       |                  | 1   |
|         | M060          | M           | 429G106     |           | Proxi Switch bracket                    |                  | 1   |
|         | M061          | M           | 418G249     | IT24      | Key                                     |                  | 1   |
|         | M064          | M           | 418G249     | IT22      | Key                                     |                  | 1   |
|         | M066          | M           | 424G111     | IT1       | Chuck Shield Plate                      |                  | 1   |
|         | M067          | M           | 424G111     | IT2       | Chuck Shield Plate                      |                  | 1   |
|         | M068          | M           | 427G225     |           | Stop Block                              |                  | 1   |
|         | M071          | M           | 424G967     |           | KTR Gear Retaining Washer               |                  | 1   |
|         | M072          | M           | 425G033     |           | SPLINED CHUCK GEAR                      |                  | 1   |
|         | M073          | M           | 426G038     |           | KTR Shiftable Linkage Mods              |                  | 1   |
|         | M074          | M           | 427G550     |           | Proxi switch bracket                    |                  | 1   |
|         | M076          | M           | 427G551     |           | Proxi switch flag                       |                  | 1   |
|         | M077          | M           | 427G650     |           | Stop Block                              |                  | 1   |
|         | M083          | M           | 428G959     |           | YOKE                                    |                  | 1   |
|         | M084          | M           | 428G967     |           | Catch Bracket                           |                  | 1   |
|         | M085          | M           | 428G971     |           | Bearing Cap                             |                  | 1   |
|         | M086          | M           | 428G966     |           | CAM PLATE                               |                  | 2   |
|         | M088          | M           | 428G970     |           | CAM                                     |                  | 2   |
|         | M089          | M           | 430G388     |           | Seal Shaft - Spline Chuck               |                  | 1   |
|         | M090          | M           | 430G389     |           | Outer Bearing Housing                   |                  | 1   |
|         | M091          | M           | 430G395     |           | Cartridge Seal Cap Rear                 |                  | 1   |
|         | M092          | M           | 430G405     |           | Shaft Seal Spacer                       |                  | 1   |
|         | M093          | M           | 430G427     |           | Bearing Spacer                          |                  | 1   |
|         | M094          | M           | 430G428     |           | Bearing Spacer                          |                  | 1   |
|         | M095          | M           | 430G429     |           | Shaft Sleeve                            |                  | 1   |
|         | M096          | M           | 430G431     |           | Retaining Ring                          |                  | 1   |
|         | M097          | M           | 430G592     |           | Oil Bottle Bracket                      |                  | 1   |
|         | M098          | M           | 430G654     |           | Lubricant Label - Chuck - Oil AWS68     |                  | 1   |
|         | S001          | M           | 421658      |           | SOCKET CAP SCREW M4 x 25 GRADE 12.9     |                  | 4   |
|         | S002          | M           | 421019      |           | Socket Head Capscrew M6 x 20 GRADE 12.9 |                  | 20  |
|         | S003          | M           | 421225      |           | SOCKET CAP SCREW M6 x 60 GRADE 12.9     |                  | 3   |
|         | S004          | M           | 421049      |           | SOCKET CAP SCREW M8 x 20 GRADE 12.9     |                  | 12  |
|         | S005          | M           | 421138      |           | SOCKET CAP SCREW M8 x 25 GRADE 12.9     |                  | 16  |
|         | S006          | M           | 421237      |           | SOCKET CAP SCREW M8 x 35 GRADE 12.9     |                  | 26  |
|         | S008          | M           | 421149      |           | SOCKET CAP SCREW M10 x 20 GRADE 12.9    |                  | 2   |
|         | S009          | M           | 421035      |           | SOCKET CAP SCREW M10 x 25 GRADE 12.9    |                  | 2   |
|         | S010          | M           | 421018      |           | SOCKET CAP SCREW M10 x 35 GRADE 12.9    |                  | 8   |
|         | S011          | M           | 421706      |           | SOCKET CAP SCREW M10 x 60 GRADE 12.9    |                  | 4   |
|         | S012          | M           | 421195      |           | SOCKET CAP SCREW M12 x 40 GRADE 12.9    |                  | 9   |
|         | S015          | M           | 421115      |           | SOCKET CAP SCREW M16 x 45 GRADE 12.9    |                  | 4   |
|         | S016          | M           | 421617      |           | SOCKET CAP SCREW M16 x 65 GRADE 12.9    |                  | 6   |
|         | S018          | M           | 421194      |           | Washer Plain M4                         |                  | 8   |
|         | S019          | M           | 421034      |           | WASHER PLAIN M6                         |                  | 11  |
|         | S020          | M           | 421054      |           | PLAIN WASHER M8                         |                  | 6   |
|         | S021          | M           | 421029      |           | PLAIN WASHER M10                        |                  | 4   |
|         | S022          | M           | 421030      |           | PLAIN WASHER M12                        |                  | 5   |
|         | S023          | M           | 421059      |           | Washer Plain M16                        |                  | 32  |
|         | S025          | M           | 421073      |           | Washer Spring M12                       |                  | 4   |
|         | S026          | M           | 421226      |           | GRUB SCREW M6x10                        |                  | 2   |
|         | S027          | M           | 4211098     |           | GRUB SCREW DOG POINT M10x15             |                  | 2   |
|         | S028          | M           | 421180      |           | Nut Full Hex M4                         |                  | 4   |
|         | S029          | M           | 421057      |           | NUT FULL HEXAGON M6                     |                  | 1   |
|         | S034          | M           | 421258      |           | DOWEL 6 dia x 40 mm LONG                |                  | 2   |
|         | S036          | M           | 421418      |           | Screw Socket Countersunk Head M6 x 16   |                  | 4   |
|         | S043          | M           | 421104      |           | SOCKET CAP SCREW M16 x 50 GRADE 12.9    |                  | 6   |
|         | S046          | M           | 421633      |           | Locknut M20                             |                  | 4   |
|         | S048          | M           | 421148      |           | GRUB SCREW M8x30                        |                  | 1   |

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| Section       | Bubble Number  | Mech / Elec | Part Number | Parameter | Part Description                               | Assembly Drawing | Qty      |
|---------------|----------------|-------------|-------------|-----------|------------------------------------------------|------------------|----------|
|               | S057           | M           | 421234      |           | SOCKET CAP SCREW M12 x 30 GRADE 12.9           |                  | 4        |
|               | S058           | M           | 421710      |           | Screw Socket M12x30 Countersunk Head           |                  | 6        |
|               | S059           | M           | 4211212     |           | SCREW, COUNTERSUNK SOCKET HEAD, M10x20         |                  | 4        |
|               | S060           | M           | 421613      |           | Screw Socket Buttonhead M10x20                 |                  | 4        |
|               | S063           | M           | 4211949G    |           | SOCKET CAP SCREW M14 x 25 GRADE 12.9           |                  | 4        |
|               | S064           | M           | 421121      |           | SOCKET CAP SCREW M6 x 16 GRADE 12.9            |                  | 21       |
|               | S065           | M           | 421467      |           | SOCKET CAP SCREW M16 x 60 GRADE 12.9           |                  | 8        |
|               | S066           | M           | 421038      |           | NUT FULL HEXAGON M16                           |                  | 8        |
|               | S067           | M           | 421425      |           | SOCKET CAP SCREW M8 x 50 GRADE 12.9            |                  | 6        |
|               | S071           | M           | 421182      |           | SOCKET CAP SCREW M8 x 16 GRADE 12.9            |                  | 12       |
|               | S076           | M           | 421051      |           | Screw Socket Countersunk Head M6 x 12          |                  | 3        |
|               | S077           | M           | 421033      |           | SOCKET CAP SCREW M6 x 12 GRADE 12.9            |                  | 2        |
|               | S078           | M           | 421245      |           | SOCKET CAP SCREW M5 x 10 GRADE 12.9            |                  | 6        |
|               | S079           | M           | 421133      |           | SOCKET CAP SCREW M5 x 16 GRADE 12.9            |                  | 4        |
| <b>05-020</b> | <b>910G241</b> |             |             |           | <b>Unwind Layon Roller Assy</b>                | <b>429G083</b>   | <b>1</b> |
|               | B001           | M           | 2061183G    |           | Linear Guide Block                             |                  | 8        |
|               | B002           | M           | 2061223G    |           | Linear Guide Rail                              |                  | 4        |
|               | B003           | M           | 200588      |           | Bearing Deep Ball Groove with Shield 6205-Z    |                  | 2        |
|               | B004           | M           | 200885G     |           | Bearing Deep Ball Groove with Shield 6206-Z    |                  | 1        |
|               | B005           | M           | 200608      |           | CARB BEARING C2206                             |                  | 1        |
|               | B006           | M           | 221920G     |           | Belt Timing Poly Chain 8M-2000-21              |                  | 2        |
|               | B007           | M           | 220659      |           | PULLEY 8M-32S-21                               |                  | 2        |
|               | B008           | M           | 239856G     |           | Seal Simmering BABSL35-52-6/72NBR              |                  | 2        |
|               | B009           | M           | 233004G     |           | CIRCLIP INTERNAL D1300-53                      |                  | 2        |
|               | B010           | M           | 234003      |           | CIRCLIP EXTERNAL 1400-0250                     |                  | 2        |
|               | B011           | M           | 233127      |           | CIRCLIP INTERNAL 1300-52                       |                  | 2        |
|               | B012           | M           | 233136      |           | CIRCLIP INTERNAL D1300-0620                    |                  | 1        |
|               | B013           | M           | 200502      |           | BEARING NO SEALS 2206E                         |                  | 1        |
|               | B014           | M           | 200578      |           | Bearing Flanged Cartridge MFC40                |                  | 1        |
|               | B016           | M           | 224190      |           | TAPERLOCK BUSH 1610-42                         |                  | 2        |
|               | B018           | M           | 239217      |           | Seal O Ring RM0245-30 Nitrile                  |                  | 1        |
|               | B019           | M           | 239246      |           | Seal O Ring RM0645-30Nitrile                   |                  | 1        |
|               | B020           | M           | 239303      |           | Seal O Ring RM1043-57 Nitrile                  |                  | 1        |
|               | B021           | M           | 242009      |           | LOCKNUT KM6                                    |                  | 1        |
|               | B022           | M           | 242010      |           | LOCKWASHER MB6                                 |                  | 1        |
|               | B023           | M           | 214025      |           | Nipple Grease M6                               |                  | 2        |
|               | B024           | M           | 201110      |           | HARDENED BUSH IR30 x 35 x 26                   |                  | 1        |
|               | B025           | E           | 675217G     |           | PROXIMITY SWITCH 18MM 8MM SENSING RANGE        |                  | 1        |
|               | M001           | M           | 429G084     | 2900B     | Unwind Layon Roller                            |                  | 1        |
|               | M003           | M           | 431G836     |           | Mounting Bracket                               |                  | 2        |
|               | M005           | M           | 423G271     |           | Carrier Plate                                  |                  | 2        |
|               | M006           | M           | 421G069     |           | Unwind Layon Bracket                           |                  | 2        |
|               | M007           | M           | 421G086     | 2900      | Layshaft                                       |                  | 1        |
|               | M008           | M           | 309G125     |           | DRIVE BELT CLAMP                               |                  | 2        |
|               | M010           | M           | 419G842     |           | BEARING COVER                                  |                  | 2        |
|               | M011           | M           | 419G828     |           | STUB SHAFT                                     |                  | 2        |
|               | M012           | M           | 400G950     |           | Timing Pulley - Modified Pilot Bored 8M-32S-21 |                  | 2        |
|               | M013           | M           | 403G387     |           | Bearing Bush                                   |                  | 2        |
|               | M014           | M           | 419G760     |           | Seal Housing                                   |                  | 1        |
|               | M015           | M           | 419G762     |           | Bearing Housing                                |                  | 1        |
|               | M016           | M           | 418G249     | IT6       | Key                                            |                  | 1        |
|               | M017           | M           | 418G248     | IT7       | Key                                            |                  | 2        |
|               | M018           | M           | 427G390     | IT1       | ROLLER MOUNTING BLOCK                          |                  | 2        |
|               | M019           | M           | 419G140     |           | DRIVE BELT BRACKET                             |                  | 2        |
|               | M020           | M           | 421G110     |           | Shaft Sleeve Cap                               |                  | 1        |
|               | M021           | M           | 421G336     |           | Proxi Switch Bracket                           |                  | 1        |
|               | M022           | M           | 421G337     |           | UW Layon Roller Stop Bracket                   |                  | 2        |
|               | M023           | M           | 421G774     |           | Spacer                                         |                  | 2        |
|               | M024           | M           | 428G924     |           | Wear Plate/Proxy Flag                          |                  | 2        |
|               | M025           | M           | 424G126     |           | Jacking Block                                  |                  | 2        |
|               | M026           | M           | 430G114     |           | End Stop                                       |                  | 2        |
|               | S001           | M           | 421049      |           | SOCKET CAP SCREW M8 x 20 GRADE 12.9            |                  | 2        |
|               | S002           | M           | 421019      |           | Socket Head Capscrew M6 x 20 GRADE 12.9        |                  | 46       |
|               | S003           | M           | 421145      |           | SOCKET CAP SCREW M6 x 30 GRADE 12.9            |                  | 3        |
|               | S004           | M           | 421138      |           | SOCKET CAP SCREW M8 x 25 GRADE 12.9            |                  | 20       |
|               | S005           | M           | 421122      |           | SOCKET CAP SCREW M10 x 30 GRADE 12.9           |                  | 3        |
|               | S006           | M           | 421114      |           | SOCKET CAP SCREW M10 x 40 GRADE 12.9           |                  | 4        |
|               | S007           | M           | 421161      |           | SOCKET CAP SCREW M10 x 50 GRADE 12.9           |                  | 8        |

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| Section       | Bubble Number  | Mech / Elec | Part Number | Parameter | Part Description                                 | Assembly Drawing | Qty      |
|---------------|----------------|-------------|-------------|-----------|--------------------------------------------------|------------------|----------|
|               | S008           | M           | 421036      |           | NUT FULL HEXAGON M8                              |                  | 4        |
|               | S009           | M           | 421027      |           | NUT FULL HEX M10                                 |                  | 4        |
|               | S010           | M           | 421054      |           | PLAIN WASHER M8                                  |                  | 16       |
|               | S011           | M           | 421029      |           | PLAIN WASHER M10                                 |                  | 8        |
|               | S012           | M           | 421059      |           | Washer Plain M16                                 |                  | 4        |
|               | S014           | M           | 421285      |           | Screw HEXHD SetscrewM10x60 GRADE 8.8             |                  | 2        |
|               | S015           | M           | 421318      |           | Screw HEXHD Setscrew M16X40 GRADE 8.8            |                  | 4        |
|               | S016           | M           | 421023      |           | Screw Countersunk Head Socket M5x16              |                  | 4        |
|               | S017           | M           | 4211655G    |           | Screw Csk Head Slotted M4 x 12                   |                  | 6        |
|               | S019           | M           | 421233      |           | SOCKET CAP SCREW M12 x 90 GRADE 12.9             |                  | 4        |
|               | S021           | M           | 421224      |           | Screw HEXHD Setscrew M8x70 GRADE 8.8             |                  | 4        |
|               | S022           | M           | 421572      |           | SOCKET CAP SCREW M12 x 20 GRADE 12.9             |                  | 4        |
| <b>05-030</b> | <b>908G486</b> |             |             |           | <b>Spreader Roller</b>                           | <b>421G100</b>   | <b>1</b> |
|               | B001           | M           | 200624      |           | BEARING 22209 E                                  |                  | 2        |
|               | B002           | M           | 423043      |           | Nipple Straight Grease 1/8" BSP                  |                  | 2        |
|               | M001           | M           | 421G101     |           | Spreader Roller Housing S.P.                     |                  | 1        |
|               | M002           | M           | 421G102     |           | Spreader Roller Housing F.P.                     |                  | 1        |
|               | M003           | M           | 421G104     | 2900B     | Spreader Roller                                  |                  | 1        |
|               | M004           | M           | 421G107     |           | Cut Spreader Shaft End Cap                       |                  | 2        |
|               | S001           | M           | 421278      |           | SOCKET CAP SCREW M8 x 45 GRADE 12.9              |                  | 8        |
|               | S002           | M           | 421526      |           | SOCKET CAP SCREW M12 x 45 GRADE 12.9             |                  | 6        |
|               | S003           | M           | 421122      |           | SOCKET CAP SCREW M10 x 30 GRADE 12.9             |                  | 2        |
|               | S004           | M           | 421073      |           | Washer Spring M12                                |                  | 6        |
|               | S005           | M           | 421103      |           | Washer Spring M8                                 |                  | 8        |
| <b>05-040</b> | <b>908G470</b> |             |             |           | <b>Load Cell Roller Unwind</b>                   | <b>420G629</b>   | <b>1</b> |
|               | B001           | M           | 200625      |           | BEARING SPHERICAL ROLLER                         |                  | 2        |
|               | B002           | M           | 233135      |           | CIRCLIP INTERNAL 1300-0800                       |                  | 4        |
|               | B003           | M           | 214025      |           | Nipple Grease M6                                 |                  | 2        |
|               | B004           | E           | 631044      |           | Pressductor Radial Tensiometer Load Cell         |                  | 2        |
|               | B005           | E           | 631045      |           | Connecting Cable c/w 90deg Plug Contact 20m lg   |                  | 2        |
|               | B006           | E           | 631123G     |           | 2 Channel Profibus Control Unit PFEA112-20       |                  | 1        |
|               | B007           | E           | 650689      |           | ET200B L2 PROFIBUS CONNECTOR 6ES7-972-0BA42-0XA0 |                  | 1        |
|               | M001           | M           | 420G630     | 2900B     | Loadcell Roller                                  |                  | 1        |
|               | M002           | M           | 420G632     |           | Support Housing                                  |                  | 1        |
|               | M003           | M           | 400G625     |           | Load Cell                                        |                  | 2        |
|               | M004           | M           | 423G765     |           | Support Housing                                  |                  | 1        |
|               | M006           | M           | 401G044     |           | CLAMPING WASHER                                  |                  | 2        |
|               | M007           | M           | 423G766     |           | Housing                                          |                  | 1        |
|               | S001           | M           | 421122      |           | SOCKET CAP SCREW M10 x 30 GRADE 12.9             |                  | 2        |
|               | S002           | M           | 421089      |           | SOCKET CAP SCREW M10 x 70 GRADE 12.9             |                  | 8        |
|               | S003           | M           | 421210      |           | SOCKET CAP SCREW M12 x 35 GRADE 12.9             |                  | 6        |
|               | S004           | M           | 421021      |           | Washer Spring M10                                |                  | 16       |
|               | S005           | M           | 421073      |           | Washer Spring M12                                |                  | 3        |
|               | S006           | M           | 421027      |           | NUT FULL HEX M10                                 |                  | 8        |
|               | S007           | M           | 421160      |           | SOCKET CAP SCREW M10 x 90 GRADE 12.9             |                  | 4        |
|               | S008           | M           | 421030      |           | PLAIN WASHER M12                                 |                  | 3        |
| <b>05-050</b> | <b>908G533</b> |             |             |           | <b>Draw Roller Assy</b>                          | <b>420G649</b>   | <b>1</b> |
|               | B001           | M           | 239870G     |           | Seal Simmerring BABSL60-80-7 FKM                 |                  | 2        |
|               | B002           | M           | 234007G     |           | CIRCLIP EXTERNAL D1400-0550                      |                  | 1        |
|               | B003           | M           | 239201      |           | Seal O Ring R0445-30 Nitrile                     |                  | 1        |
|               | B004           | M           | 201100      |           | BEARING INNER RACE LR 50 x 60 x 28 EGS           |                  | 1        |
|               | B005           | M           | 239357      |           | Seal O Ring RM1095-30 Nitrile                    |                  | 1        |
|               | B006           | M           | 239351      |           | Seal O Ring RM1645-30 Nitrile                    |                  | 1        |
|               | B007           | M           | 200864G     |           | SPHERICAL ROLLER BEARING 22211E                  |                  | 2        |
|               | B008           | M           | 423036      |           | Adaptor Male 8mm-1/4"BSPT 36-0530-11             |                  | 2        |
|               | B009           | M           | 290008      |           | Cork Covering 1/16" Thk x 2" Wide 61 Metre Roll  |                  | 1        |
|               | B010           | M           | 423038      |           | Tubing Nut 8 OD                                  |                  | 2        |
|               | B011           | M           | 233037      |           | CIRCLIP 1300-82                                  |                  | 2        |
|               | B012           | M           | 423039      |           | Tubing Sleeve 8 OD                               |                  | 2        |
|               | B013           | M           | 234019      |           | CIRCLIP EXTERNAL D1400-0500                      |                  | 1        |
|               | M001           | M           | 420G650     | 2900B     | Draw Roller                                      |                  | 1        |
|               | M002           | M           | 420G651     |           | Bearing Housing Inboard                          |                  | 1        |
|               | M003           | M           | 420G652     |           | Seal Housing                                     |                  | 1        |
|               | M004           | M           | 420G653     |           | Bearing Housing Outboard                         |                  | 1        |
|               | M005           | M           | 418G248     | IT13      | Key                                              |                  | 1        |
|               | M006           | M           | 420G662     |           | Labyrinth Seal                                   |                  | 1        |
|               | M007           | M           | 421G367     |           | LABYRINTH SEAL                                   |                  | 1        |
|               | M008           | M           | 420G663     |           | LABYRINTH SEAL                                   |                  | 1        |

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| Section       | Bubble Number  | Mech / Elec | Part Number | Parameter | Part Description                                     | Assembly Drawing | Qty      |
|---------------|----------------|-------------|-------------|-----------|------------------------------------------------------|------------------|----------|
|               | M009           | M           | 421G698     |           | Spacer                                               |                  | 2        |
|               | S001           | M           | 421579      |           | SOCKET CAP SCREW M6 x 45 GRADE 12.9                  |                  | 3        |
|               | S002           | M           | 421210      |           | SOCKET CAP SCREW M12 x 35 GRADE 12.9                 |                  | 7        |
|               | S003           | M           | 421030      |           | PLAIN WASHER M12                                     |                  | 6        |
|               | S004           | M           | 245205      |           | Grease Nipple M6 - Obsolete Use 214025               |                  | 2        |
| <b>05-060</b> | <b>910G491</b> |             |             |           | <b>Kickert Manual Varibow Roller-Apex Drive Assy</b> | <b>430G107</b>   | <b>1</b> |
|               | B001           | M           | 426G040     |           | Kickert Free Running Pulley WG160/21                 |                  | 1        |
|               | M001           | M           | 430G106     | 2900WT    | Kickert Manual Varibow Roller-Apex Drive             |                  | 1        |
|               | M002           | M           | 425G681     |           | Adaptor Plate                                        |                  | 1        |
|               | M003           | M           | 425G682     |           | Adaptor Plate                                        |                  | 1        |
|               | S001           | M           | 421234      |           | SOCKET CAP SCREW M12 x 30 GRADE 12.9                 |                  | 6        |
|               | S002           | M           | 421467      |           | SOCKET CAP SCREW M16 x 60 GRADE 12.9                 |                  | 8        |
|               | S003           | M           | 421059      |           | Washer Plain M16                                     |                  | 8        |
| <b>05-070</b> | <b>909G520</b> |             |             |           | <b>Water Cooled Drum 700 Dia</b>                     | <b>425G876</b>   | <b>1</b> |
|               | B001           | M           | 200401      |           | BEARING SPHERICAL ROLLER 23024CC/W33                 |                  | 2        |
|               | B002           | M           | 233003G     |           | INTERNAL CIRCLIP D1300-1520                          |                  | 2        |
|               | B004           |             | 253M070     |           | EXTERNAL CIRCLIP - D1400 1200                        |                  | 1        |
|               | B005           | M           | 239925G     |           | Seal Simmerring BABSL130-150-8/72 NBR 902            |                  | 2        |
|               | B006           | M           | 208204G     |           | ROTARY UNION 2 1/2" BSPRH -DEUBLIN 755-707-330-139   |                  | 1        |
|               | B007           | M           | 214025      |           | Nipple Grease M6                                     |                  | 2        |
|               | B008           | M           | 239216      |           | Seal O Ring RM 0246-24 Nitrile                       |                  | 2        |
|               | B009           | M           | 239291      |           | Seal O Ring RM1145-30 Nitrile                        |                  | 1        |
|               | B010           | M           | 239350      |           | Seal O Ring RM1845-30 Nitrile                        |                  | 1        |
|               | B011           | M           | 239213      |           | Seal O Ring RM 2393-57 Nitrile                       |                  | 1        |
|               | B012           | M           | 200861G     |           | Bearing Inner Race LR 120-130-30 EGS                 |                  | 1        |
|               | B013           | M           | 423036      |           | Adaptor Male 8mm-1/4"BSPT 36-0530-11                 |                  | 2        |
|               | B014           | M           | 423038      |           | Tubing Nut 8 OD                                      |                  | 2        |
|               | B015           | M           | 423039      |           | Tubing Sleeve 8 OD                                   |                  | 2        |
|               | B017           | M           | 2442378G    |           | GREASE NIPPLE M6 LONG                                |                  | 1        |
|               | B018           | M           | 242078      |           | LOCK WASHER MB24                                     |                  | 1        |
|               | B019           | M           | 242077      |           | LOCKNUT KM24                                         |                  | 1        |
|               | B020           | M           | 4212007G    |           | GRUB SCREW M6x30                                     |                  | 1        |
|               | M001           | M           | 425G877     | 2900B     | Water Cooled Drum 700 Dia Machining Details          |                  | 1        |
|               | M002           | M           | 424G747     |           | Bearing Housing                                      |                  | 1        |
|               | M003           | M           | 424G751     |           | Seal Housing                                         |                  | 1        |
|               | M004           | M           | 419G783     |           | Bearing Housing - Outboard - Water Cooled Drum       |                  | 1        |
|               | M005           | M           | 419G796     |           | Labryinth Seal                                       |                  | 2        |
|               | M006           | M           | 421G198     |           | Labyrinth Seal                                       |                  | 1        |
|               | M008           | M           | 418G248     | IT25      | Key                                                  |                  | 1        |
|               | M009           | M           | 421G127     | ITA       | Water Cooled Drum Centre Tubing                      |                  | 1        |
|               | M010           | M           | 421G119     |           | Spacer                                               |                  | 2        |
|               | M011           | M           | 303G508     |           | 1/2" BSP PLUG                                        |                  | 2        |
|               | M013           | M           | 421G179     |           | LIFTING BRACKET WATER COOLED DRUM                    |                  | 2        |
|               | M014           | M           | 430G259     |           | Rotary Union 2 1/2" Modification                     |                  | 1        |
|               | S001           | M           | 421278      |           | SOCKET CAP SCREW M8 x 45 GRADE 12.9                  |                  | 3        |
|               | S002           | M           | 421526      |           | SOCKET CAP SCREW M12 x 45 GRADE 12.9                 |                  | 6        |
|               | S003           | M           | 421054      |           | PLAIN WASHER M8                                      |                  | 3        |
|               | S004           | M           | 421030      |           | PLAIN WASHER M12                                     |                  | 4        |
|               | S005           | M           | 421103      |           | Washer Spring M8                                     |                  | 3        |
|               | S006           | M           | 421073      |           | Washer Spring M12                                    |                  | 5        |
|               | S009           | M           | 421303      |           | SOCKET CAP SCREW M16 x 40 GRADE 12.9                 |                  | 8        |
|               | S010           | M           | 421116      |           | Washer Spring M16                                    |                  | 8        |
|               | S011           | M           | 4211154     |           | Locknut M6                                           |                  | 1        |
| <b>05-080</b> | <b>908G507</b> |             |             |           | <b>Water Cooled Roller Assy 250 Dia -25C</b>         | <b>421G238</b>   | <b>1</b> |
|               | B001           | M           | 200624      |           | BEARING 22209 E                                      |                  | 1        |
|               | B002           | M           | 200174      |           | BEARING 22217CC/W33                                  |                  | 1        |
|               | B003           | M           | 208205G     |           | Rotary Union 1 1/2" BSPLH - 555-385-199              |                  | 1        |
|               | B004           | M           | 4231404G    |           | ELBOW - DEUBLIN 451-171                              |                  | 1        |
|               | B005           |             | 252M072     |           | INTERNAL CIRCLIP - D1300-1120                        |                  | 2        |
|               | B006           | M           | 234003      |           | CIRCLIP EXTERNAL 1400-0250                           |                  | 2        |
|               | B007           | M           | 234050      |           | CIRCLIP EXTERNAL 1400-0800                           |                  | 1        |
|               | B008           |             | 253M059     |           | EXTERNAL CIRCLIP - D1400-0850                        |                  | 1        |
|               | B009           | M           | 234001      |           | CIRCLIP EXTERNAL 1400-0450                           |                  | 1        |
|               | B010           | M           | 214025      |           | Nipple Grease M6                                     |                  | 2        |
|               | B011           | M           | 418G249     | IT5       | Key                                                  |                  | 1        |
|               | B012           | M           | 239230      |           | Seal O Ring RM 0151-16 Nitrile                       |                  | 2        |
|               | B013           | M           | 239198      |           | Seal O Ring RM0745-30 Nitrile                        |                  | 1        |
|               | B014           | M           | 239323      |           | Nitrile O-Ring RM1545-30                             |                  | 1        |



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| Section       | Bubble Number  | Mech / Elec | Part Number | Parameter | Part Description                                | Assembly Drawing | Qty      |
|---------------|----------------|-------------|-------------|-----------|-------------------------------------------------|------------------|----------|
|               | B015           | M           | 239324      |           | Nitrile O-Ring RM2093-57                        |                  | 1        |
|               | B016           | M           | 239858G     |           | Seal Simmerring BABSL90-110-7-5-72NBR           |                  | 2        |
|               | B017           | M           | 200863G     |           | Bearing Inner Race LR 80-90-35 EGS              |                  | 1        |
|               | B018           | M           | 423036      |           | Adaptor Male 8mm-1/4"BSPT 36-0530-11            |                  | 2        |
|               | B019           | M           | 423038      |           | Tubing Nut 8 OD                                 |                  | 2        |
|               | B020           | M           | 423039      |           | Tubing Sleeve 8 OD                              |                  | 2        |
|               | B021           | M           | 2442381G    |           | TESA BAND 4863 25 MTR X 120mm WIDE              |                  | 1        |
|               | B022           | M           | 2442378G    |           | GREASE NIPPLE M6 LONG                           |                  | 1        |
|               | M001           | M           | 421G239     | 2900B     | Water Cooled Roller 250Dia (DTR1)               |                  | 1        |
|               | M002           | M           | 421G253     |           | Bearing Housing RH WCR                          |                  | 1        |
|               | M003           | M           | 421G254     |           | Seal Housing                                    |                  | 1        |
|               | M004           | M           | 421G281     |           | Bearing Housing LH WCR                          |                  | 1        |
|               | M005           | M           | 423G908     |           | Brass Centre Tube Bush                          |                  | 1        |
|               | M006           | M           | 421G287     |           | Labyrinth Seal                                  |                  | 1        |
|               | M007           | M           | 400G935     |           | LABYRINTH SEAL LH                               |                  | 2        |
|               | M008           | M           | 400G415     |           | PLUG                                            |                  | 2        |
|               | M009           | M           | 421G965     |           | Centre Tube                                     |                  | 1        |
|               | S001           | M           | 421524      |           | SOCKET CAP SCREW M8 x 55 GRADE 12.9             |                  | 3        |
|               | S002           | M           | 421526      |           | SOCKET CAP SCREW M12 x 45 GRADE 12.9            |                  | 3        |
|               | S003           | M           | 421402      |           | SOCKET CAP SCREW M12 x 60 GRADE 12.9            |                  | 3        |
|               | S004           | M           | 421030      |           | PLAIN WASHER M12                                |                  | 6        |
|               | S005           | M           | 421073      |           | Washer Spring M12                               |                  | 6        |
| <b>05-090</b> | <b>908G512</b> |             |             |           | <b>Ebonite Roller Assembly (DTR2)</b>           | <b>421G467</b>   | <b>1</b> |
|               | B001           | M           | 200864G     |           | SPHERICAL ROLLER BEARING 22211E                 |                  | 2        |
|               | B002           | M           | 252A059     |           | D1300-0820                                      |                  | 2        |
|               | B003           | M           | 234019      |           | CIRCLIP EXTERNAL D1400-0500                     |                  | 1        |
|               | B004           | M           | 234007G     |           | CIRCLIP EXTERNAL D1400-0550                     |                  | 1        |
|               | B005           | M           | 239870G     |           | Seal Simmerring BABSL60-80-7 FKM                |                  | 2        |
|               | B006           | M           | 239201      |           | Seal O Ring R0445-30 Nitrile                    |                  | 1        |
|               | B007           | M           | 239357      |           | Seal O Ring RM1095-30 Nitrile                   |                  | 1        |
|               | B008           | M           | 239351      |           | Seal O Ring RM1645-30 Nitrile                   |                  | 1        |
|               | B009           | M           | 423036      |           | Adaptor Male 8mm-1/4"BSPT 36-0530-11            |                  | 2        |
|               | B010           | M           | 423039      |           | Tubing Sleeve 8 OD                              |                  | 2        |
|               | B011           | M           | 201100      |           | BEARING INNER RACE LR 50 x 60 x 28 EGS          |                  | 1        |
|               | B012           | M           | 245205      |           | Grease Nipple M6 - Obsolete Use 214025          |                  | 2        |
|               | B013           | M           | 423038      |           | Tubing Nut 8 OD                                 |                  | 2        |
|               | B014           | M           | 290008      |           | Cork Covering 1/16" Thk x 2" Wide 61 Metre Roll |                  | 1        |
|               | M001           | M           | 421G468     | 2900B     | Ebonite Roller (DTR2)                           |                  | 1        |
|               | M002           | M           | 420G651     |           | Bearing Housing Inboard                         |                  | 1        |
|               | M003           | M           | 420G652     |           | Seal Housing                                    |                  | 1        |
|               | M004           | M           | 420G653     |           | Bearing Housing Outboard                        |                  | 1        |
|               | M005           | M           | 418G248     | IT13      | Key                                             |                  | 1        |
|               | M006           | M           | 420G662     |           | Labyrinth Seal                                  |                  | 1        |
|               | M007           | M           | 420G663     |           | LABYRINTH SEAL                                  |                  | 1        |
|               | M008           | M           | 421G367     |           | LABYRINTH SEAL                                  |                  | 1        |
|               | M009           | M           | 421G698     |           | Spacer                                          |                  | 2        |
|               | S001           | M           | 421579      |           | SOCKET CAP SCREW M6 x 45 GRADE 12.9             |                  | 3        |
|               | S002           | M           | 421210      |           | SOCKET CAP SCREW M12 x 35 GRADE 12.9            |                  | 7        |
|               | S003           | M           | 421030      |           | PLAIN WASHER M12                                |                  | 6        |
| <b>05-100</b> | <b>910G554</b> |             |             |           | <b>Idler Roller - No Load Cells</b>             | <b>430G582</b>   | <b>1</b> |
|               | B001           | M           | 200625      |           | BEARING SPHERICAL ROLLER                        |                  | 2        |
|               | B002           | M           | 233135      |           | CIRCLIP INTERNAL 1300-0800                      |                  | 2        |
|               | B003           | M           | 214025      |           | Nipple Grease M6                                |                  | 2        |
|               | B004           | M           | 2442381G    |           | TESA BAND 4863 25 MTR X 120mm WIDE              |                  | 1        |
|               | M001           | M           | 420G630     | 2900B     | Loadcell Roller                                 |                  | 1        |
|               | M002           | M           | 420G632     |           | Support Housing                                 |                  | 1        |
|               | M003           | M           | 430G583     |           | Roller Adaptor                                  |                  | 2        |
|               | M004           | M           | 423G765     |           | Support Housing                                 |                  | 1        |
|               | M005           | M           | 423G766     |           | Housing                                         |                  | 1        |
|               | M006           | M           | 401G044     |           | CLAMPING WASHER                                 |                  | 2        |
|               | S001           | M           | 421122      |           | SOCKET CAP SCREW M10 x 30 GRADE 12.9            |                  | 2        |
|               | S002           | M           | 421706      |           | SOCKET CAP SCREW M10 x 60 GRADE 12.9            |                  | 8        |
|               | S003           | M           | 421210      |           | SOCKET CAP SCREW M12 x 35 GRADE 12.9            |                  | 6        |
|               | S004           | M           | 421021      |           | Washer Spring M10                               |                  | 8        |
|               | S005           | M           | 421073      |           | Washer Spring M12                               |                  | 3        |
|               | S006           | M           | 421030      |           | PLAIN WASHER M12                                |                  | 3        |
|               | S007           | M           | 421160      |           | SOCKET CAP SCREW M10 x 90 GRADE 12.9            |                  | 4        |
| <b>05-110</b> | <b>908G471</b> |             |             |           | <b>Load Cell Roller Rewind</b>                  | <b>424G405</b>   | <b>1</b> |

# V18G019 - Bill of Materials



| Section       | Bubble Number  | Mech / Elec | Part Number | Parameter | Part Description                                 | Assembly Drawing | Qty      |
|---------------|----------------|-------------|-------------|-----------|--------------------------------------------------|------------------|----------|
|               | B001           | M           | 200625      |           | BEARING SPHERICAL ROLLER                         |                  | 2        |
|               | B002           | M           | 233135      |           | CIRCLIP INTERNAL 1300-0800                       |                  | 4        |
|               | B003           | M           | 214025      |           | Nipple Grease M6                                 |                  | 2        |
|               | B004           | E           | 631044      |           | Pressductor Radial Tensiometer Load Cell         |                  | 2        |
|               | B005           | E           | 631045      |           | Connecting Cable c/w 90deg Plug Contact 20m lg   |                  | 2        |
|               | B006           | E           | 631123G     |           | 2 Channel Profibus Control Unit PFEA112-20       |                  | 1        |
|               | B007           | E           | 650689      |           | ET200B L2 PROFIBUS CONNECTOR 6ES7-972-0BA42-0XA0 |                  | 1        |
|               | M001           | M           | 420G630     | 2900B     | Loadcell Roller                                  |                  | 1        |
|               | M002           | M           | 420G632     |           | Support Housing                                  |                  | 1        |
|               | M003           | M           | 400G625     |           | Load Cell                                        |                  | 2        |
|               | M005           | M           | 420G973     |           | Support Housing                                  |                  | 1        |
|               | M006           | M           | 401G044     |           | CLAMPING WASHER                                  |                  | 2        |
|               | S001           | M           | 421122      |           | SOCKET CAP SCREW M10 x 30 GRADE 12.9             |                  | 2        |
|               | S002           | M           | 421089      |           | SOCKET CAP SCREW M10 x 70 GRADE 12.9             |                  | 8        |
|               | S003           | M           | 421210      |           | SOCKET CAP SCREW M12 x 35 GRADE 12.9             |                  | 5        |
|               | S004           | M           | 421021      |           | Washer Spring M10                                |                  | 16       |
|               | S005           | M           | 421073      |           | Washer Spring M12                                |                  | 5        |
|               | S006           | M           | 421027      |           | NUT FULL HEX M10                                 |                  | 8        |
|               | S007           | M           | 421093      |           | Screw HEXHD Setscrew M12x35 GRADE 8.8            |                  | 1        |
| <b>05-120</b> | <b>908G491</b> |             |             |           | <b>Rewind Layon Assembly</b>                     | <b>429G076</b>   | <b>1</b> |
|               | B001           | M           | 428G122     | 2900B     | Kickert Bowed Roller Driven Apex Rubber Cover    |                  | 1        |
|               | B002           | M           | 200588      |           | Bearing Deep Ball Groove with Shield 6205-Z      |                  | 2        |
|               | B003           | M           | 200885G     |           | Bearing Deep Ball Groove with Shield 6206-Z      |                  | 1        |
|               | B004           | M           | 200502      |           | BEARING NO SEALS 2206E                           |                  | 1        |
|               | B005           | M           | 200608      |           | CARB BEARING C2206                               |                  | 1        |
|               | B006           | M           | 242010      |           | LOCKWASHER MB6                                   |                  | 1        |
|               | B007           | M           | 242009      |           | LOCKNUT KM6                                      |                  | 1        |
|               | B008           | M           | 200485      |           | Bearing Inner Race                               |                  | 1        |
|               | B009           | M           | 200578      |           | Bearing Flanged Cartridge MFC40                  |                  | 1        |
|               | B010           | M           | 2061183G    |           | Linear Guide Block                               |                  | 8        |
|               | B011           | M           | 2061222G    |           | Linear Guide Rail (Hiwin EGR25R890C)             |                  | 4        |
|               | B012           | M           | 239856G     |           | Seal Simmerring BABSL35-52-6/72NBR               |                  | 2        |
|               | B013           | M           | 221942G     |           | TIMING BELT POWERGRIP GT 2200-8M-20              |                  | 2        |
|               | B014           | M           | 220756G     |           | TIMING PULLEY POLYCHAIN GT 8M-40S-21             |                  | 2        |
|               | B015           | M           | 224190      |           | TAPERLOCK BUSH 1610-42                           |                  | 2        |
|               | B016           | E           | 675217G     |           | PROXIMITY SWITCH 18MM 8MM SENSING RANGE          |                  | 1        |
|               | B018           | M           | 239217      |           | Seal O Ring RM0245-30 Nitrile                    |                  | 1        |
|               | B019           | M           | 239246      |           | Seal O Ring RM0645-30Nitrile                     |                  | 1        |
|               | B020           | M           | 239303      |           | Seal O Ring RM1043-57 Nitrile                    |                  | 1        |
|               | B021           | M           | 233004G     |           | CIRCLIP INTERNAL D1300-53                        |                  | 2        |
|               | B022           | M           | 233136      |           | CIRCLIP INTERNAL D1300-0620                      |                  | 1        |
|               | B023           | M           | 234003      |           | CIRCLIP EXTERNAL 1400-0250                       |                  | 2        |
|               | B024           | M           | 421437      |           | Shoulder Screw Socket Head 16mm (M12) x 30 LG    |                  | 2        |
|               | B026           | M           | 233014      |           | INTERNAL CIRCLIP D1300-0520                      |                  | 2        |
|               | B027           | M           | 290008      |           | Cork Covering 1/16" Thk x 2" Wide 61 Metre Roll  |                  | 1        |
|               | M001           | M           | 419G911     | 2900B     | REWIND LAYON ROLLER                              |                  | 1        |
|               | M002           | M           | 429G079     |           | Roller Bracket (Layon)                           |                  | 1        |
|               | M003           | M           | 419G760     |           | Seal Housing                                     |                  | 1        |
|               | M004           | M           | 419G762     |           | Bearing Housing                                  |                  | 1        |
|               | M005           | M           | 429G077     |           | Roller Bracket (Layon)                           |                  | 1        |
|               | M006           | M           | 419G842     |           | BEARING COVER                                    |                  | 2        |
|               | M007           | M           | 419G843     | IT1       | ROLLER MOUNTING BLOCK                            |                  | 2        |
|               | M008           | M           | 419G860     | 2900      | TIEBAR                                           |                  | 1        |
|               | M009           | M           | 420G727     |           | Rewind Layon Bracket                             |                  | 2        |
|               | M010           | M           | 420G739     |           | Pulley Shaft                                     |                  | 2        |
|               | M011           | M           | 421G110     |           | Shaft Sleeve Cap                                 |                  | 1        |
|               | M012           | M           | 421G180     |           | Bracket - Proximity Switch                       |                  | 1        |
|               | M013           | M           | 421G086     | 2900      | Layshaft                                         |                  | 1        |
|               | M014           | M           | 309G125     |           | DRIVE BELT CLAMP                                 |                  | 2        |
|               | M015           | M           | 403G387     |           | Bearing Bush                                     |                  | 2        |
|               | M016           | M           | 416G979     |           | PULLEY                                           |                  | 2        |
|               | M017           | M           | 429G768     |           | Bracket                                          |                  | 2        |
|               | M018           | M           | 421G224     | IT1       | Packing Plate                                    |                  | 2        |
|               | M019           | M           | 418G248     | IT23      | Key                                              |                  | 2        |
|               | M020           | M           | 421G774     |           | Spacer                                           |                  | 2        |
|               | M021           | M           | 418G248     | IT30      | Key                                              |                  | 1        |
|               | M022           | M           | 429G078     | IT1       | Roller Support Plate                             |                  | 1        |
|               | M023           | M           | 429G078     | IT2       | Roller Support Plate                             |                  | 1        |

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| Section       | Bubble Number  | Mech / Elec | Part Number | Parameter | Part Description                          | Assembly Drawing | Qty      |
|---------------|----------------|-------------|-------------|-----------|-------------------------------------------|------------------|----------|
|               | M024           | M           | 403G527     |           | Jacking Block                             |                  | 4        |
|               | S001           | M           | 421047      |           | Screw HEXHD Setscrew M8x60 GRADE 8.8      |                  | 4        |
|               | S002           | M           | 421019      |           | Socket Head Capscrew M6 x 20 GRADE 12.9   |                  | 44       |
|               | S003           | M           | 421145      |           | SOCKET CAP SCREW M6 x 30 GRADE 12.9       |                  | 16       |
|               | S005           | M           | 421579      |           | SOCKET CAP SCREW M6 x 45 GRADE 12.9       |                  | 3        |
|               | S006           | M           | 421138      |           | SOCKET CAP SCREW M8 x 25 GRADE 12.9       |                  | 12       |
|               | S007           | M           | 421195      |           | SOCKET CAP SCREW M12 x 40 GRADE 12.9      |                  | 6        |
|               | S008           | M           | 421122      |           | SOCKET CAP SCREW M10 x 30 GRADE 12.9      |                  | 3        |
|               | S009           | M           | 421114      |           | SOCKET CAP SCREW M10 x 40 GRADE 12.9      |                  | 4        |
|               | S010           | M           | 421161      |           | SOCKET CAP SCREW M10 x 50 GRADE 12.9      |                  | 2        |
|               | S011           | M           | 421160      |           | SOCKET CAP SCREW M10 x 90 GRADE 12.9      |                  | 4        |
|               | S012           | M           | 421234      |           | SOCKET CAP SCREW M12 x 30 GRADE 12.9      |                  | 2        |
|               | S013           | M           | 421402      |           | SOCKET CAP SCREW M12 x 60 GRADE 12.9      |                  | 8        |
|               | S014           | M           | 421303      |           | SOCKET CAP SCREW M16 x 40 GRADE 12.9      |                  | 4        |
|               | S015           | M           | 421794      |           | SCREW, COUNTERSUNK SOCKET HEAD, M4x12     |                  | 6        |
|               | S016           | M           | 421345      |           | Screw HEXHD Setscrew M10x100 GRADE 8.8    |                  | 2        |
|               | S017           | M           | 421341      |           | Screw HEXHD Setscrew M10x120 GRADE 8.8    |                  | 2        |
|               | S018           | M           | 421036      |           | NUT FULL HEXAGON M8                       |                  | 4        |
|               | S019           | M           | 421027      |           | NUT FULL HEX M10                          |                  | 4        |
|               | S020           | M           | 421034      |           | WASHER PLAIN M6                           |                  | 2        |
|               | S021           | M           | 421054      |           | PLAIN WASHER M8                           |                  | 12       |
|               | S022           | M           | 421029      |           | PLAIN WASHER M10                          |                  | 4        |
|               | S023           | M           | 421030      |           | PLAIN WASHER M12                          |                  | 12       |
|               | S024           | M           | 421059      |           | Washer Plain M16                          |                  | 6        |
|               | S025           | M           | 245205      |           | Grease Nipple M6 - Obsolete Use 214025    |                  | 2        |
| <b>05-122</b> | <b>908G497</b> |             |             |           | <b>Rewind Layon Cable Coiler Assembly</b> | <b>421G205</b>   | <b>1</b> |
|               | B001           | E           | 619736G     |           | Plastic Energy Chain - Igus               |                  | 2        |
|               | M001           | M           | 421G184     |           | Coiler Bracket                            |                  | 1        |
|               | M002           | M           | 421G185     |           | ENERGY CHAIN BRACKET                      |                  | 1        |
|               | S001           | M           | 421033      |           | SOCKET CAP SCREW M6 x 12 GRADE 12.9       |                  | 2        |
|               | S002           | M           | 421019      |           | Socket Head Capscrew M6 x 20 GRADE 12.9   |                  | 2        |
|               | S003           | M           | 421947      |           | Screw CSK HD M3x12                        |                  | 8        |
|               | S004           | M           | 421034      |           | WASHER PLAIN M6                           |                  | 4        |
|               | S005           | M           | 421118      |           | Nut Full Hex M3                           |                  | 8        |
|               | S006           | M           | 421117      |           | Washer Plain M3                           |                  | 8        |
| <b>05-124</b> | <b>910G565</b> |             |             |           | <b>Rewind Layon Gas Wedge Assembly</b>    | <b>430G648</b>   | <b>1</b> |
|               | B001           | M           | 239270      |           | Seal O Ring RM0251-16V Viton              |                  | 2        |
|               | B002           | M           | 239237      |           | Seal O Ring RM0296-24 Nitrile             |                  | 2        |
|               | B003           | M           | 423090      |           | Elbow Male Adaptor                        |                  | 3        |
|               | B004           | M           | 423038      |           | Tubing Nut 8 OD                           |                  | 14       |
|               | B005           | M           | 423039      |           | Tubing Sleeve 8 OD                        |                  | 14       |
|               | B006           | M           | 423471      |           | Tee Connector 8 OD                        |                  | 1        |
|               | B007           | M           | 423053      |           | Straight Connector 8mm 36-0503-05         |                  | 2        |
|               | B008           | M           | 423058      |           | Elbow Connector 8mm O/D                   |                  | 2        |
|               | B009           | M           | 423056      |           | Pipe Clip 8dia Enots 34-0218-05           |                  | 8        |
|               | B010           | M           | 423046      |           | Tube Half Hard Copper 8mm O/D CS7008003   |                  | 6        |
|               | B011           | M           | 427427G     |           | Union 1/4" x 1/4" BRASS B-400-6           |                  | 2        |
|               | B012           | M           | 427603G     |           | Swagelok Reducing Union 8mm to 1/4"       |                  | 2        |
|               | B013           | M           | 4232530G    |           | Nylon Tube Natural 8mm OD                 |                  | 4        |
|               | B014           | M           | 422G489     |           | 2962 Gas Wedge Extrusion                  |                  | 1        |
|               | M001           | M           | 430G649     |           | 2900 Gas Wedge                            |                  | 1        |
|               | M002           | M           | 403G088     |           | Plug                                      |                  | 2        |
|               | M003           | M           | 429G830     |           | GAS WEDGE END CAP                         |                  | 2        |
|               | M004           | M           | 427G776     | IT2       | Gas Nozzle                                |                  | 78       |
|               | M005           | M           | 427G776     | IT5       | Gas Nozzle                                |                  | 78       |
|               | M006           | M           | 418G015     |           | 1/4" Leadthrough                          |                  | 2        |
|               | M007           | M           | 430G650     | IT1       | Gas Wedge Mounting Plate                  |                  | 1        |
|               | M008           | M           | 421G821     |           | Gas Wedge Support Bracket                 |                  | 1        |
|               | M009           | M           | 431G518     |           | Gas Wedge Support Bracket                 |                  | 1        |
|               | M010           | M           | 431G520     |           | Gas Wedge Mounting Plate                  |                  | 1        |
|               | S001           | M           | 421364      |           | SOCKET CAP SCREW M4 x 8 GRADE 12.9        |                  | 8        |
|               | S002           | M           | 421019      |           | Socket Head Capscrew M6 x 20 GRADE 12.9   |                  | 8        |
|               | S003           | M           | 421138      |           | SOCKET CAP SCREW M8 x 25 GRADE 12.9       |                  | 12       |
|               | S004           | M           | 421249      |           | M8 X 20MM SOCKET HEAD COUNTERSUNK SCREW   |                  | 4        |
|               | S005           | M           | 421036      |           | NUT FULL HEXAGON M8                       |                  | 4        |
|               | S006           | M           | 421034      |           | WASHER PLAIN M6                           |                  | 8        |
|               | S007           | M           | 421054      |           | PLAIN WASHER M8                           |                  | 16       |
|               | S008           | M           | 421227      |           | Washer Spring M4                          |                  | 8        |

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| Section       | Bubble Number  | Mech / Elec | Part Number | Parameter | Part Description                                   | Assembly Drawing | Qty      |
|---------------|----------------|-------------|-------------|-----------|----------------------------------------------------|------------------|----------|
|               | S009           | M           | 421969      |           | GRUB SCREW M8x10                                   |                  | 78       |
| <b>05-140</b> | <b>910G547</b> |             |             |           | <b>Splined Chuck With Sidelay (KTR Coupling)</b>   | <b>430G562</b>   | <b>1</b> |
|               | B001           | M           | 235266G     |           | Compression Spring LHC 162N OBS                    |                  | 2        |
|               | B003           | M           | 422008      |           | Toggle Clamp DE-STA-CO Model 341                   |                  | 2        |
|               | B004           | E           | 675226G     |           | Proximity Switch 12mm IFS205                       |                  | 6        |
|               | B005           | M           | 203355G     |           | PLAIN GLACIER BEARING MB2010DU                     |                  | 2        |
|               | B006           | M           | 203233      |           | BUSH GLACIER MB2230DU                              |                  | 2        |
|               | B007           | M           | 242005      |           | LOCKNUT KM15                                       |                  | 1        |
|               | B008           | M           | 242006      |           | LOCKWASHER MB15                                    |                  | 1        |
|               | B010           | M           | 239135      |           | O' RING RM 0176-24                                 |                  | 2        |
|               | B011           | M           | 239661G     |           | Seal O Ring RM0345-30                              |                  | 2        |
|               | B013           | M           | 239255      |           | Seal O Ring RM0795-30 Nitrile                      |                  | 1        |
|               | B015           | M           | 239351      |           | Seal O Ring RM1645-30 Nitrile                      |                  | 1        |
|               | B016           | M           | 239340      |           | Seal O Ring RM1745-30 Nitrile                      |                  | 1        |
|               | B017           | M           | 239386      |           | Seal O Ring RM3593-57 Nitrile                      |                  | 1        |
|               | B018           | M           | 4231561G    |           | 1/2" BSP Plug C/w O-ring seal                      |                  | 2        |
|               | B020           | M           | 2061182G    |           | Linear Guide Block                                 |                  | 4        |
|               | B021           | M           | 2061250G    |           | Linear Rail Hiwin RGR35-R-200-H                    |                  | 4        |
|               | B022           | M           | 239380      |           | SEAL SIMRIT ROD NI300 20-30-10                     |                  | 2        |
|               | B023           | E           | 658001G     |           | Linear Transducer (Balluff)BIW1-A310-M0075-P1-S115 |                  | 1        |
|               | B024           | M           | 2442367G    |           | Precision Screwjack, 25kN, 60mm stroke             |                  | 1        |
|               | B027           | M           | 423155      |           | Nipple 45 DEG Grease 1/8"BSP GN1/8BSP45            |                  | 1        |
|               | B029           | M           | 239381      |           | Simmerring Wiper Seal AUAS 20-30-10                |                  | 2        |
|               | B030           | M           | 232455G     |           | KTR Radex Coupling NN Size 70 (65-70 BORE H7)      |                  | 1        |
|               | B032           | M           | 664512G     |           | SIEMENS 1PH8184-1DB13-2BD1 - 400RPM - 16.3Kw 49A   |                  | 1        |
|               | B034           | M           | 240020G     |           | Spring Plunger WDS 606-5712                        |                  | 1        |
|               | B036           | M           | 200M570     |           | BEARING 61828-2RS1                                 |                  | 1        |
|               | B037           | M           | 206T094     |           | KB 2045                                            |                  | 1        |
|               | B038           | M           | 204M090     |           | FLANGE BUSH GLACIER FMB 3016DU                     |                  | 2        |
|               | B042           | M           | 2061225G    |           | THK-LBF85-UUCL+350L                                |                  | 1        |
|               | B043           | M           | 232474G     |           | Gearax 'SD' 20 Hub                                 |                  | 1        |
|               | B044           | M           | 232487G     |           | Gearax 'SD' 20 sleeve                              |                  | 1        |
|               | B045           | M           | 232476G     |           | KTR Shiftable Linkage Size 3                       |                  | 1        |
|               | B046           | M           | 234010G     |           | External Circlip D1400-1400 (140mm Dia)            |                  | 1        |
|               | B047           | M           | 239926G     |           | Seal Simmerring BABSL90-110-7-5-72 Viton           |                  | 2        |
|               | B048           | M           | 234050      |           | CIRCLIP EXTERNAL 1400-0800                         |                  | 1        |
|               | B049           | M           | 200632      |           | CYLINDRICAL ROLLER BEARING N216 ECP                |                  | 1        |
|               | B050           | M           | 239198      |           | Seal O Ring RM0745-30 Nitrile                      |                  | 2        |
|               | B051           | M           | 239373      |           | Seal O Ring RM1245-30 Nitrile                      |                  | 1        |
|               | B052           | M           | 239368      |           | O' RING RM 1495-30                                 |                  | 1        |
|               | B053           | M           | 200704      |           | BEARING 3216A                                      |                  | 1        |
|               | B054           | M           | 423036      |           | Adaptor Male 8mm-1/4"BSPT 36-0530-11               |                  | 3        |
|               | B055           | M           | 423039      |           | Tubing Sleeve 8 OD                                 |                  | 5        |
|               | B056           | M           | 423038      |           | Tubing Nut 8 OD                                    |                  | 5        |
|               | B057           | M           | 201006G     |           | Bearing Inner Race IR80 x 90 x 25 - EGS            |                  | 1        |
|               | B058           | M           | 201007G     |           | Bearing Inner Race IR80 x 90 x 54 - EGS            |                  | 1        |
|               | B059           | M           | 423044      |           | Elbow Stem 8 OD                                    |                  | 2        |
|               | B060           | M           | 425003      |           | Oil Reservoir 1/4"BSP (Ref OR 7730)                |                  | 1        |
|               | B061           | M           | 423046      |           | Tube Half Hard Copper 8mm O/D CS7008003            |                  | 1        |
|               | M001           | M           | 424G339     |           | Bearing Support IB-Splined Chuck                   |                  | 1        |
|               | M002           | M           | 424G220     |           | Bearing Clamp IB-Splined Chuck                     |                  | 1        |
|               | M003           | M           | 424G420     |           | Spring Pin                                         |                  | 2        |
|               | M004           | M           | 427G719     |           | Splined Chuck Mounting Block                       |                  | 2        |
|               | M005           | M           | 428G777     |           | Bracket                                            |                  | 1        |
|               | M006           | M           | 425G045     |           | Pivot Pin                                          |                  | 2        |
|               | M007           | M           | 426G250     |           | SideShift Drive Block                              |                  | 1        |
|               | M008           | M           | 424G437     |           | Proximity Switch Bracket                           |                  | 2        |
|               | M009           | M           | 421G534     |           | Stop Block                                         |                  | 3        |
|               | M010           | M           | 427G214     |           | Stop Block                                         |                  | 1        |
|               | M014           | M           | 426G122     |           | Latch                                              |                  | 2        |
|               | M015           | M           | 418G653     |           | Spacer - Clamp                                     |                  | 2        |
|               | M017           | M           | 421G446     |           | Ferrofluidic Seal Housing                          |                  | 1        |
|               | M021           | M           | 424G407     |           | Bearing Block OB                                   |                  | 1        |
|               | M024           | M           | 428G949     |           | Chuck Support Bracket (Non Drive Side)             |                  | 1        |
|               | M025           | M           | 424G222     |           | Splined Chuck Bearing Clamp OB                     |                  | 1        |
|               | M026           | M           | 424G100     |           | Chuck Guard                                        |                  | 2        |
|               | M027           | M           | 424G106     |           | Nylon Wear Plate                                   |                  | 2        |
|               | M030           | M           | 427G707     |           | Push Rod                                           |                  | 1        |

# V18G019 - Bill of Materials



| Section | Bubble Number | Mech / Elec | Part Number | Parameter | Part Description                        | Assembly Drawing | Qty |
|---------|---------------|-------------|-------------|-----------|-----------------------------------------|------------------|-----|
|         | M031          | M           | 422G706     |           | Push Rod Housing                        |                  | 1   |
|         | M032          | M           | 403G630     |           | Seal Ring                               |                  | 1   |
|         | M033          | M           | 403G364     |           | Seal Housing                            |                  | 1   |
|         | M034          | M           | 417G144     |           | Bearing Retainer                        |                  | 1   |
|         | M035          | M           | 420G837     |           | Linear Transducer Bracket               |                  | 1   |
|         | M036          | M           | 420G838     |           | Linear Potentiometer Square Pin         |                  | 1   |
|         | M037          | M           | 430G612     |           | Splined Chuck Bell Housing              |                  | 1   |
|         | M039          | M           | 421G456     |           | Ball Spline Shaft                       |                  | 1   |
|         | M047          | M           | 424G252     |           | Ballspline Adaptor Housing              |                  | 1   |
|         | M056          | M           | 422G699     |           | Proximity Bracket                       |                  | 1   |
|         | M060          | M           | 429G106     |           | Proxi Switch bracket                    |                  | 1   |
|         | M061          | M           | 418G249     | IT24      | Key                                     |                  | 1   |
|         | M064          | M           | 418G249     | IT22      | Key                                     |                  | 1   |
|         | M066          | M           | 424G111     | IT1       | Chuck Shield Plate                      |                  | 1   |
|         | M067          | M           | 424G111     | IT2       | Chuck Shield Plate                      |                  | 1   |
|         | M068          | M           | 427G225     |           | Stop Block                              |                  | 1   |
|         | M071          | M           | 424G967     |           | KTR Gear Retaining Washer               |                  | 1   |
|         | M072          | M           | 425G033     |           | SPLINED CHUCK GEAR                      |                  | 1   |
|         | M073          | M           | 426G038     |           | KTR Shiftable Linkage Mods              |                  | 1   |
|         | M074          | M           | 427G550     |           | Proxi switch bracket                    |                  | 1   |
|         | M076          | M           | 427G551     |           | Proxi switch flag                       |                  | 1   |
|         | M077          | M           | 427G650     |           | Stop Block                              |                  | 1   |
|         | M083          | M           | 428G959     |           | YOKE                                    |                  | 1   |
|         | M084          | M           | 428G967     |           | Catch Bracket                           |                  | 1   |
|         | M085          | M           | 428G971     |           | Bearing Cap                             |                  | 1   |
|         | M086          | M           | 428G966     |           | CAM PLATE                               |                  | 2   |
|         | M088          | M           | 428G970     |           | CAM                                     |                  | 2   |
|         | M089          | M           | 430G388     |           | Seal Shaft - Spline Chuck               |                  | 1   |
|         | M090          | M           | 430G389     |           | Outer Bearing Housing                   |                  | 1   |
|         | M091          | M           | 430G395     |           | Cartridge Seal Cap Rear                 |                  | 1   |
|         | M092          | M           | 430G405     |           | Shaft Seal Spacer                       |                  | 1   |
|         | M093          | M           | 430G427     |           | Bearing Spacer                          |                  | 1   |
|         | M094          | M           | 430G428     |           | Bearing Spacer                          |                  | 1   |
|         | M095          | M           | 430G429     |           | Shaft Sleeve                            |                  | 1   |
|         | M096          | M           | 430G431     |           | Retaining Ring                          |                  | 1   |
|         | M097          | M           | 430G592     |           | Oil Bottle Bracket                      |                  | 1   |
|         | M098          | M           | 430G654     |           | Lubricant Label - Chuck - Oil AWS68     |                  | 1   |
|         | S001          | M           | 421658      |           | SOCKET CAP SCREW M4 x 25 GRADE 12.9     |                  | 4   |
|         | S002          | M           | 421019      |           | Socket Head Capscrew M6 x 20 GRADE 12.9 |                  | 20  |
|         | S003          | M           | 421225      |           | SOCKET CAP SCREW M6 x 60 GRADE 12.9     |                  | 3   |
|         | S004          | M           | 421049      |           | SOCKET CAP SCREW M8 x 20 GRADE 12.9     |                  | 12  |
|         | S005          | M           | 421138      |           | SOCKET CAP SCREW M8 x 25 GRADE 12.9     |                  | 16  |
|         | S006          | M           | 421237      |           | SOCKET CAP SCREW M8 x 35 GRADE 12.9     |                  | 26  |
|         | S008          | M           | 421149      |           | SOCKET CAP SCREW M10 x 20 GRADE 12.9    |                  | 2   |
|         | S009          | M           | 421035      |           | SOCKET CAP SCREW M10 x 25 GRADE 12.9    |                  | 2   |
|         | S010          | M           | 421018      |           | SOCKET CAP SCREW M10 x 35 GRADE 12.9    |                  | 8   |
|         | S011          | M           | 421706      |           | SOCKET CAP SCREW M10 x 60 GRADE 12.9    |                  | 4   |
|         | S012          | M           | 421195      |           | SOCKET CAP SCREW M12 x 40 GRADE 12.9    |                  | 9   |
|         | S015          | M           | 421115      |           | SOCKET CAP SCREW M16 x 45 GRADE 12.9    |                  | 4   |
|         | S016          | M           | 421617      |           | SOCKET CAP SCREW M16 x 65 GRADE 12.9    |                  | 6   |
|         | S018          | M           | 421194      |           | Washer Plain M4                         |                  | 8   |
|         | S019          | M           | 421034      |           | WASHER PLAIN M6                         |                  | 11  |
|         | S020          | M           | 421054      |           | PLAIN WASHER M8                         |                  | 6   |
|         | S021          | M           | 421029      |           | PLAIN WASHER M10                        |                  | 4   |
|         | S022          | M           | 421030      |           | PLAIN WASHER M12                        |                  | 5   |
|         | S023          | M           | 421059      |           | Washer Plain M16                        |                  | 32  |
|         | S025          | M           | 421073      |           | Washer Spring M12                       |                  | 4   |
|         | S026          | M           | 421226      |           | GRUB SCREW M6x10                        |                  | 2   |
|         | S027          | M           | 4211098     |           | GRUB SCREW DOG POINT M10x15             |                  | 2   |
|         | S028          | M           | 421180      |           | Nut Full Hex M4                         |                  | 4   |
|         | S029          | M           | 421057      |           | NUT FULL HEXAGON M6                     |                  | 1   |
|         | S034          | M           | 421258      |           | DOWEL 6 dia x 40 mm LONG                |                  | 2   |
|         | S036          | M           | 421418      |           | Screw Socket Countersunk Head M6 x 16   |                  | 4   |
|         | S043          | M           | 421104      |           | SOCKET CAP SCREW M16 x 50 GRADE 12.9    |                  | 6   |
|         | S046          | M           | 421633      |           | Locknut M20                             |                  | 4   |
|         | S048          | M           | 421148      |           | GRUB SCREW M8x30                        |                  | 1   |
|         | S057          | M           | 421234      |           | SOCKET CAP SCREW M12 x 30 GRADE 12.9    |                  | 4   |
|         | S058          | M           | 421710      |           | Screw Socket M12x30 Countersunk Head    |                  | 6   |